

The Good, the Bad and the Plasticity Genes: The Role of both Positive
and Negative Environmental Factors and the SERT KO Genotype on the
Development of ASD-Like Symptoms in a Rodent Model

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Abstract

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental disorder with a number of contributing genetic and environmental factors. While a number of studies have looked at the negative environmental factors underpinning ASD, few have investigated the ameliorating effects of a combination of positive environmental treatments and genetic factors. The current study investigated whether 24 hours of maternal deprivation (MD, constituting a negative environment) on PND9, in combination with the serotonin transporter knockout (SERT KO) genotype would increase the autistic-like characteristics of male Wistar rats, and additionally, whether early handling (EH, constituting a positive environment) from PND2-14 in combination with the same genotype would decrease these behaviours. Five experiments were used: social interaction, social approach and avoidance, scent marking, hole dipping and reversal learning. The results showed that SERT KO MD rats have autistic-like social impairments, while the SERT KO EH rats show improvements on social and communicative tasks, indicating that, in some behavioural measures, both positive and negative environmental factors have a significant influence on the SERT KO genotype. These results highlight the importance of serotonin and early-life stress in the development of ASD, as well as the role of positive environmental factors in the treatment of this disorder.

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The Good, the Bad and the Plasticity Genes: The Role of both Positive and Negative Environmental Factors and the SERT KO Genotype on the Development of ASD-Like Symptoms in a Rodent Model 1

Autism Spectrum Disorder (ASD) is a multifaceted neurodevelopmental disorder that is characterised by impairments in social and communicative behaviours, along with stereotyped repetitive behaviours and restricted interests (American Psychiatric Association (APA), 2013). It is estimated that one percent of children are diagnosed with autism in New Zealand (Ministry of Education, 2015), with males being four to five times more likely to receive this diagnosis (CDC, 2014). Social and communicative impairments in ASD include: abnormal social and emotional reciprocity, impaired verbal and non-verbal communication, and reduced ability to develop and maintain relationships. Stereotyped behaviours, on the other hand, include repetitive motor movements, insistence on sameness, restricted interests and adherence to routines (APA, 2013). These abnormal behaviours are usually first observed in early childhood, prior to the age of three (APA, 2013; Corsello, 2005; DiCicco-Bloom et al., 2006; Levy, Mandell, & Schultz, 2009), and are present throughout an individual's life, often with an apparent increased severity during adolescence, most likely as a result of increased social and educational demands (CDC, 2014). Individuals that receive behavioural intervention prior to 48 months of age show significantly improved outcomes, such as an increased likelihood of independent living in adulthood, compared to those that receive intervention at a later stage (Corsello, 2005; DiCicco-Bloom et al., 2006; Harris & Weiss, 1998; Levy et al., 2009). As ASD is one of the most severe developmental disorders, in relation to prevalence, outcome, family impact and societal cost (DiCicco-Bloom et al., 2006), further understanding of the etiology of autism is vital, in order to increase the effectiveness of screening, early intervention and treatment (Corsello, 2005; Levy et al., 2009).

The majority of autism related research has focused on the social and communicative² impairments that are characteristic of ASD, resulting in a severe lack of information concerning the repetitive behaviours that also occur (Langen, Durston, Kas, van Engeland, & Staal, 2011b; Lewis & Bodfish, 1998; Lewis, Tanimura, Lee, & Bodfish, 2007; Powell et al., 1999; Rutter, 1996; Wotruba & Scattoni, 2013). Furthermore, of the studies that have investigated stereotypical behaviours, most have concerned drug-induced, rather than spontaneous repetitive behaviours (Powell et al., 1999). Stereotypic behaviours are invariant repeated movements or actions that do not appear to have an apparent purpose or function, and are usually perceived as being inappropriate for the environment (APA, 2013; Goldman et al., 2009; Lewis & Bodfish, 1998; Mason, 2006). Repetitive behaviours occur in typical development, as they are a crucial means of learning new skills in infancy, however, while these developmental stereotypies are adaptive, the stereotypies that occur in obsessive compulsive disorder (OCD) and ASD result in significant impairments to the day-to-day functioning of the individual (Langen et al., 2011b). Examples of stereotypies in humans include: hand flapping, tapping, twisting, clapping and waving, chin rubbing, body rocking, echolalia (repetition of speech), head tilting and nodding, jumping, pacing and self-injurious behaviours (Goldman et al., 2009; Langen et al., 2011b; Lewis & Bodfish, 1998; Lewis et al., 2007; Powell et al., 1999). These repetitive behaviours are often referred to as *stimming*, due to their self-stimulatory nature (Goldman et al., 2009). Along with repetitive motor movements, individuals with ASD also show cognitive rigidity, including: insistence on sameness, distress at unexpected changes to regular routines, and specialized interests (Goldman et al., 2009; Langen et al., 2011b; Lewis et al., 2007). Very few studies have been conducted concerning these higher order cognitive impairments, perhaps due to the more salient nature of the motor stereotypies that also occur (Langen et al., 2011b). The lack of research on stereotypies is concerning, as these repetitive behaviours are associated with

poorer outcomes for autistic individuals (Levy et al., 2009). These behaviours are also 3
extremely salient and disruptive, particularly in a classroom environment, which often results
in teasing from peers, and increased social isolation in individuals that already struggle with
social situations (Goldman et al., 2009; Langen et al., 2011b; Levy et al., 2009). Because of
these issues, further research into stereotypies is essential, requiring the development of
ecologically valid animal models of these behaviours.

Genetics of ASD

Further research concerning the genetics of ASD is also very important, as
autism is a highly heritable disorder (Brown, 2009; DiCicco-Bloom et al., 2006).
Monozygotic twins show a concordance of 60% for ASD specifically, and 80-90% in terms
of the broad autistic phenotype, which includes the general presence of social impairments
and cognitive abnormalities (Bailey et al., 1995; Hallmayer et al., 2011). While this high
concordance rate does indicate that the development of ASD has a strong genetic basis, this
disorder is also highly heterogeneous and complex, with a large degree of variation in the
type and severity of behaviour that is observed in each individual (Brown, 2009; DiCicco-
Bloom et al., 2006). For example, some individuals diagnosed with ASD have very severe
symptoms, including a complete lack of speech and reduced intelligence, while others, who
would receive the diagnosis of Asperger's Disorder under the previous version of the
Diagnostic and Statistical Manual (DSM-IVR), have impairments in social behaviours, but
average to above-average intelligence (Brown, 2009). This phenotypic heterogeneity, and the
fact that many different impairments can lead to similar symptom presentation, indicates that
autism does not result from a single genetic mutation (Bleuler, 1996; DiCicco-Bloom et al.,
2006; Levy et al., 2009; Narita et al., 2010). Instead, it appears that a number of risk factors
interact to produce the autistic phenotype, including both gene-gene and gene-environment
interactions (Brown, 2009; DiCicco-Bloom et al., 2006; Levy et al., 2009; Lewis et al., 2007).

While there is no single genetic cause of ASD, a large number of genetic risk factors ⁴ have been associated with this disorder (Brown, 2009). As symptom onset occurs prior to the age of three, ASD appears to be a result of disruptions to prenatal and early postnatal brain development (DiCicco-Bloom et al., 2006), which may explain the heterogeneous symptomology, as there are a wide variety of different genetic and environmental factors that are involved in neurodevelopment (Courchesne, 2004; Levy et al., 2009). Known chromosomal abnormalities, copy number variations (CNVs) and point mutations are estimated to account for the development of autism in 20% of subjects (Brown, 2009). In addition to these genetic impairments, a disruption to the regulation of serotonin has also been linked to the development of ASD.

Serotonin

Disruptions to the serotonin system, specifically the serotonin re-uptake transporter, have been consistently linked to the development of ASD and are a promising target for current research (Iwata et al., 2014; Kalueff et al., 2010; Levy et al., 2009; Rapport, Green & Page, 1948). Serotonin is a neuromodulator that is highly distributed throughout the brain, and subsequently plays an important role in a number of processes, including; the sleep-wake cycle, mood, motivation, emotion regulation, inhibitory control, appetite and motor function (Fox, Chuang, & Brotchie, 2009; Golden & Gilmore, 1990; Heils et al., 1996; Kalueff et al., 2010; Murphy et al., 1998). Serotonin also plays an important role in neurodevelopment, as it is involved in processes such as synaptogenesis and cell proliferation (Lauder, 1990), which may explain why abnormal serotonin levels have been implicated in neurodevelopmental disorders, including ASD (Iwata et al., 2014). Further evidence for the role of serotonin in ASD includes: increased blood serotonin levels found in autistic subjects when compared to controls (Anderson et al., 1987), the finding that stereotypical behaviour can be induced via stimulation of postsynaptic serotonin receptors (Curzon, 1990), and that there has been some

success using selective serotonin reuptake inhibitors (SSRIs) to reduce repetitive behaviours 5 in autism (Kolevzon, Matthewson, & Hollander, 2006). In fact, the two FDA approved drugs for the treatment of autism, risperidone and aripiprazole, have a strong influence on the serotonergic system, and have been found to reduce stereotypical behaviours in autistic individuals (McPheeters et al., 2011). In relation to repetitive behaviours, abnormal serotonin levels have also been implicated in the development of movement disorders, such as Parkinson's Disease (PD) (Fox et al., 2009), most likely as a result of the modulatory effects of serotonin on other neurotransmitter systems, such as dopamine, GABA and glutamate (Nicholson & Brotchie, 2002). Specifically, serotonergic drugs appear to mediate the severity of tremors in PD (Fox et al., 2009; Gordon, Pullman, Louis, Frucht, & Fahn, 2002), indicating that serotonin might also play a role in other such abnormal motor symptoms, including the repetitive motor movements that are observed in ASD (Beatty, Staton, Weir, Monson, & Whitaker, 2007; Cools, van den Bercken, Horstink, van Spaendonck, & Berger, 1984; Diaz-Santos et al., 2015).

The role of SERT and the short/long polymorphism. As serotonin has been implicated in the development of ASD, a promising research target is the mechanism by which extracellular serotonin levels are controlled; the serotonin reuptake transporter (SERT) (Caspi et al., 2002; Heils et al., 1996; Iwata et al., 2014; Murphy, Lerner, Rudnick, & Lesch, 2004). SERT regulates extracellular serotonin levels by controlling the reuptake of serotonin from the synaptic cleft to the presynaptic neuron, where it is then either re-used or degraded (Murphy et al., 1998). As the serotonin transporter is the main mechanism of removing serotonin from the synaptic cleft, a disruption in the SERT gene, located on chromosome 17q in humans, leads to an excess of extracellular serotonin (Greenberg et al., 1999; Homberg et al., 2007b; Kalueff et al., 2010; Murphy et al., 2004; Olivier et al., 2010). This further results in decreased SERT receptor density and desensitisation of the SERT receptors (Albert &

Lemondé, 2004; Albert, Lembo, Storrington, Charest, & Saucier, 1996). Approximately 30% of Caucasians have a 44 base pair deletion in a highly repetitive GC base pair sequence in the promoter region of the *SLC6A4* serotonin transporter gene, making this polymorphism a significant risk factor at a population level (Caspi et al., 2002; Heils et al., 1996; Lesch et al., 1996). This short allele (s) polymorphism in the serotonin transporter-linked polymorphic region (5HTTLPR), reduces the transcriptional efficiency of the SERT gene, resulting in a 40% decrease in serotonin reuptake, which subsequently alters the serotonergic response to environmental factors, such as stress (Caspi et al., 2002; Greenberg et al., 1999; Homberg et al., 2007b). This is particularly the case if individuals are homozygous for this mutation, and subsequently have two copies of the s allele (Heils et al., 1996; Lesch et al., 1996). Because of the vital role of SERT in regulating serotonin levels, and the importance of serotonin in a variety of processes, abnormal SERT function has been implicated in a number of psychiatric disorders, including; schizophrenia, substance abuse, anxiety, and depression (Golden & Gilmore, 1990; Heils et al., 1996; Homberg et al., 2007b; Murphy et al., 1998; Murphy et al., 2004). In relation to ASD, hyperserotonemia has been found in autistic subjects when compared to controls (Anderson et al., 1987). Furthermore, autistic subjects have been found to have a disruption in the expression of SERT binding proteins, such as *N*-ethylmaleimide-sensitive factor (NSF), which leads to a reduction in the number of SERT proteins present at the presynaptic membrane, resulting in an excess of serotonin remaining in the synaptic cleft, due to an impairment to this re-uptake mechanism (Iwata et al., 2014). There is also some evidence of an association between the s allele variant of the SERT gene and ASD (Cook et al., 1997), however these findings have not been consistently replicated, most likely due to the heterogeneous nature of ASD (Amaral, Geschwind & Dawson, 2011). In addition, several other genetic variations in the SERT have been linked to ASD, in particular the STin2 polymorphism, which involves a variable number of tandem repeats in the second intron,

with the number of repeats (nine and ten) being positively associated with ASD (Warrier, 7
Chee, Smith, Chakrabarti, & Baron-Cohen, 2015).

Environmental Factors in ASD

While it was initially estimated that the heritability for autism was around 90% (Folstein & Rosen-Sheidley, 2001; Freitag, 2007), more recent twin research proposes an estimate of around 50-60% heritability of strict autism for monozygotic males, and 20% for dizygotic male twins, with a slightly higher heritability for females (Hallmayer et al., 2011). This suggests that environmental factors are partially responsible for the development of autism, and that both genetic and environmental factors interact to produce the autistic phenotype (Brown, 2009; Crawley, 2007; DiCicco-Bloom et al., 2006; Levy et al., 2009). As with genetic factors, there does not seem to be a single environmental risk factor that accounts for the development of ASD (Dietert, Dietert, & Dewitt, 2011). For example, advanced parental age has been shown to be an ASD risk factor, due to increased likelihood of pregnancy complications in females (Croen, Najjar, Fireman, & Grether, 2007). Maternal stress also appears to be an environmental risk factor for ASD (Kinney, Miller, Crowley, Huang, & Gerber, 2008a; Kinney, Munir, Crowley, & Miller, 2008b; Narita et al., 2010), as mothers that experienced severe hurricanes during the middle or end of their pregnancy, were found to have a four times increased risk of having an autistic child (Kinney et al., 2008a). Other examples of early environmental risk factors include; prenatal exposure to the antinausea drug thalidomide (Narita et al., 2010; Stromland, Nordin, Miller, Akerstrom, & Gillberg, 1994), treatment with the anticonvulsant drug valproate (VPA) during pregnancy (Bromley, Mawer, Clayton-Smith, & Baker, 2008; Narita et al., 2010; Rodier, Ingram, Tisdale, Nelson, & Romano, 1996) and gestational exposure to viral or bacterial infections, such as influenza (Zerbo et al., 2013).

development of ASD, there are also particular periods of development that are more critical, meaning that environmental influences have a stronger effect during these periods (Bleuler, 1966; Hensch, 2004; LeBlanc & Fagiolini, 2011). It is likely that early environmental factors play a critical role in the development of ASD due to the malleability of the brain during the important prenatal and early postnatal developmental stages (Courchesne, 2004; Dietert et al., 2011; Hensch, 2004). For instance, maternal stress during gestation results in the release of stress hormones, which are able to cross the blood brain barrier in the foetus. This can result in permanent alterations to the Hypothalamus-Pituitary-Adrenal (HPA) axis, increasing anxiety in the child, and potentially leading to the abnormal social behaviours seen in ASD (Brown, 2009; Huizink, De Medina, Mulder, Visser, & Buitelaar, 2002). As the HPA axis is important in the maintenance of homeostasis, and is involved in directing energy to vital organs in times of stress, disruptions to this system during early life can result in a number of issues in later life, such as learning difficulties, anxiety and depression (De Kloet, Vreugdenhil, Oitzl, & Joels, 1998; Huizink et al., 2002; Levine, 2000; Weinstock, 2008). It has also been theorised that maternal stress during important developmental periods results in an increase in testosterone levels, which further results in the development of an *extreme male brain*, and later, leads to the development of autism (Baron-Cohen, 2002). This theory provides a possible explanation as to why males are four times more likely to develop autism than females (Brown, 2009).

In sum, it appears that a number of different genetic and environmental factors are involved in the development of the heterogeneous phenotype of autism, meaning that an abundant amount of research is required in order to gain better insight into these factors, and whether interactions between these factors occur. Although there is a severe lack of clinical literature concerning gene-environment interactions in ASD, Nijmeijer and colleagues (2010)

did find that maternal smoking during pregnancy, in combination with the s/s allele SERT 9 genotype, resulted in an increase in ASD-like behaviours in children with ADHD, providing some support for the role of gene-environment interactions in ASD.

The Rat Model of ASD

In order to gain further insight into the underlying factors that are involved in the development of autism, animal models can be used, to better study the behavioural effects of genetic and environmental manipulations of interest (Belzung, Leman, Vourc'h, & Andreas, 2005; DiCicco-Bloom et al., 2006; Smits et al., 2006; Wohr & Scattoni, 2013). Animal models are also useful in terms of looking at the causal relationship between early life events and subsequent development of disorders, which is difficult to determine in clinical studies for ethical reasons (Marco et al., 2015). There are a number of techniques that can be used to create animal models with a disease phenotype, including pharmacological, genetic and environmental manipulations (Kalueff et al., 2010; Smits et al., 2006; Wohr & Scattoni, 2013). The underlying genetic factors involved in disorders of interest need to be isolated and investigated in order for effective and targeted treatment to be developed, and for this reason, genetic knockout models are often used (Smits et al., 2006; Wohr & Scattoni, 2013). Many different techniques exist for altering the genotype of an animal, including homologous recombination, ENU mutagenesis, zinc finger nucleases and the CRISP/CAS technique (Hogan, Constantini, & Lacy, 1994; Wohr & Scattoni, 2013). It would be well beyond the scope of this thesis to discuss these techniques, suffice to say that each has advantages as well as limitations.

Rodents are the most commonly used animal models in the behavioural neuroscience field, with rats being preferred over mice, as their larger size is more practical for experimentation, and their physiology is more similar to that of humans (Belzung et al., 2005; "Rats!", 2010; Sengupta, 2014). An important aspect in the translation between rodents and

humans, especially for childhood onset disorders such as ASD, is the difference in the developmental timeline between rats and mice on one hand, and humans on the other hand. Thus, while rodents are born prematurely compared to humans, the development that occurs after birth is much more rapid. Therefore, it is estimated that the equivalent of human “birth” occurs around postnatal day (PND) 12, with adulthood occurring around PND63 (Sengupta, 2014).

As there have been difficulties integrating animal and clinical research, it is important that animal studies meet particular clinical criteria, to enable the findings to be interpreted across species (Powell et al., 1999). Due to the heterogeneous nature of ASD, it is not possible to produce a single animal model that accurately reflects the many genetic and environmental factors that give rise to this disorder (DiCicco-Bloom et al., 2006). However, despite this difficulty, it is essential that animal models of autism have similar behavioural characteristics to what is found in ASD in humans, by displaying the triad of impairments in social, communicative and repetitive behaviours (Crawley, 2007; DiCicco-Bloom et al., 2006). Rats display a wide range of social behaviours, such as: juvenile rough and tumble play, parental care, and sexual and aggressive behaviours in adulthood (Belzung et al., 2005; Crawley, 2007; Wohr & Scattoni, 2013). This extensive variety of social behaviours is particularly important when the neurodevelopmental disorder of interest is predominantly characterised by deficits in socialisation, as is the case with ASD. A large number of social behavioural measures can then be used, to better enable researchers to determine whether genetic and environmental factors have in fact given rise to autistic-like behaviours (Wohr & Scattoni, 2013). Behavioural measures such as the social interaction paradigm and the three chambered social approach task are commonly used to measure whether autistic-like social impairments are present in rat models (Crawley, 2007; DiCicco-Bloom et al., 2006). In addition to the intricate social behaviours shown, rats also show complex communication, in

the form of ultrasonic vocalisations (USVs) (Belzung et al., 2005; Crawley, 2007; 11 McIntosh, Barfield, & Geyer, 1978; Wohr & Scattoni, 2013). This is an important characteristic in terms of ASD research, as the impairment in communication that occurs in autism is notoriously difficult to study in animal models, due to the lack of human-like language expression (Belzung et al., 2005). Additionally, scent marking behaviour is an excellent measure of social communication in rodents, as it is a means of both territorial and reproductive communication (Achiraman et al., 2011; Arakawa, Blanchard, Arakawa, Dunlap, & Blanchard, 2008; Wohr & Scattoni, 2013). Finally, rats are also capable of showing complex ritualistic and repetitive behaviours, including repetitive motor movements, such as grooming, meandering and head-weaving, as well as *perseveration*, where a behaviour is continued despite the lack of an appropriate stimulus (Sandson & Albert; 1984), as measured by reversal learning and hole-dipping tasks (Belzung et al., 2005; Langen et al., 2011b; Lewis et al., 2007; Wohr & Scattoni, 2013).

Stereotypy

As with repetitive behaviours in humans, stereotypies can be adaptive in rodents, and are an important part of learning skills which can aid with survival, however, stereotypies of a persistent and abnormal nature can also develop (Langen et al., 2011a; Lewis et al., 2007). Repetitive behaviours of interest in the rat include: excessive self-grooming, rearing, rotation, head weaving and meandering (Langen et al., 2011a; Lewis et al., 2007; Wohr & Scattoni, 2013). As well as these lower level stereotypies, which can be used to reflect stimming in humans with autism, there are also more complex restricted and repetitive behaviours that can occur, which mimic the cognitive rigidity that occurs in ASD in humans (Lewis et al., 2007; Wohr & Scattoni, 2013). These higher order perseverative behaviours can be modelled in rats through tasks such as reversal learning, where the subject is provided with a reward at a particular location over a number of consecutive trials until the subject has learned the

location of the target (Lewis & Bodfish, 1998; Stephens, Weidmann, Quartermain, & 12
Sarter, 1985; Wang, Li, Du, Shao, & Wang, 2015). Once the location of the target has been
learned, this location is then changed, and the subject must then learn the new target location.
Rats with highly restricted behaviours will continue to visit the original target location for a
significantly longer period than control rats, showing their cognitive inflexibility and fixation
with following particular routines (Wang et al., 2015). Another paradigm that is used to
measure perseveration in rodent models, and will be used as a task in the current study, is
hole-board dipping (File & Wardill, 1975; Mekanjuola, Hill, Maben, Dow, & Ashcroft, 1977;
Wang et al., 2015). In this task, rats are placed on a hole-board apparatus floor, containing 16
holes, and the number of times they poke their nose into each hole is then counted. Rats with
stereotypies show increased re-entries into the holes when compared to controls, who show a
more random pattern of hole-visits (Mekanjuola et al., 1977). By studying the presence and
extent of these repetitive disorder-like behaviours in rats, valuable insight can be made into
the underlying genetics of ASD in rodents, which can then be translated into a human clinical
model.

SERT in Rodents

SERT function has been found to be relatively conserved in rats when compared to
humans, and N-ethyl-N-nitrosourea (ENU) mutagenesis has previously been used to create
SERT knockout (KO) rat models that reflect the human condition (Homberg et al., 2007b;
Kalueff et al., 2010; Smits et al., 2006). ENU targeted mutagenesis induces genome-wide
mutations, which can result in premature stop codons being produced in genes of interest,
such as the SERT gene (Smits et al., 2006). These SERT KO rodents have autistic-like
behaviours, including an increase in stereotypy, as shown by meandering and self-grooming
behaviours, and a decrease in social approach behaviour, as shown by a reduction in the
amount of time spent with the novel mouse in the three chamber social approach task

(Kalueff et al., 2010; Moy et al., 2009). Both homozygous and heterozygous SERT KO 13 rats can be used to mimic autism in humans, although they show a gene-dosage effect, with the heterozygous knockouts showing a reduction in SERT expression and function compared with controls, and the homozygous knockouts having a complete lack of functional SERT protein (Homberg et al., 2007a; Homberg et al., 2008). Although the homozygous SERT knockout rats have the most severe genetic vulnerability in terms of the development of autistic symptoms, with a nine fold increase in extracellular serotonin levels, the heterozygous knockouts are a more reliable representation of the genetic susceptibility in the human condition, as their reduced SERT levels are comparable to humans with the *s-allele* genotype (Homberg et al., 2007a; Homberg et al., 2007b; Kalueff et al., 2007).

Environmental Factors in the Rodent Model of ASD

As well as genetic disturbances, the role of environmental factors in the development of ASD has also been studied in animals. In fact, it is often easier to determine the role of environmental factors in an animal model, as environmental influences can be more easily controlled than in humans (Levine, Chevalier, & Korchin, 1956). A classic example of the behavioural effects of environmental risk factors in animals is the maternal separation study conducted by Harlow and colleagues (1965). To mimic social isolation in humans, the researchers kept rhesus monkeys in empty wire cages from birth onwards. They found that maternal deprivation in the macaques led to similar adverse behavioural symptoms in later life as what occurs in humans, including anxiety, impaired socialisation and stereotypical movements, such as persistent rocking motions. However, whilst the Harlow study provided valuable insight into the effects of early isolation on behavioural symptoms, the complete isolation of the monkeys was a very extreme example of an adverse environmental condition, as the subjects were isolated from both their mothers and their peers. Less severe environmental manipulations have also been conducted in an attempt to investigate the role

of these factors on the development of neurodevelopmental disorders. For instance, based 14
on the finding that thalidomide and valproate treatment during pregnancy increases the risk of
the unborn child later developing autism, these treatments have also been used in rodent
models (Narita et al., 2010; Scheider & Przewlocki, 2005). Schneider and Przewlocki (2005)
exposed pregnant dams to valproic acid on day 12.5 of gestation, and found that the juveniles
of these mothers exhibited autistic-like behaviours including a decrease in socialisation,
decreased awareness of social chemosignals and increased repetitive behaviours. A rat model
of thalidomide exposure has also been created, where rats were exposed to thalidomide
(THAL) on gestational day nine, mimicking the critical period of exposure in humans (Narita
et al., 2010). These THAL rats showed increased hyperactivity in unfamiliar environments,
and reduced exploratory behaviours. Importantly, these models show that environmental
factors that increase the risk of ASD in humans can also act as risk factors in animals (Narita
et al., 2010).

Environmental conditions and the development of stereotypy. While stereotypical
behaviours are usually associated with neurological and developmental disorders in humans,
in animals, these behaviours can often arise as a result of negative environmental conditions
(Dantzer, 1991; Mason & Rushen, 2008; Powell et al., 1999). In fact, the development of
stereotypies is the most common behavioural impairment observed in confined animals
(Dantzer, 1991; Lewis et al., 2007). It was originally suggested that these behaviours arose
purely as a result of limited space (Ridley, 1994), however, it now seems more likely that
these stereotypies are a means of coping with stress resulting from negative environmental
factors, and are also due to permanent changes to the central nervous system (Latham &
Mason, 2008; Weinstock, 2008). For example, mice housed in normal cages show
significantly more stereotypical behaviours than those housed in larger and enriched
enclosures (Powell et al., 1999) and the primates in the Harlow (1965) study that experienced

maternal deprivation showed high amounts of body rocking and tail biting behaviours. It is also likely that some forms of stereotypical behaviours arise as a form of adjustment to lack of maternal care, such as the self-clutching and rocking stereotypies that developed in Harlow's macaques (Harlow et al., 1965; Latham & Mason, 2008). Finally, some stereotypies appear to develop as a means of attempting to escape, such as the bar-biting that occurs in lab mice, as biting occurs more frequently in sections of the cage that are opened more often (Wurbel & Stauffacher, 1997). It appears that these environmental restrictions result in permanent neurobiological changes, as although environmental enrichment can reduce the severity of stereotypies, these behaviours are usually very robust once developed, particularly when they are a result of early social deprivation (Langen et al., 2011b; Mason & Rushen, 2008). Consequently, it appears that environmental stress plays an important role in the development of abnormal stereotypies in both humans and animals (Langen et al., 2011b).

Environmental Manipulations: Maternal Deprivation and Early Handling

Maternal Deprivation

Although the aforementioned environmental stressors (such as confined space) can clearly lead to increased stereotyped behaviour, they are less relevant for the present research, as ASD develops very early in life. Hence, the current study will investigate the effects of two early life stressors, namely maternal deprivation (MD) and early handling (EH) on SERT KO rats, as such early life events are known to lead to the development of behavioural alterations in later life (Belsky et al., 2009; Caspi et al., 2002; Levine et al., 1956; Mason, 2006). In humans, maternal deprivation occurs when a child's attachment to their mother is impaired in some way, whether due to loss, or lack of care (Rutter, 1972). This deprivation can lead to a number of issues, such as psychopathy, increased stress responsivity, anxiety and depression (Eisenberg & Earls, 1975; Fisher et al., 1997; Kuhn & Schanberg, 1998; Mason, 2006). A classic example of maternal deprivation in humans is the case of the

Romanian orphans, who developed intellectual, communicative, motor and social impairments, after experiencing prolonged periods of isolation from adults and other children (Fisher et al., 1997; Mason, 2006). Some individuals, particularly those that had been institutionalised for a number of years before being adopted, continued to display these behavioural impairments long after their adoption, indicating that the length of deprivation plays a critical role in symptom severity (Fisher et al., 1997).

Maternal deprivation in rodents is not intended to mimic maternal deprivation in humans as such, particularly because the manipulation on PND9 occurs at a developmental period associated with the perinatal period in humans, rather than early childhood (Marco et al., 2015; Sengupta, 2014). However, similarly to maternal deprivation in humans, MD in rats is a highly stressful event during a critical developmental period, which leads to significant alterations in behaviour in adulthood (Marco et al., 2015; Plotsky & Meaney, 1993). In terms of the length of the MD in rodents, when the mother is removed from the home cage for a relatively short (one hour) period, there is no significant increase in stress sensitivity when the pups are later tested, however, there is a moderate increase in the stress response when the mother is removed for an eight hour period, and a dramatic increase after a 24 hour period of deprivation (Stanton, Gutierrez, & Levine, 1988). This effect is particularly severe when the period of deprivation occurs on postnatal day nine (PND9), as this is a critical period of neurodevelopment where the rodent brain is especially sensitive to the influence of stress (Ellenbroek & Riva, 2003; Levine, 1994; Marco et al., 2015; Narita et al., 2010). When MD rats are tested in adulthood, they show a number of behavioural impairments, including: a reduction in the amount of time spent in the center of the open field when compared to controls, an indicator of anxiety (Meaney, 2001), reduced acoustic startle habituation (Ellenbroek, de Bruin, van Den Kroonenburg, van Luijtelaar, & Cools, 2004) as well as a reduction in weight, which may be an indicator of impaired serotonergic activity in these rats,

as serotonin plays an important role in appetite control (Blundell, 1984; Rentesi et al., 17 2010). MD rats also have increased serotonin 1A receptors in the hippocampus, and increased serotonin 2A receptors in the cortex, when compared to controls, again highlighting the importance of serotonin in the development of these behavioural impairments (Vazquez, Lopez, Van Hoers, Watson, & Levine, 2000).

It is theorised that the long-term negative behavioural consequences of maternal deprivation in rats are due to three factors: reduction in temperature when the mother is removed, as the pups have poor thermoregulation at PND9, reduction in nutrients, as the pups have not yet been weaned at this stage, and lastly, reduction in tactile stimulation and maternal care (Kuhn & Schanberg, 1998; Marco et al., 2015; Rotta et al., 2003; Stanton et al., 1988). While all three of these factors interact to produce the behavioural impairments seen in MD, lack of maternal care appears to be the most important of these three factors, as pups that are placed in heat chambers during MD, or provided with nutrients, continue to show behavioural impairments, however these impairments are reduced when the pups are provided with passive contact with an adult female rat (Kuhn & Schanberg, 1998; Marco et al., 2015; Stanton et al., 1988).

MD rats show increased stress sensitivity, due to a hyper-reactivity of the HPA axis, which is usually in a hypo-responsive state during the first two weeks of development in rats (Levine, 2000; Liu et al., 1997; Stanton et al., 1988). This stress hypo-responsive period (SHRP) is the direct result of maternal mediation, where the greater the maternal care the pup receives, the more the stress response is reduced (Liu et al., 1997; Marco et al., 2015; Smotherman & Bell, 1980). For instance, pups with mothers that show higher levels of licking, grooming and arched back nursing behaviours towards their litter, which are indications of high maternal care, have been shown to have a reduction in HPA activity as well as reduced stress sensitivity when tested in later life (Levine, 2000; Liu et al., 1997).

Therefore, by removing the mother from the litter for an extended period, the lack of maternal care results in an increased HPA activation, and the subsequent development of an increased stress response (Levine, 2000). While an increased stress response may be adaptive in a high-stress environment, as it increases the likelihood of survival under threat conditions, prolonged stress exposure can result in a number of issues, such as; insulin resistance, heart damage, poorer episodic memory and increased risk of pathology (Arnsten, 1998; Brindley & Rolland, 1989; Koob, Heinrichs, Menzaghi, Pich, & Britton, 1994; Meaney, 2001). These alterations to the HPA axis stress response also have downstream effects on the serotonergic system (Vasquez et al., 2000). As the SERT KO rats in the current study already have serotonergic impairments, due to the lack of serotonin reuptake transporter, it is predicted that the additional serotonergic dysfunction as a result of the maternal deprivation environmental treatment, will result in salient behavioural impairments.

Maternal deprivation and stereotypy. Maternal deprivation is a common occurrence in captive animals in zoos and laboratories, as it is often more efficient in these environments for the animals to be removed from their mothers before they would naturally be weaned (Latham & Mason, 2008). When this separation is abrupt and occurs at a critical developmental stage, this can result in the development of behavioural impairments, the most common of which is stereotypy (Latham & Mason, 2008; Mason & Latham, 2004). In farming environments, where high animal turnover makes it more economically efficient to remove offspring from their mothers as soon as possible, these repetitive behaviours occur in a number of animals, such as pigs, who show high levels of repetitive belly nosing and sucking, which mimic suckling behaviours (Bench, 2005), and cows, who also show repetitive suckling behaviours (Hammell, Metz, & Mekking, 1988). In the study conducted by Harlow and colleagues (1965), when the macaques were separated from their mothers at birth and then completely isolated for months, severe stereotypies were observed, such as

whole-body rocking, self-clasping, and even self-injurious behaviours (Latham & Mason, 19 2008). In research labs, rodents are usually removed from their mothers from PND21-28, whereas they would naturally stop suckling at PND35-42 (Latham & Mason, 2004; Redman & Sweeney, 1976; Williams, Rosenblatt, & Hall, 1979). This early weaning results in stereotypic behaviours, such as repetitive bar-biting in mice (Wurbel & Stauffacher, 1997). Perseveration is also seen in subjects that have experienced maternal deprivation, as they are much slower at performing extinction tasks, and continue to display particular behaviours, despite the lack of reward (Gluck & Sackett, 1976). As is the case with the Romanian orphans, the effects of maternal separation are graded, where the more extreme the separation event is, the more severe and long-lasting the stereotypical behaviour becomes (Latham & Mason, 2008).

Early Handling

Along with negative environmental factors, such as maternal deprivation, that increase the likelihood of the development of adverse behaviours, it is also important to determine whether positive environmental factors can *reduce* the likelihood of these behaviours developing, by increasing resilience and stress tolerance (Powell et al., 1999; Rainekei, Lucion, & Weinberg, 2014). For instance, in humans, high levels of parental care have been found to improve the outcome of individuals with chronic illnesses, and increase resilience and stress resistance (Rutter, 1987; Smith & Prior, 1995). Furthermore, high levels of parental care have been found to counterbalance the effects of negative environmental factors, such as poverty (Eisenberg & Earls, 1975). The outcome of prenatal and postnatal stress is dependent on a number of factors, including the timing, intensity and duration of the stress (Weinstock, 2008). In relation to timing, as previously stated, the stress hypo-responsive period (SHRP) from PND4-14, before the rats are weaned, is a critical period of neurodevelopment, where stress can play a major role in the development of subsequent

adverse behaviours (Levine et al., 1956; Plotsky & Meaney, 1993; Stanton et al., 1988). 20

The stress response involves a series of neuroendocrine and behavioural alterations, such as an increase in energy availability and hypervigilance, after the presentation of an adverse stimulus (Ursin & Olf, 1993). As discussed, due to the negative effects of prolonged stress exposure, stress resistance is an important behavioural adaptation in individuals that reside in largely non-threatening environments, as it enables healthy emotional functioning following a stressful event, and decreases the risk of illness and psychological harm (Meaney, 2001; Parker, Buckmaster, Sundlass, Schatzberg, & Lyons, 2006).

While the removal of the mother for an extended period during the SHRP increases stress sensitivity in the pups, conversely, when the mother is removed from the litter for a short period of 3-15 minutes daily, during the first two weeks of development, the opposite effect has been found: the HPA axis becomes further hypoactive, and the stress sensitivity of the pups is decreased when tested later in adulthood (Levine et al., 1956; Levine, Haltmeyer, Karas, & Denenberg, 1967; Meaney et al., 1985). This technique, termed *postnatal* or *early handling* (EH) was first used by Levine and colleagues (1956) who were interested in the well-established finding that early life trauma results in permanent behavioural changes in later life. They exposed a group of rats to small shocks daily from postnatal day 1-20. They also used two control groups: one consisted of rats that were not handled during this period, and the other group was placed on the shock plate daily, but did not receive shocks. To their surprise, when they tested these groups in adulthood, the handled rats that did not receive shocks showed significantly more adaptive behavioural characteristics than the other groups, and the non-handled group had the worst performance of all groups. The poor performance of the non-handled group was attributed to lack of stimulation during early development. As maternal care has been previously implicated in regulating the pup HPA axis, the superior behavioural performance of the EH rats is attributed to an increased level of maternal care

when the mother is returned to the litter (Liu et al., 1997). Indeed, increased licking and grooming behaviours have been found in EH mothers when compared to those in the non-handling (NH) condition, supporting this theory (Zimmerberg, Kim, Davidson & Rosenthal, 2003a). Furthermore, both pups that experience EH, and those with high levels of maternal care, show reduced adrenocorticotrophic hormone, adrenal corticosterone, and corticotropin-releasing-factor levels in adulthood compared to NH rats, when exposed to a variety of stressful stimuli (Ader & Grotta, 1969; Hess, Denenberg, Zarrow, & Pfeifer, 1969; Liu et al., 1997; Plotsky & Meaney, 1993). EH and high maternal care rats also show an increase in glucocorticoid receptor expression in the hippocampus and frontal cortex, which increases the glucocorticoid negative feedback response on the HPA axis, further suppressing its activity and reducing the stress response (Meaney et al., 1985). Finally, as with humans, where parental care can counterbalance the adverse effects of negative environmental factors, such as poverty, EH in rats has been found to rescue the phenotype of rats exposed to prenatal stress, by increasing the amount of time spent in the open arm of the elevated plus maze, and increasing hippocampal neurogenesis (Bogoch, Biala, Linial, & Weinstock, 2007; Rainekei et al., 2014; Wakshlak & Marta, 1990). In summary, due to the plasticity of the neuroendocrine stress response during the early developmental period in rats, disruptions to maternal care, such as maternal deprivation, can result in increased HPA activity during this sensitive period, resulting in increased stress sensitivity in the pups. However, if the period of maternal separation is reduced, the HPA axis can be further suppressed, resulting in an increased stress tolerance in the pups (Levine, 2000). In this way, maternal care mediates the effects of environmental challenges on neural development in rat pups, with maternal deprivation and early handling having opposing effects on the development of stress sensitivity in rats (Meaney, 2001).

According to the diathesis-stress model (Bleuler, 1966), an individual's likelihood of developing a disorder, such as depression, anxiety, and ASD, is dependent on a combination of genetic risk factors, such as the *s* allele polymorphism in the human SERT gene, and adverse environmental conditions, including; threatening situations, loss, and maternal stress (Caspi et al., 2002; Monroe & Simons, 1991). The diathesis-stress model was proposed because of a number of schizophrenia related findings that indicated that genetics alone could not account for the development of this disorder. These findings include: the high occurrence of schizophrenia-like behaviours in relatives of patients with the disorder, increased prevalence of schizophrenia following a tragedy, and concordance rates in monozygotic and dizygotic twin studies (Bleuler, 1966). The diathesis-stress model has since been one of the most influential theories, not only with respect to schizophrenia, but with much wider implications. Caspi and colleagues (2002) found that participants with the *s/s* SERT genotype that experienced three or more stressful life events, were significantly more likely to have developed depression than those with the *l/l* genotype that experienced the same number of adverse events, supporting the diathesis-stress theory that some individuals are more vulnerable to adversity than others.

However, while these studies have mostly concentrated on the interaction between specific genes and a negative environment, they have largely ignored the (potential) influence of a more positive environment. It seems somewhat unusual that negative environmental factors would have an influence on an individuals' phenotype, while positive environmental factors would have little or no effect at all (van IJzendoorn et al., 2012). Moreover, from an evolutionary point of view, it seems unlikely that particular genetic mutations, such as the *s* allele of the SERT (which is also found in other species, such as rhesus monkeys), would be maintained in the population if they only resulted in adverse effects (Homberg & Lesch,

2011). Consequently, the *plasticity* model was proposed, which predicts that particular genetic mutations, such as the s allele polymorphism, increase an individual's general susceptibility to environmental factors, irrespective of whether these factors are positive or negative (Belsky et al., 2009). Therefore, while individuals with such "*plasticity*" genes will suffer most when exposed to negative environmental conditions (as with the diathesis-stress model), they will benefit most from positive environmental conditions (van IJzendoorn et al., 2012).

Evidence in favour of the plasticity theory comes from the finding that males with a mutation that inactivated the monoamine oxidase (MAOA) enzyme, who had also experienced childhood adversity, showed significantly greater levels of antisocial behaviours than those without the MAOA mutation (Caspi et al., 2002), while those that *did not* experience childhood adversity, actually showed the lowest antisocial scores of any of the groups, including the controls, indicating that this gene is not inherently a risk factor, and can lead to positive outcomes if the environment is favourable (Belsky et al., 2009). Further support for the *plasticity* theory includes the finding that the s allele mutation is present in such a high proportion of the population (Caspi et al., 2002; Heils et al., 1996), which seems unlikely, unless this gene were to convey some type of adaptive advantage (Belsky et al., 2009). Indeed, whilst individuals with the s/s genotype show significantly higher rates of depression when exposed to large amounts of stress during their lifetime, compared to those with the s/l or l/l genotypes, those with the s/s genotype that have not experienced such adverse life events, show the lowest depressive scores overall (Belsky et al., 2009; Caspi et al., 2002). Finally, van IJzendoorn and colleagues (2012) conducted a meta-analysis to determine whether the s/s and s/l genotypes were primarily influenced by negative environmental conditions, as predicted in the diathesis-stress theory, or if there was an equal influence of positive and negative environmental factors on the developmental outcome of

these individuals, compared to those with the *l/l* genotype, as predicted in the plasticity theory. In the Caucasian sample, individuals with the *s* allele were highly influenced by *both* positive and negative environmental factors, providing support for the plasticity theory. However, as previously mentioned, gene*environment studies in humans are correlational in nature and do not allow conclusions on the causality of the effect. Therefore, these theories will be tested in the current study by investigating the phenotypic effects of both negative and positive environmental treatments on genetic knockout rats.

Experimental Tasks and Predictions

As little is known about the development of repetitive behaviours in autistic children, this has led to difficulties in improving treatment and early intervention techniques (Lewis et al., 2007). By thoroughly investigating repetitive behaviours in a rat model, it is hoped that further insight can be gained in relation to the underlying processes involved in the development of these behaviours, which, in turn, can lead to better screening, intervention and treatment techniques (Levy et al., 2009). Many previous studies have demonstrated that an individual's reaction to environmental factors is moderated by genotype (Bleuler, 1966; Caspi et al., 2002; Monroe & Simons, 1991). However, there is some debate as to whether this is due to increased vulnerability to the effects of negative environmental factors, as predicted in the *diathesis-stress/vulnerability* model (Bleuler, 1966), or because particular genotypes are more susceptible to the effects of both positive and negative environmental factors, as outlined by the *plasticity* theory (Belsky et al., 2009). What has made it challenging to distinguish between these two theories is that researchers appear to be predisposed to focus on the negative environmental factors, making it difficult to determine whether positive factors result in particular individuals flourishing (Belsky et al., 2009). Therefore, in the current research we focused on both positive environmental conditions, in the form of early handling, and negative factors, in the form of maternal deprivation, in both

heterozygous and homozygous SERT knockout rats, as well as wildtype controls. These 25 rats will complete five behavioural tasks in order to determine the presence and severity of ASD-like symptoms. These tasks include: social interaction, social approach and avoidance, scent marking, hole-dipping and reversal learning experiments.

It is predicted that, if the *vulnerability* model is correct, then the SERT KO MD rats will still show the most severe ASD-like symptoms, however there should be little to no improvement of the behavioural impairments present in the SERT KO EH subjects compared to the WT controls (see *Figure 1*). If, however, the *plasticity* model is correct, the SERT KO rats in the negative environmental condition, maternal deprivation, will show the most severe ASD-like behavioural characteristics in these tasks, while the SERT KO rats in the positive environmental condition, early handling, will show the fewest ASD-like symptoms (see *Figure 2*). It is uncertain whether the heterozygous or homozygous SERT KO rats will show the greatest impairment in the MD condition and improvement in the EH condition, as it is possible that the complete lack of SERT protein present in the homozygous SERT KO rats will result in such severe behavioural impairments that the phenotype cannot be further impaired.

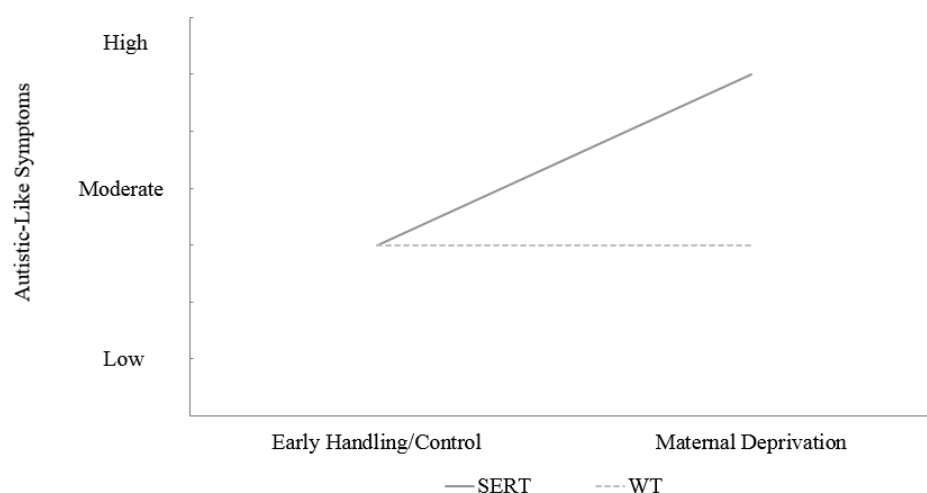


Figure 1. Line graph depicting the predicted severity of ASD-like symptoms if hypothesis one, which follows the *plasticity* theory, is supported

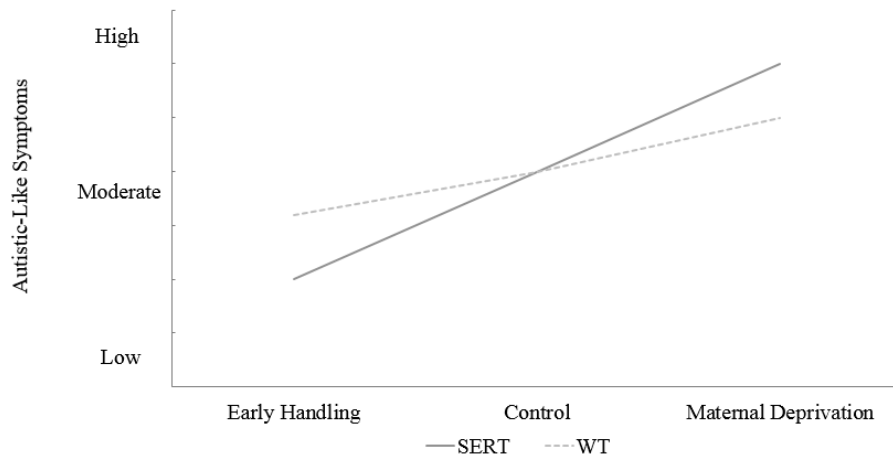


Figure 2. Line graph depicting the predicted severity of ASD-like symptoms if hypothesis two, which follows the *plasticity* theory, is supported

Materials and Methods

A three (genotype: HOM SERT KO, HET SERT KO and controls) by three (treatment: maternal deprivation (MD), early handling (EH) and control) between subjects experimental design was used. The experiments were approved by the Victoria University of Wellington Animal Ethics Committee.

Subjects

A total of 77 rats were used. Females were housed with males until they were visibly pregnant, at which point they were separated into individual cages during gestation. The KO rats were the progeny of the original SERT KO rats produced by Smits and colleagues (2006), who used N-ethyl-N-nitrosourea (ENU) driven mutagenesis to induce mutations, where a premature stop codon results in the production of a non-functional protein (for a more detailed description of this method, see Smits et al., 2006). As the SERT mutation was originally found in the Wistar strain, this strain was also used for this research.

Rats were placed on a 12-12 hour light-dark cycle, with the light phase commencing at 07:00. Food and water were freely available throughout the experiments, except one week prior to and during the reversal learning paradigm, where food was restricted to 10-15 grams per rat per day in order to motivate the animals to perform the task and obtain the reward. On

postnatal day (PND) 21, rats were weaned and genotyped, by taking small samples from 27 the ears. Subjects were subsequently housed in same-genotype groups of three to five. As ASD is four times more prevalent in males (Dawson et al., 2002), only male rats were tested in this experiment. Subjects were at least two months old when experimentation began, as adulthood is estimated to occur around PND63 (Sengupta, 2014), and were provided with at least one week break between experiments. Although ASD starts relatively early in life, in most cases it persists into adulthood. For this reason we focused only on adult animals in the present study.

Environmental Treatment

The litters were randomly allocated to one of three environmental treatments: MD (five litters), EH (10 litters) or the control condition (five litters) (see *Table 1* for number of rats in each treatment). The negative environmental treatment (MD, $N=24$ male pups), involved the dam being removed from the home cage at 11:30 for a 24 hour period on PND9, and then returned to the home cage at 11:30 on PND10. The mother was placed in the same room as the home cage, in an empty polycarbonate cage with clean bedding, as well as freely available food and water. The pups remained in the home cage for this period. After the mother was returned on PND10, the litter was observed for several minutes at two hour intervals to ensure that the mother continued to take care of her pups after reunification. All MD mothers continued to show maternal care once returned to their home cage. Although some researchers individually remove the pups from the home cage during MD treatment, infants show increased distress when mothers are removed from the home environment, compared to when they are removed themselves, indicating that the removal of the mother acts as a stronger environmental manipulation (Hinde & Spencer-Booth, 1970; Stanton et al., 1988). A 24 hour period of separation was used, as previous studies have found that the stress response in the pups is graded according to the length of deprivation, with an eight hour

period producing moderate effects, and a 24 hour period resulting in pronounced effects 28
(Stanton et al., 1988).

The positive environmental treatment, early handling (EH) ($N=27$ male pups), involved the dam being removed from the home cage at 10:30 for a 15 minute period every day from PND2 to PND14. The mothers were handled in the same room as the home cage by the researchers during this separation period. This handling period did not result in a loss of maternal care for the rat pups, as mothers usually spend periods of 20-30 minutes away from their pups several times per day (Jans & Woodside, 1990). Rats in the control environmental group ($N=26$ male pups) were exposed to standard laboratory care procedures, involving regular bedding, food and water replenishment.

Table 1

Number of Subjects

Genotype	Environmental Treatment			Total
	Control	EH	MD	
WT	10	9	11	30
HET	6	10	8	24
HOM	10	8	5	23
Total	26	27	24	77

Experiments

As the behavioural tests that were used did not cause excessive stress, the order in which they were carried out was randomly decided, as determined by the availability of rats. Due to the need for food restriction, and the length of the reversal learning task, this was performed last by all groups tested.

The majority of ASD related research has focused on abnormal social behaviours, most likely due to the fact that this is the behavioural characteristic that is most commonly associated with autism, as well as being relatively easy to assess in animals (Crawley, 2007; Wohr & Scattoni, 2013). One of the behavioural paradigms that has been used to measure the presence of social impairments in rodents is the social interaction test (File & Hyde, 1978). In our version of the task, the experimental rat is placed in an 80cm diameter round open field with an unfamiliar experimental rat of the same gender, treatment, genotype and general size. The behaviour is recorded for a total of 30 minutes, with the number of interactions between the nose, tail and center of the rats being analysed using EthoVision version 9.0 software (with the Social Interaction Module, Noldus Information Technology, Netherlands). In order to enable EthoVision to distinguish between the experimental and non-experimental rats, the rats were first dyed with either red or green hairspray (125ml: Lenan Corporation Pty Ltd., Australia). Hairspray was used as a non-toxic spray, to reduce the risk of excessive grooming, leading to baldness. Under the vulnerability model, it was predicted that the HET and HOM MD rats would show the least amount of social interaction, as shown by the lowest duration and frequency of contact between subjects. Following the plasticity model, it was further predicted that the HET and HOM EH rats would show the greatest amount of social interaction.

Analysis

The measures of social interaction were split into active contact, where the body points (nose, center and tail) of the subjects were in contact while the subjects were moving, and passive, where the body points were in contact while the rat was not moving (File, 1980). A (2(environment: EH or MD, control) x 3 (genotype: WT, HET, HOM) x 2 (duration/frequency: active, passive)) multivariate analysis of variance (MANOVA) was

calculated to determine the duration (in seconds) and frequency of active and passive social interaction. 30

Measures of Communication: Ultrasonic Vocalisations (USVs)

Ultrasonic vocalisations (USVs) were recorded concurrently with the social interaction task, using the UltraVox program (Noldus Information Technology, Netherlands). Rodents communicate via USVs, which differ in terms of the frequency and duration, although the precise meaning of these differentiations is still relatively unknown (Wohr & Scattoni, 2013). A number of studies have been conducted investigating the USVs emitted by pups when they have been separated from their mothers, and these studies have found that pups vocalise increasingly in the 30-40kHz range when separated, compared to when they are with their littermates in the nest, in order to regain food and warmth (Noirot, 1972; Zimmerberg, Kim, Davidson & Rosenthal, 2003a; Zimmerberg, Rosenthal & Stark, 2003b). As well as separation calls, rats also vocalise in social situations, such as during juvenile play, interacting with a novel conspecific during adulthood, and during male to female reproductive interactions (22kHz, 50-60kHz) (McIntosh, Barfield & Geyer, 1978; Sales, 1972; Wohr & Scattoni, 2013). As previous research has determined that an increase in vocalisation is associated with an increase in social motivation and reward (Burgdorf, Knutson, Panksepp, 2000; Knutson, Burgdorf, & Panksepp, 1999), it was predicted under both the vulnerability and plasticity models, that HET and HOM MD rats would show a reduction in overall number of USVs measured during the social interaction task compared to the control rats. The plasticity model further predicts that the HET and HOM EH rats should show the greatest number of USVs.

Experiment Two: Social Approach and Avoidance

The social approach and avoidance paradigm is based on the three chamber social approach task described by Crawley (2007), however a 125cm by 75cm black Plexiglas T

maze was used in place of the three chambered apparatus (see *Figure 3*). The experimental 31 rat was first placed in the vertical arm of the maze, which mimics the central chamber in the original task. The subject was given ten minutes to freely explore and habituate to the maze, to determine whether there were random biases to either the left or the right arm. To reduce the likelihood of biases affecting the results, the location of the novel rat in phase two and three of the experiment was alternated between subjects. Next, in phase two of the experiment, the experimental rat was removed from the apparatus and an overturned wire cup (10.5cm diameter) was placed on the left and right side of the horizontal arm. One of the wire cups was empty (termed the *novel object*), and the other contained an unfamiliar juvenile male rat (termed the *novel rat*). Juvenile rats under the age of two months were selected, both to reduce the likelihood of aggression between the experimental and non-experimental rats, and to ensure that the rats could fit comfortably underneath the overturned wire cup. The experimental rat was then reintroduced to the apparatus and given five minutes to freely explore the maze, with the time spent in close proximity to the novel object and the novel rat being calculated using EthoVision analysis. Due to the socially inquisitive nature of rats, it was expected that HET and HOM EH rats would spend significantly more time with the novel rat than the novel object under the plasticity model, whereas HET and HOM MD rats should show decreased interest in the novel rat when compared to the controls, according to both the vulnerability and plasticity theories. In phase three, a novel rat was placed under the previously empty wire cup, while the novel rat from phase two remained under the other wire cup, making this rat the *familiar rat* in this phase. The experimental rat was again given five minutes to freely explore, and the time spent in close proximity with the familiar and novel rat was measured. According to the plasticity model, it was predicted that HET and HOM EH rats would spend more time with the novel rat than the familiar rat. Both the vulnerability and

plasticity models predict that the HET and HOM MD rats would show equal preference for 32 the familiar and novel rat.

Analysis

Habituation, phase two and phase three of the social approach and avoidance task were recorded for later analysis using EthoVision software. Duration of time spent in close proximity (defined as the nose point being within 5cm distance of the cup) to the novel rat and object in phase two, and the novel and familiar rat in phase three were measured, as well frequency of (nose point) visits to these locations. A (2 (environment: EH or MD, control) x3 (gene: WT, HET, HOM) x2 (duration/frequency: novel rat, novel object/familiar rat))

MANOVA was used to analyse these data.

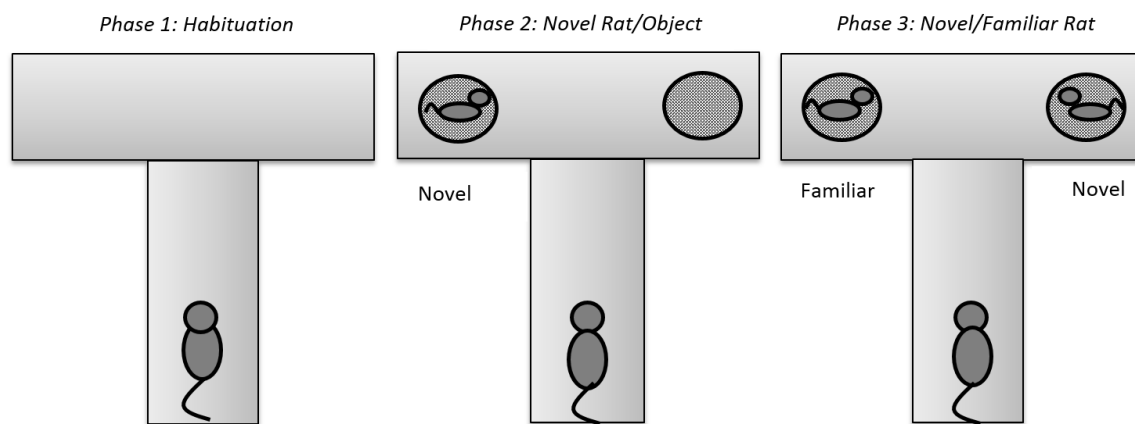


Figure 3. Simplified diagram of the social approach/avoidance task. The location of the novel rat alternated between subjects.

Experiment Three: Scent Marking

Olfactory Communication

Olfactory communication, which involves using scent marking behaviours to convey information such as territorial boundaries between male rats, and sexual receptivity in females, is an important behaviour in rodents, and is thus an ecologically valid measure of social communication (Achiraman et al., 2011; Arakawa et al., 2011; Wohr & Scattoni, 2013). Urinary scent, in particular, provides information about the species, gender and strain of an individual (Arakawa et al., 2011; Wohr & Scattoni, 2013). Female rat urine also acts as

a signal to males that the female is in the oestrus stage of the reproductive cycle, when 33
ovulation occurs, and is thus sexually receptive for mating (Achiraman et al., 2011). As males
show an increased interest in the urine of females in the oestrus phase, in this experiment,
males were exposed to the urine of females in oestrus (or in a transitioning phase), and the
number of scent markings around this odour were compared to the scent markings around a
non-social odour, lemon essence.

Collection of female urine and analysis. While there are certain visual differences in
the appearance of the rat vagina in the different stages of the reproductive cycle, such as an
increased redness and relaxed appearance during the oestrus phase (Champlin, Dorr, & Gates,
1973), these visual characteristics are often very subtle and difficult to determine. For this
reason, vaginal swabs were taken from the female rats before urine collection, to ensure that
urine was only used from females in the oestrus phase, as this stage is when the male rat has
the most heightened interest in the female urine scent, for reproductive purposes (Achiraman
et al., 2011). First, the tip of a plastic pipette containing 0.2mL of saline solution was
shallowly inserted two to five millimeters into the entrance of the vagina, while the rat was
restrained using one hand around the neck, and the other restraining the tail. Next, the top of
the pipette was gently squeezed and released two to three times, to flush the fluid in and out
of the vagina. The cell suspension was then placed in the center of a glass slide and covered
with a cover slip for microscope inspection. The slides were then examined for the presence
of *cornified cells*, which are large, irregularly shaped and non-nucleated, and are present in
the highest amount during the oestrus phase of the reproductive cycle (see *Appendix A*) (Long
& Evans, 1922; Marcondes, Bianchi, & Tanno, 2002). Once it had been determined that a
particular rat was in the oestrus phase, urine was then collected for use in the scent marking
experiment, by holding the rat over a ramekin and stroking its stomach to encourage

urination. The urine that was collected was not kept for use for longer than an hour in any 34
of the experiments.

Procedure

Socialisation phase. To begin, the male rats were individually placed with a female rat of the same strain in a polycarbonate cage lined with bedding. This phase lasted five minutes, and was conducted to ensure that the male rats were familiarised with the female scent, in preparation for the scent marking task.

Habituation phase. Secondly, the rats were individually placed in an 80cm diameter black Plexiglass open field for 15 minutes. The habituation phase was carried out six to nine days after the socialisation phase, for the purpose of familiarising the rats with the open field, and subsequently reducing their anxiety in this unfamiliar environment. Once they had completed the habituation phase, the rats were removed from the open field, which was then cleaned with 70% ethanol solution and Virkon S disinfectant.

Scent marking phase. Two pieces of circular filter paper (32cm diameter) were placed inside the open field, with 30 microlitres of lemon essence pipetted into the middle of one, as the non-social odour, and 30 microliters of urine from a female in the oestrus phase (the social odour) pipetted onto the other. The rat was again placed in the open field, and given five minutes to freely explore. The subject was then removed and returned to its home cage. The filter papers were also removed from the field, labelled, and then left for an hour to dry. After each session, the open field was cleaned with a 70% ethanol, 30% water preparation, and wiped with paper towels, to sterilise the environment for the next subject.

Analysis

Once dried, the filter paper was sprayed under a fume hood with ninhydrin spray (Beijing Bulant Police Equipment Co., Ltd.; Beijing, China), a substance that detects the ammonia in urine, along with amines (Laskar, Bhattacharya, & Basak, 1991). This meant that

scent markings, urine pools, footprints, human fingerprints and sneeze droplets were visualised as a purple colour. The filter paper was sprayed under a fume hood, to prevent inhalation, and gloves were worn, to prevent the spray coming into contact with the skin. After being sprayed, the filter paper was left over night, to enable visualisation to occur. After a 24 hour period, a transparent one by one centimeter grid was used to count the number of scent markings surrounding the lemon and urine odours. Scent markings were determined as purple circular formations that covered an area smaller than four one centimeter grids that did not have the visual characteristics of footprints, which were a collection of uniformly shaped markings, or sneezes, which were considerably smaller than a single grid, and often had a slight red colouration. Marks that were larger than four one centimeter grids were labelled as urine pools and not included in the analysis (see *Appendix B*). 35

Following the vulnerability model, it was predicted that the HET and HOM MD rats would show fewer overall scent markings when compared with controls, as well as an equal interest in the social and non-social odour. The plasticity model further predicts that the HET and HOM EH rats would show increased overall scent markings, as well as increased scent marking around the social odour, compared with the non-social odour.

For data analysis, the percentage social scent markings was first calculated, where the number of scent markings around the social odour was divided by the total number of scent markings for each subject. A (2(environment: EH or MD, control) x 3 (genotype: WT, HET, HOM) x 2 (odour (social, non-social)) MANOVA was then performed, to determine whether EH subjects were scent marking more around the social (female urine) than the non-social (lemon essence) odour as compared with the control group.

Experiment Four: Hole Dipping

The hole-board paradigm is used as a measure of novelty seeking behaviour (Brown & Nemes, 2008), and can also be used as a measure of repetitive behaviours in rodents

(Mankanjuola et al., 1977; Moy et al., 2008). Hole-dipping behaviour is used as a measure 36 of perseveration in rodents, as rats with restricted interests have been found to visit particular holes repeatedly, whereas control rats show more explorative behaviours, visiting more novel holes (Mankanjuola et al., 1977).

Procedure

Rats were removed from their home cages and then individually placed in standard sound and light attenuating locomotor boxes (45 x 45 x 35cm) (Med Associates Inc., USA; model ENV-515) which were fitted with a metal hole-board insert containing 16 holes (Med Associates Inc., USA), as well as two rows of infrared photocells that were used to detect movement via photocell beam breaks. Rats were placed in the boxes for a period of 15 minutes, during which time their general locomotor activity was automatically recorded using the accompanying computer software (Med Associates Inc, USA). The tracking system also recorded the number of nose dips that occurred into each hole, resulting in a record of the total number of hole dips, as well as the number of repeat entries. After the task was completed, the apparatus was cleaned with a 70% ethanol solution as well as Virkon S disinfectant.

Analysis

A (2 (environment: EH or MD, control) x3 (gene: WT, HET, HOM) x2 (nose-poke: novel, repeat)) MANOVA was used to determine whether there was a difference in the number of novel and repeat nose-pokes/hole-dips in the environmental treatment groups compared with the controls.

Experiment Five: Reversal Learning Paradigm

Due to time constraints, only four of the nine experimental groups were run on this paradigm: HOM controls, WT controls, WT EH and WT MD.

restriction, where they were given 15 grams of food per day, to ensure they would reduce to 85-90% of their free feeding weight once the experiment commenced. This restriction continued throughout the experimental period, which lasted a maximum of 35 days (see *Figure 4* for a general outline of the procedure). The food restriction was necessary to ensure that the subjects would be sufficiently hungry for the sugar pellets used in the paradigm to be highly rewarding, and result in behavioural motivation. The rats were also introduced to the chocolate chips and sugar pellets during the food restriction period, so that they would be familiar with this food during the experimental paradigm.

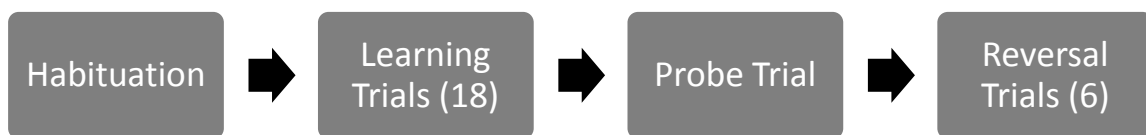


Figure 4. The reversal learning paradigm procedure. The experiment took a maximum of 35 days.

Procedure

Habituation phase. Subjects were first habituated to the 180x200cm rectangular open field. The open field contained five different cues for the rats to use as reference points for spatial guidance. These cues included: a plastic sunflower in the lower right hand corner, a white stand decorated with black circles in the upper right hand corner, a black and white striped piece of paper in the center of the left hand wall, two metal bells, and a small plastic trophy in the center of the lower portion of the maze (see *Figure 5* for layout). The habituation phase involved a single ramekin being placed in the center of the right hand wall of the maze. Bedding was sprinkled on and around the ramekin to help familiarise the rats, and chocolate chips were placed near the ramekin, as well as inside. Chocolate chips were used as the target initially to enable the rats to use scent to guide them. After two habituation trials, the chocolate chips were paired with sugar pellets, and finally the sugar pellets were used in isolation, so that the rats were not able to find the target using odour as a guide. It was

determined that the rats had completed the habituation phase of the experiment when they 38 were able to locate the sugar pellets in the triple stacked ramekin placed at the back wall of the maze, within 30 seconds in three consecutive trials. In order to ensure the rats used a spatial strategy, they were released from three different entry points into the maze.

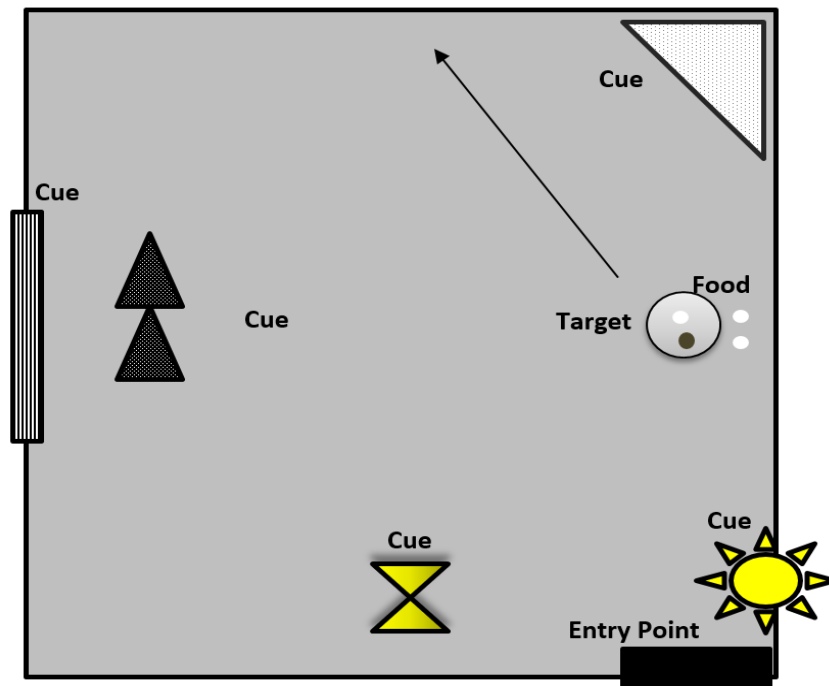


Figure 5. The habituation phase of the reversal learning paradigm in the 180x200cm rectangular open field. Subjects were initially released from the lower right hand side of the open field, with a single ramekin containing chocolate and sugar pellets as the target. In the final phases of the task, three ramekins were stacked in the center of the back wall, and the rats were released from the left, center and right hand side.

Learning phase. The learning phase involved 12 sets of ramekins, each stacked three ramekins high, arranged in groups of three in the four quadrants of the square maze (see *Figure 6* for layout). Each rat was given a target ramekin, either on the left or right hand side of the back section of the maze. The target ramekin was baited with five sugar pellets. The target remained the same for each rat, but was switched between rats to ensure that they could not be guided to the target by using the scent of the previous subject. Subjects were given a maximum of ten minutes to find the target, after which time they were marked as having failed to complete the trial. Five consecutive failures resulted in the subject being excluded from subsequent trials.

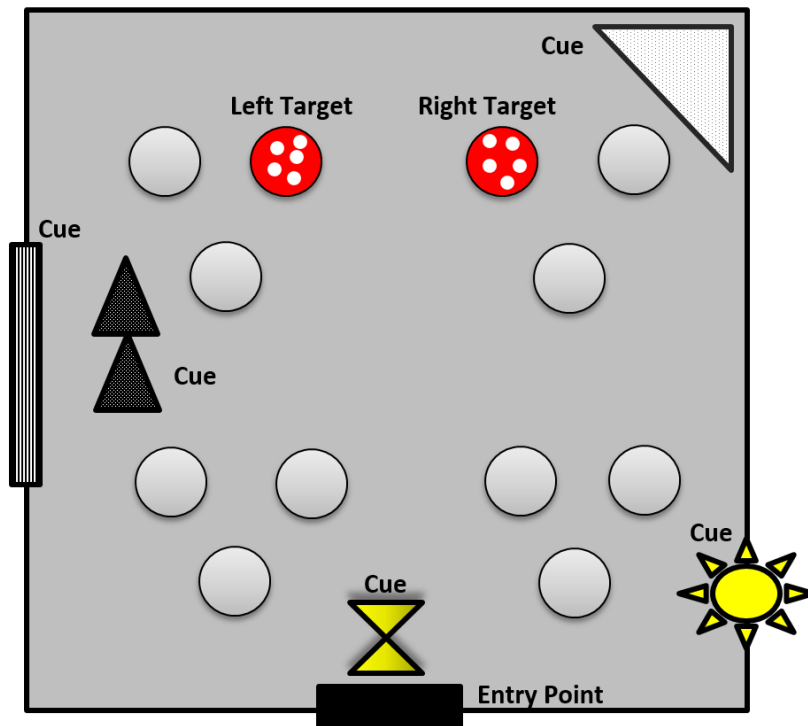


Figure 6. The Learning Phase: subjects were randomly allocated either the left or the right side target, which was kept consistent throughout the learning trials. The trial was completed once the subject successfully located the target.

Probe trial. Once performance on the learning phase appeared to have stabilised, after 18 learning trials, a probe trial was conducted, where the rats were placed in the maze with the ramekins and cues in place as would occur in a learning trial, but without the sugar pellet reward in the target ramekin. For analysis, the maze was divided into a target quadrant, and an opposite quadrant, and the duration of time spent in each of these quadrants was measured. The probe trial was stopped when the subject had made ten dips into the ramekins. If the subject had learned the location of the target, it was predicted that they would spend significantly longer in the target quadrant when compared to the opposite quadrant.

Reversal phase. After the probe trial was complete, the rats were provided with one additional learning trial, to ensure that they were aware that there was still a target ramekin that they were required to locate. Next, the target was switched to the opposite quadrant for each subject (i.e. from the left hand ramekin to the right hand ramekin), and the trials continued until the rat consistently found the new target location under a distance of 400cm.

Three different types of measures were used to analyse performance in the reversal learning paradigm: learning, anxiety and stereotypy. The measures of learning performance included the probe trial, distance travelled and number of ramekin visits. Distance travelled was used as the measure of spatial learning, as a decrease in distance over trials indicates that the subject is learning the location of the target as they are moving in a more direct route (Wang et al., 2015). Visits to each ramekin were calculated, as a visit to a ramekin that was not the target was an indication of a reference memory error, whereas a repeat visit to a ramekin was considered a working memory error. The measures of anxiety were: duration of body elongation (above 80%) and contraction (below 30%), percentage of time in the center of the field (thigmotaxis), and duration of time spent immobile (less than 20% movement). Finally, in terms of stereotypy, the variables of interest included: head weaving and body rotation (frequency of .25 rotation with a threshold at 50 degrees).

Repeated measures ANOVAs were performed to determine whether the EH, MD and HOM rats differed from the controls on the measures of learning, anxiety and stereotypy.

Results

Experiment One: Social Interaction

Effects of EH

The duration and frequency of active and passive social interaction was first calculated by determining whether the contact that occurred between the two subjects occurred while they were moving (active) or still (passive).

There were no significant differences in relation to the duration of active social interaction when the EH and control subjects were compared (see *Appendix C*).

When comparing the duration of passive social interaction, there was a significant main effect of genotype ($F(2, 46)=10.52, p<.001$), with post-hoc Tukey tests revealing that

the HETs ($M=43.29$, $SD=30.90$) spent significantly less time in passive social contact with each other, compared to the WT ($M=194.78$, $SD=120.79$) ($p=.002$) and the HOMs ($M=227.35$, $SD=145.53$) ($p<.001$) (see Figure 7). There was no significant main effect of environmental treatment, or interaction between genotype and environment.

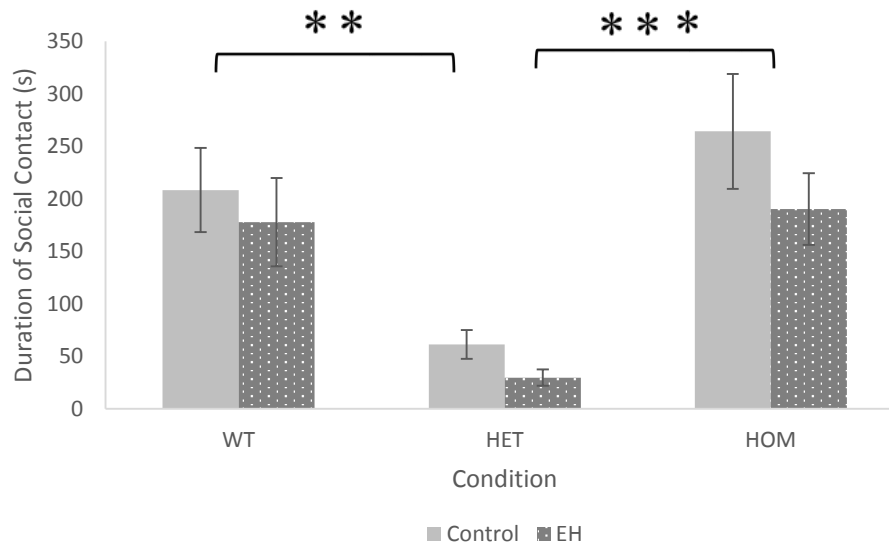


Figure 7. Bar graph depicting the time spent (in seconds) in *passive* social contact in the EH and control groups during the social interaction task. Error bars represent ± 1 standard error of the mean.

Note: * $p<.05$, ** $p<.01$, *** $p<.001$

When frequency of active social interaction was measured, there was a significant main effect of genotype ($F(2, 46)=5.05$, $p=.011$), with post-hoc Tukey tests revealing that the HOMs ($M=1051.67$, $SD=843.33$) had a significantly higher frequency of active social interaction than the WT ($M=708.94$, $SD=433.75$) ($p=.010$) (see Figure 8). There was no significant main effect of environmental treatment, or interaction between genotype and environment.

When frequency of passive social interaction was analysed, there was also a significant main effect of genotype ($F(2, 46)=5.89$, $p=.005$), with post-hoc Tukey tests revealing that the HOMs ($M=1305.95$, $SD=1041.26$) had a significantly higher frequency of passive interactions than both the WT ($M=578.39$, $SD=354.22$) ($p=.009$) and the HETs ($M=592.64$, $SD=461.55$) ($p=.018$) (see Figure 9). There were no significant findings in terms of environmental treatment or a genotype-environmental treatment interaction.

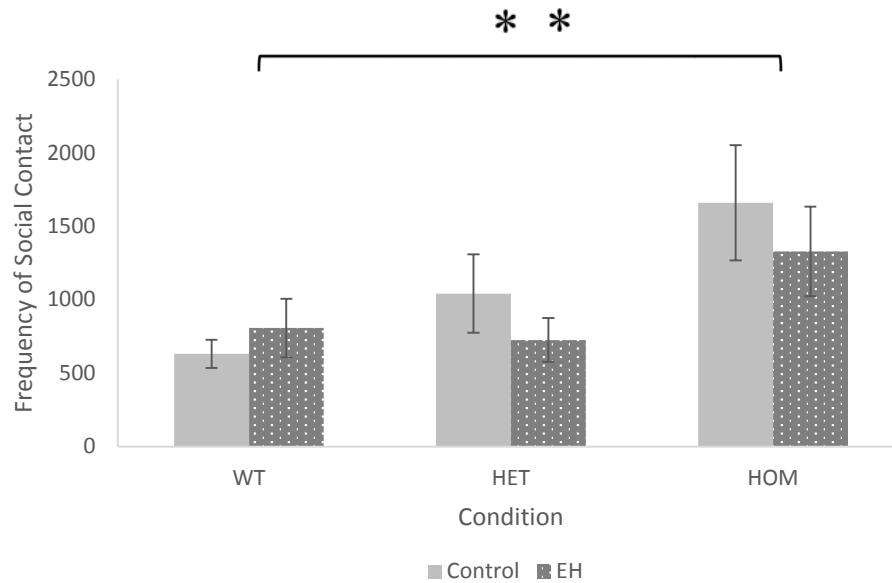


Figure 8. Bar graph depicting the frequency of *active* social contact in the EH and control groups during the social interaction task. Error bars represent ± 1 standard error of the mean.

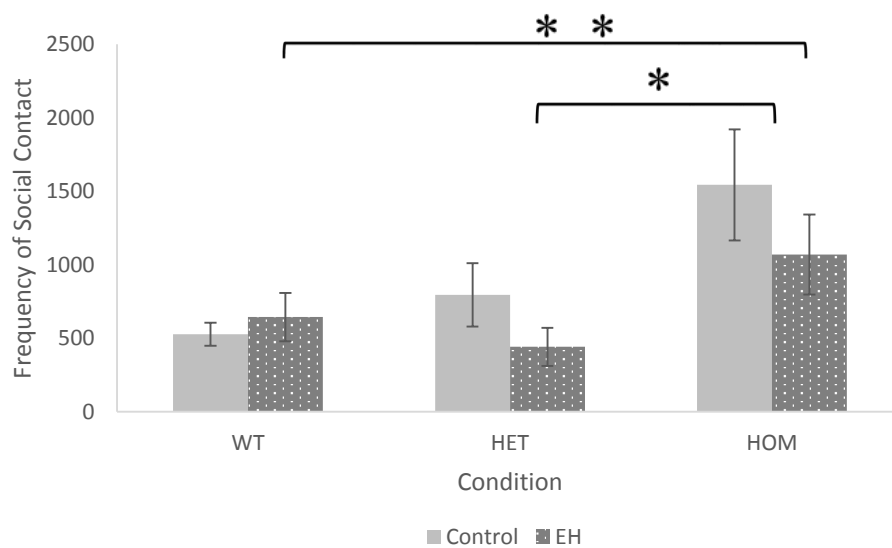


Figure 9. Bar graph depicting the frequency of *passive* social contact in the EH and control groups during the social interaction task. Error bars represent ± 1 standard error of the mean.

Effects of MD

There were no significant findings for duration of active social interaction when the MD and controls were compared (see *Appendix C*). For duration of passive social interaction, there was a significant main effect of genotype ($F(2, 38)=4.98, p=.012$), with post-hoc Tukey tests revealing that the WT ($M=234.10, SD=132.35$) spent a significantly longer

duration in passive contact when compared to the HETs ($M=87.03$, $SD=114.44$) ($p=.013$). 43

Although there was no main effect of environment, there was a significant genotype-environmental treatment interaction was also found when MD and control subjects were compared on duration of passive social interaction ($F(2, 38)=4.98$, $p=.012$), with post-hoc independent samples t-tests showing that the HOM MD subjects ($M=26.97$, $SD=25.62$) showed a significantly reduced duration of passive interaction when compared with the HOM control subjects ($M=264.32$, $SD=173.19$) ($t(9.92)=4.22$, $p=.002$) (see *Figure 10*).

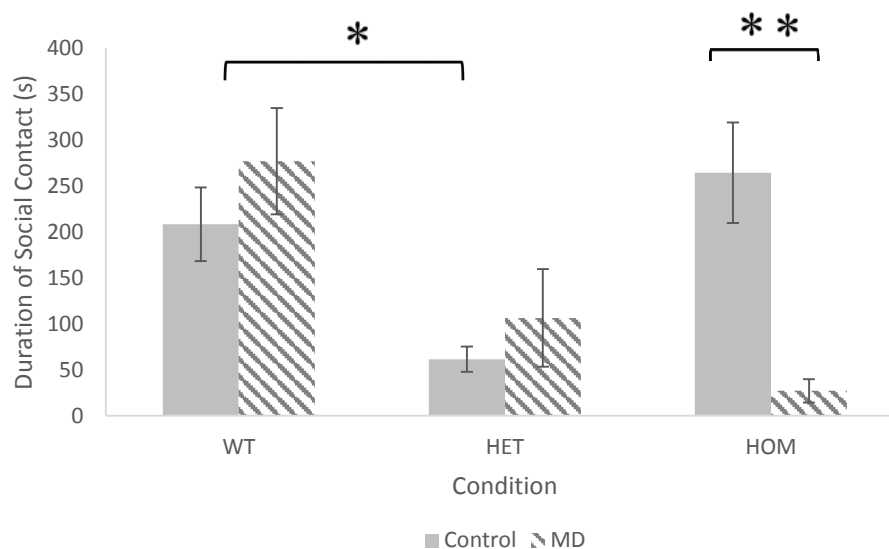


Figure 10. Bar graph depicting the duration of *passive* social contact in the MD and control groups during the social interaction task. Error bars represent ± 1 standard error of the mean.

Lastly, there were no significant findings for genotype, environmental treatment or an interaction between these factors when the MD and control groups were compared on frequency of active and passive social interaction (see *Appendix C*).

USV Data

Due to a large amount of extraneous background noise during recording, the USV data could not be analysed satisfactorily.

Experiment Two: Social Approach and Avoidance

Effects of EH

repeated measures ANOVA was conducted to determine whether the EH subjects spent significantly longer with the novel rat versus the object in phase two, and the novel rat compared to the familiar rat in phase three. There was a significant difference between the novel rat and object in phase two in the EH group, with the HETs (novel rat: $M=178.01$, $SD=127.98$; novel object: $M=62.75$, $SD=111.82$) and HOMs (novel rat: $M=238.30$, $SD=97.93$; novel object: $M=42.58$, $SD=98.48$) showing a preference for the novel rat ($F(1, 26)=8.07$, $p=.009$) (see *Figure 11*). There were no significant findings for the control group, nor were any differences present in phase three.

When frequency of visits in phase two was analysed, there was a significant interaction between genotype and stimulus in the EH group ($F(2, 24)=3.87$, $p=.035$), whereby the HOM EH subjects showed a significant increase in frequency of visits to the novel rat ($M=83.00$, $SD=65.34$) compared with the novel object ($M=9.38$, $SD=14.28$) (see *Figure 12*). There were no significant differences in the control group in phase two, nor were there any differences found in frequency of visits to the novel versus the familiar rat in phase three.

When the EH ($M=172.96$, $SD=96.71$) and control groups ($M=164.39$, $SD=116.42$) were compared on duration near the novel rat, a significant effect of genotype was found ($F(2, 47)=4.93$, $p=.011$), with post-hoc Tukey tests revealing a significant difference between the WT ($M=132.82$, $SD=88.15$) and HOM ($M=229.28$, $SD=108.45$) subjects ($p=.023$), where the HOMs spent significantly more time with the novel rat than the WT did (see *Figure 11*). There were no significant differences found for environmental treatment or genotype-environment interaction for time spent with the novel rat in phase two, nor were there any significant findings for genotype, environment or genotype-environment interaction

when the EH and control groups were compared on time spent with the novel object during⁴⁵ phase two.

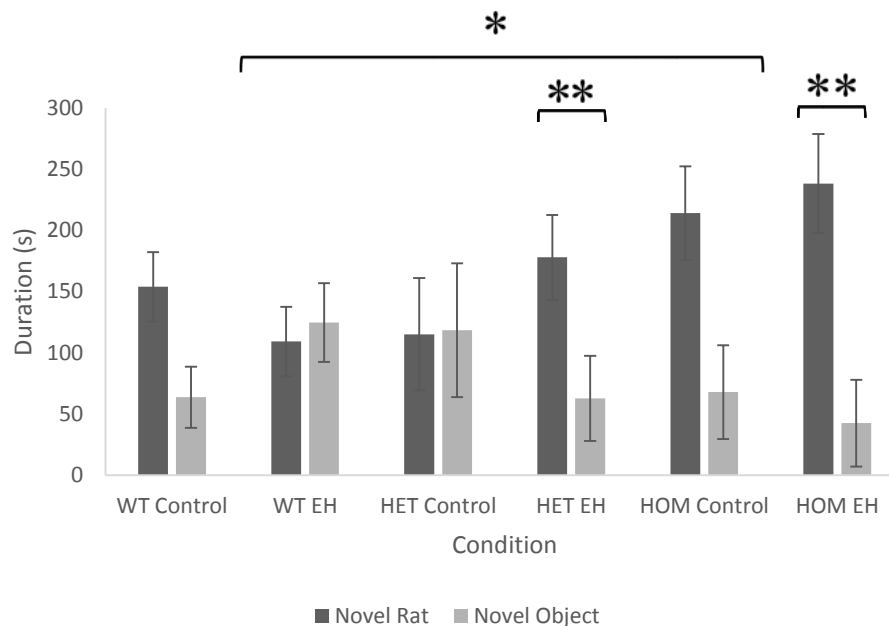


Figure 11. Bar graph depicting the time spent (in seconds) in close proximity with the novel rat and novel object during phase two of the social approach and avoidance task in the EH and control groups. Error bars represent ± 1 standard error of the mean.

There were no significant findings for genotype, environment, or genotype-environment interaction when the EH group was compared with the controls on duration spent with the novel versus the familiar rat in phase three (see *Appendix D*).

When frequency of visits to the novel rat was measured in the EH and the control groups, there was a trend towards significance for the main effect of genotype ($F(2, 47)=3.07$, $p=.056$) and environmental treatment ($F(1, 47)=3.88$, $p=.055$). A significant genotype-environment interaction was found ($F(2, 47)=3.68$, $p=.033$). Post-hoc independent samples t-tests were then conducted, showing significant differences between the HOM groups ($t(7.83)=2.61$, $p=.032$), with the HOM EHs having a significantly higher frequency of visits to the novel rat ($M=83$, $SD=65.33$) than the HOM controls ($M=20.90$, $SD=17.76$) (see *Figure 12*). There were no significant differences found in the WT or HET groups.

In relation to the frequency of visits to the novel object in phase two of the approach/avoidance task in the EH versus the control group, there was a significant effect of

genotype ($F(2,47)=6.91, p=.002$), with post-hoc Tukey tests revealing that the WT_s ($M=42.63, SD=59.25$) had a significantly higher frequency of visits to the novel object than the HOM_s ($M=5.72, SD=10.25$) ($p=.003$). There was a significant effect of environmental treatment ($F(1, 47)=4.53, p=.039$), where the EH group ($M=33.04, SD=11.65$) had a significantly higher frequency of visits to the novel object than the controls ($M=11.65, SD=18.60$). Finally, there was also a significant genotype-environment interaction ($F(2,47)=9.48, p<.001$), whereby the WT EH_s ($M=79.78, SD=69.29$) had a significantly higher frequency of visits to the novel object than the WT controls ($M=9.20, SD=11.41$) ($t(8.39)=3.02, p=.016$) (see *Figure 12*).

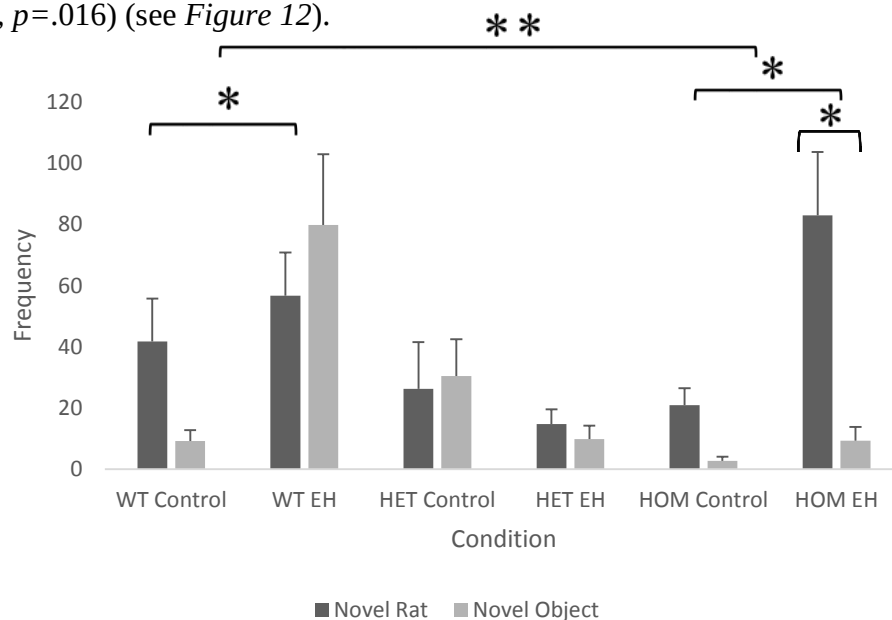


Figure 12. Bar graph showing the frequency of visits to the location of the novel rat and novel object in phase two of the social approach and avoidance paradigm in the EH and control groups. Error bars represent ± 1 standard error of the mean.

Finally, when the EH and control group were compared on frequency of visits to the novel versus the familiar rat in phase three, a significant main effect of environment was found ($F(1, 47)=4.09, p=.049$), whereby the EH group ($M=75.00, SD=131.35$) had a significantly higher number of visits to the familiar rat location than the control group ($M=8.60, SD=10.86$) (see *Figure 13*). There were no significant effects of genotype or a genotype-environment interaction.

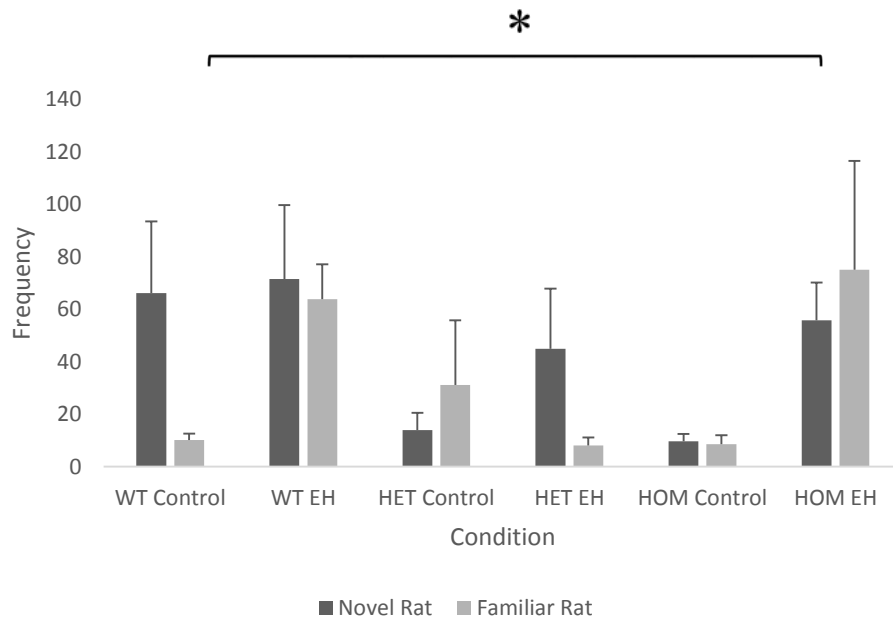


Figure 13. Bar graph showing the frequency of visits to the location of the novel and familiar rat in phase three of the social approach and avoidance paradigm in the EH and control groups. Error bars represent +1 standard error of the mean.

Effects of MD

A (3 (genotype: WT, HET, HOM) x2(stimulus: novel rat, object/familiar rat) repeated measures ANOVA was conducted to determine whether the MD subjects spent significantly longer with the novel rat versus the object in phase two, and the novel rat compared to the familiar rat in phase three. In duration of visits to the novel rat versus the novel object in phase two, the WTs (novel rat: $M=209.21$, $SD=87.85$; novel object $M=34.52$, $SD=45.97$), HETs (novel rat: $M=155.62$, $SD=109.36$; novel object: $M=59.24$, $SD=69.98$) and HOMs (novel rat: $M=228.69$, $SD=109.59$; novel object: $M=62.97$, $SD=122.94$) all spent significantly longer with the novel rat than the novel object ($F(1, 21)=16.33$, $p=.001$) (see *Figure 14*). There were no significant differences found in duration of visits in phase three, nor were there any significant differences in frequency of visits in either phase (see *Appendix D*).

When the MD and control groups were compared on duration spent in close proximity to the novel rat in phase two of the approach avoidance task, a significant result was found for genotype ($F(2, 44)=3.48$, $p=.040$), with post-hoc Tukey tests revealing that the HET

group ($M=125.61$, $SD=112.22$) spent significantly less time near the novel rat than did the 48 HOMs (224.27 , $SD=113.34$) ($p=.037$) (see *Figure 14*). There were no significant differences found for environmental treatment or genotype-environment interaction for time spent with the novel rat in phase two, nor were there any significant findings for genotype, environment or genotype-environment interaction when the MD and control groups were compared on time spent with the novel object during phase two.

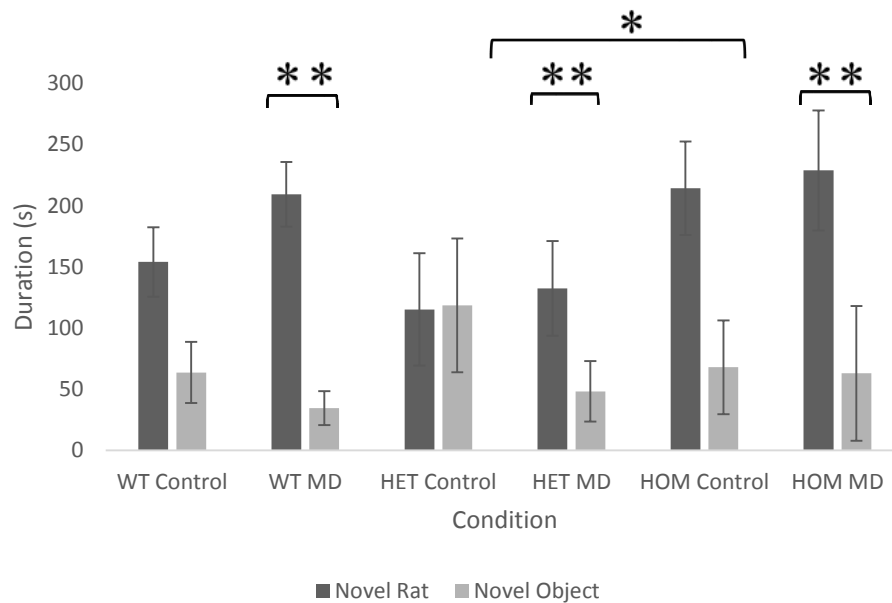


Figure 14. Bar graph depicting the time spent (in seconds) in proximity with the novel rat and novel object during phase two of the social approach and avoidance task in the MD and control groups. Error bars represent ± 1 standard error of the mean.

There were no significant findings for genotype, environment, or genotype environment interactions when the MD and control subjects were compared on duration of visits in phase three, or frequency of visits in phase two and three (see *Appendix D*).

Experiment Three: Scent Marking

Effects of EH

When the EH and control groups were compared on number of social scent markings, there was a significant effect of genotype for social scent ($F(2, 46)= 4.92$, $p=.012$), with post-hoc Tukey tests revealing that that the HETs ($M=57.88$, $SD=46.53$) ($p=.010$) and the HOMs ($M=52.33$, $SD=46.17$) ($p=.033$) had significantly more scent markings than the WTs ($M=28$,

$SD=21.42$). There was also a significant effect of environment ($F(1, 46)=26.27, p<.001$), with the EH subjects ($M=65.26, SD=44.60$) scent marking more than the controls ($M=23.68, SD=19.62$). Finally, there was a significant genotype-environmental treatment interaction ($F(2, 46)=7.64, p<.001$), with independent samples t-tests revealing that the HET EHs ($M=80.40, SD=43.52$) had significantly more social scent markings than the HET controls ($M=20.33, SD=19.53$) ($t(14)=3.16, p=.007$), and the HOM EH subjects also showed more social scent markings ($M=90.50, SD=41.19$) than the HOM controls ($M=21.80, SD=19.41$) ($t(16)=4.69, p<.001$) (see Figure 15).

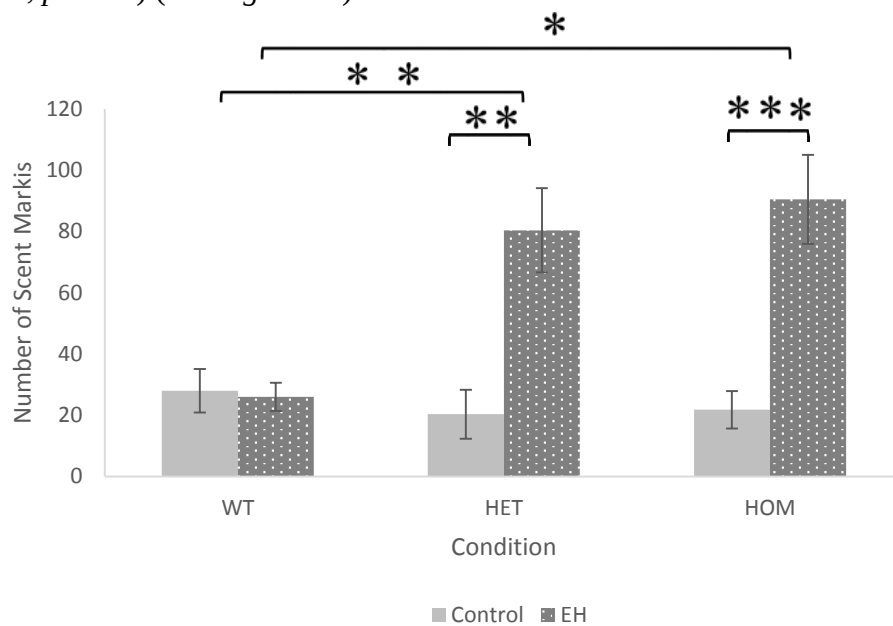


Figure 15. Bar graph showing the number of social scent markings in the EH and control groups. Error bars represent ± 1 standard error of the mean.

In relation to non-social scent markings, a similar pattern was seen, with a significant effect of genotype ($F(2, 46)= 13.39, p<.001$), where post-hoc Tukey tests showed that the HOMs ($M=45.72, SD=30.13$) had significantly more non-social scent markings than the HETs ($M=29.38, SD=20.68$) ($p=.010$) and the WTs ($M=24.06, SD=15.92$) ($p<.001$). There was a significant effect of environmental treatment ($F(1, 46)=20.60, p<.001$), with the EH subjects ($M=41.56, SD=27.84$) showing more non-social scent markings than the controls ($M=24.16, SD=16.64$). Lastly, there was a genotype-environmental treatment interaction ($F(2, 46)=21.66, p<.001$). Independent samples t-tests revealed that the HET EHs ($M=38.50,$

$SD=43.52$) had significantly more non-social scent markings than the HET controls 50

($M=14.17$, $SD=19.36$) ($t(14)=2.72$, $p=.017$), and the HOM EHs ($M=74.25$, $SD=19.42$) had more non-social scent markings than the HOM controls ($M=22.90$, $SD=10.92$) ($t(16)=7.11$, $p<.001$) (see Figure 16).

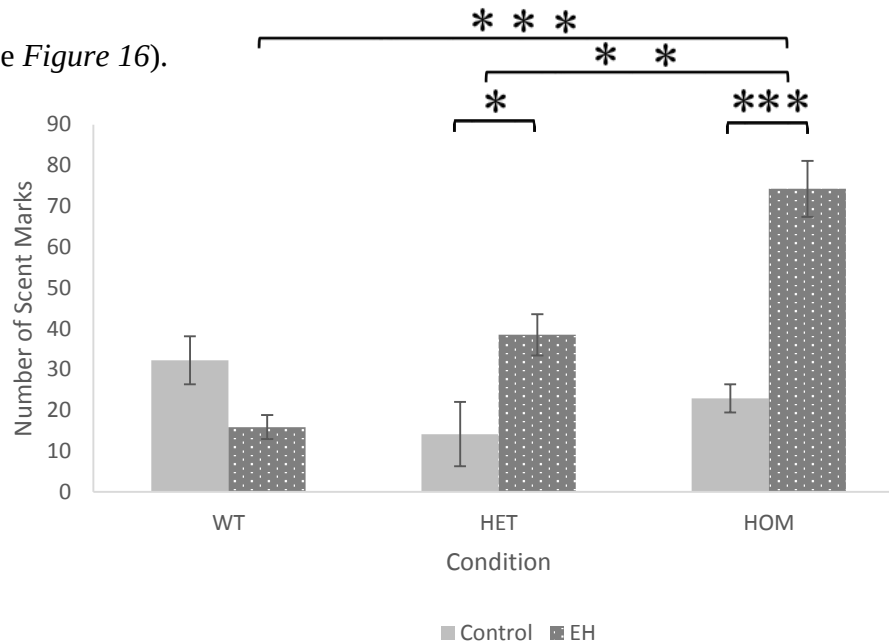


Figure 16. Bar graph showing the number of non-social scent markings in the EH and control groups. Error bars represent ± 1 standard error of the mean.

Finally, in relation to total scent marking, there was a significant effect of genotype ($F(2, 46)=11.11$, $p<.001$) with post-hoc Tukey tests revealing that the WTs ($M=51.06$, $SD=25.01$) had significantly fewer overall scent markings when compared with both the HETs ($M=87.25$, $SD=60.01$) ($p=.009$) and the HOMs ($M=98.06$, $SD=69.03$) ($p<.001$). There was also a significant effect of environment ($F(1,46)=41.83$, $p<.001$), with the EH subjects ($M=106.81$, $SD=62.41$) showing more scent markings overall compared with the controls ($M=47.84$, $SD=29.51$). Lastly, independent samples t-tests showed that the HET EHs ($M=118.90$, $SD=49.14$) had significantly more total scent markings than the HET controls ($M=34.50$, $SD=33.36$) ($t(14)=3.70$, $p=.002$), and the HOM EHs ($M=164.75$, $SD=41.34$) also had significantly more total scent markings than the HOM controls ($M=44.70$, $SD=23.52$) ($t(16)=7.77$, $p<.001$) (see Figure 17).

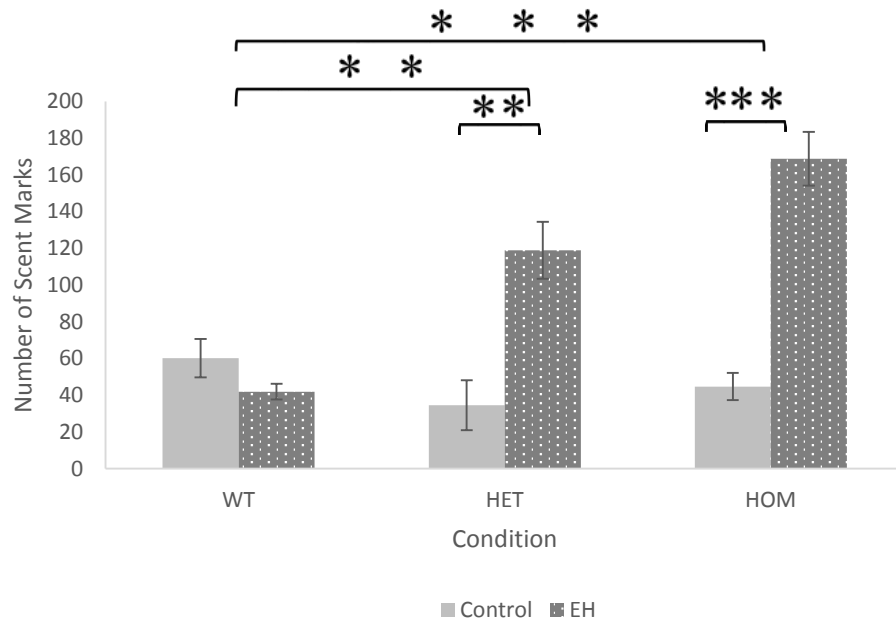


Figure 17. Bar graph showing the number of total scent markings in the EH and control groups. Error bars represent ± 1 standard error of the mean.

When the EH and control groups were compared on percentage social scent markings, there was no significant main effect of genotype, however there was a significant main effect of environmental treatment ($F(1, 46) = 3.90, p = .05$), where the EH group ($M = 59.98, SD = 17.81$) had a significantly higher percentage of social markings than the control group ($M = 46.68, SD = 22.99$) (see *Figure 18*). There was no significant interaction between genotype and environmental treatment.

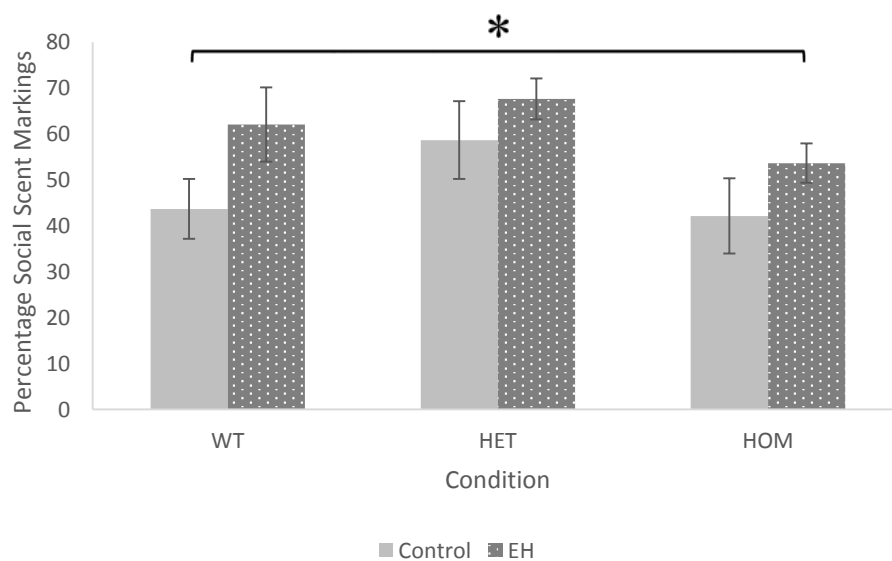


Figure 18. Bar graph showing the difference in social scent marking percentage between the EH and control groups. Error bars represent ± 1 standard error of the mean.

There were no significant findings for genotype, environmental treatment or an interaction between these factors when the MD and control group were compared on percentage social or total number of scent markings (see *Appendix E*).

Experiment Four: Hole Dipping

Effects of EH

In relation to novel hole-dips, there was a significant main effect of genotype when the EH and control groups were compared ($F(2, 44)=4.85, p=.012$), with post-hoc Tukey tests revealing that the HOMs ($M=11.89, SD=8.18$) had significantly more novel hole-dips than the HETs ($M=6.94, SD=3.82$) ($p=.014$). There was also a significant main effect of environmental treatment ($F(1, 44)= 11.06, p=.002$), where the EH rats ($M=11.63, SD=7.26$) had a significantly higher number of novel nose-pokes than the control group ($M=7.35, SD=3.24$). Finally, there was a significant interaction between genotype and environmental treatment ($F(2, 44)= 5.47, p=.008$). Post-hoc independent samples t-tests revealed that the WT EH group ($M=12.33, SD=2.74$) had significantly more novel nose-pokes than the WT control subjects ($M=6.71, SD=3.64$) ($t(14)=3.53, p=.003$). The HOM EH rats ($M=17.38, SD=9.43$) also showed significantly more novel hole-dips than the HOM controls ($M=7.50, SD=3.03$) ($t(8.16)=2.85, p=.021$) (see *Figure 19*).

In relation to the number of repeat nose pokes, there was a significant effect of environmental treatment ($F(1, 44)= 6.30, p=.016$), where the EH subjects ($M=11.44, SD=8.40$) had a significantly higher number of repeat nose pokes than the controls ($M=6.35, SD=6.13$) (see *Figure 20*). There were no significant findings in terms of genotype or genotype-environment interaction.

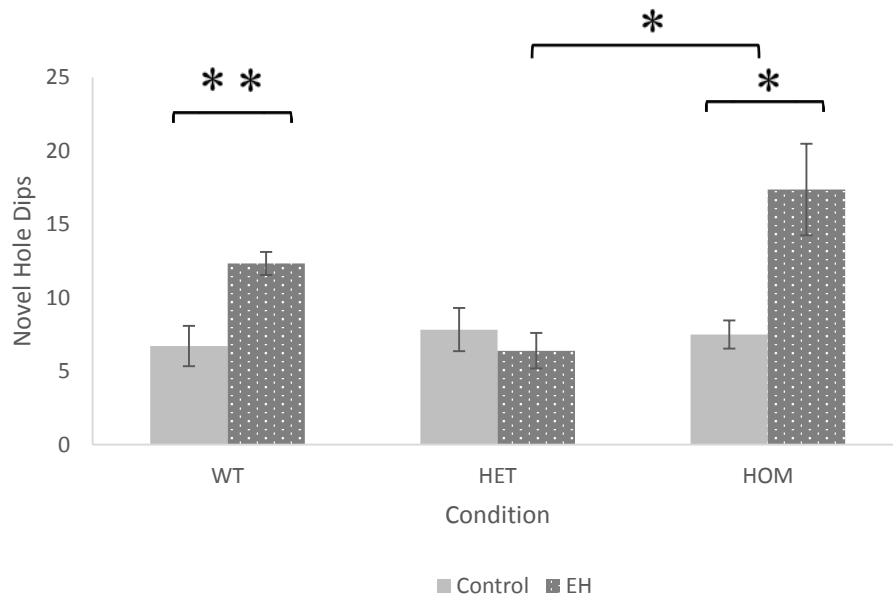


Figure 19. Bar graph depicting the number of *novel* nose pokes/ hole dips in the EH and control groups. Error bars represent ± 1 standard error of the mean.

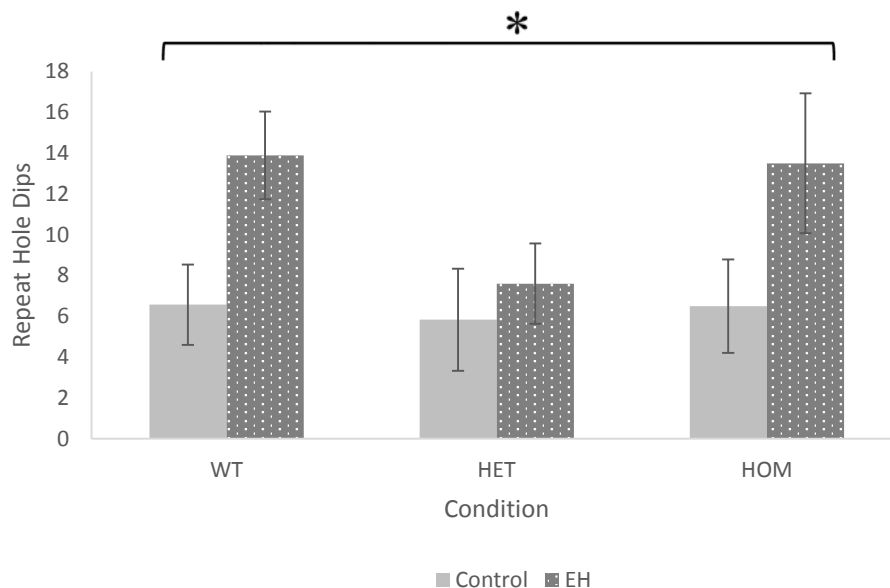


Figure 20. Bar graph depicting the number of *repeat* nose pokes/ hole dips in the EH and control groups. Error bars represent ± 1 standard error of the mean.

Effects of MD

When the MD and control subjects were compared on number of novel nose-pokes, there was a significant main effect of environmental treatment ($F(1, 41)=13.23, p=.001$), where the MD subjects ($M=12.46, SD=5.85$) had significantly more novel nose-pokes than the controls (see Figure 21). There were no significant findings for genotype or a genotype-

environmental treatment interaction. There were no significant results for number of repeat ⁵⁴ hole-dips when the MD group were compared to the controls (see *Appendix F*).

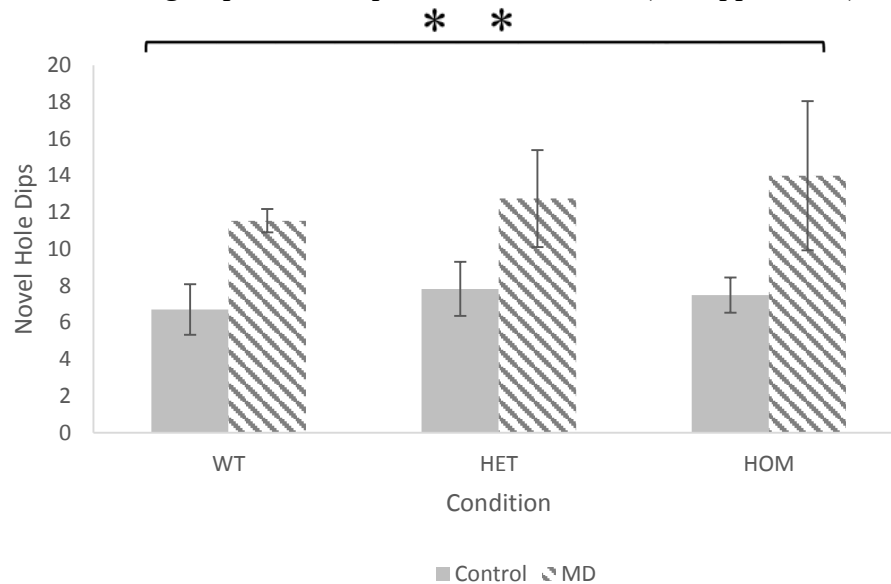


Figure 21. Bar graph depicting the number of *novel* nose pokes/ hole dips in the MD and control groups. Error bars represent ± 1 standard error of the mean.

Experiment Five: Reversal Learning Paradigm

Three rats were excluded from the SERT HOM control group, one for being unwilling to move from the corner of the maze throughout habituation, one for refusing to eat the sugar pellets, and one for five consecutive failures during the learning trials. No rats were excluded from the WT control, EH or MD groups.

Learning-Related Parameters

HOM Controls. Distance Travelled. When the HOM controls ($M=841.66$, $SD=1591.25$) were compared with the WT controls ($M=1396.40$, $SD=2255.40$) on distance travelled (in centimeters) during the learning phase, there was a significant effect of trial in the learning ($F(17, 255)=8.50$, $p<.001$) and reversal ($F(5, 75)=2.45$, $p=.042$) phases, with a decrease in distance over trials. There was also a trend of genotype, with a decrease in distance in the HOM group compared with controls ($F(1, 15)=4.23$, $p=.058$) (see *Figure 22*). There was no significant interaction between trial and genotype.

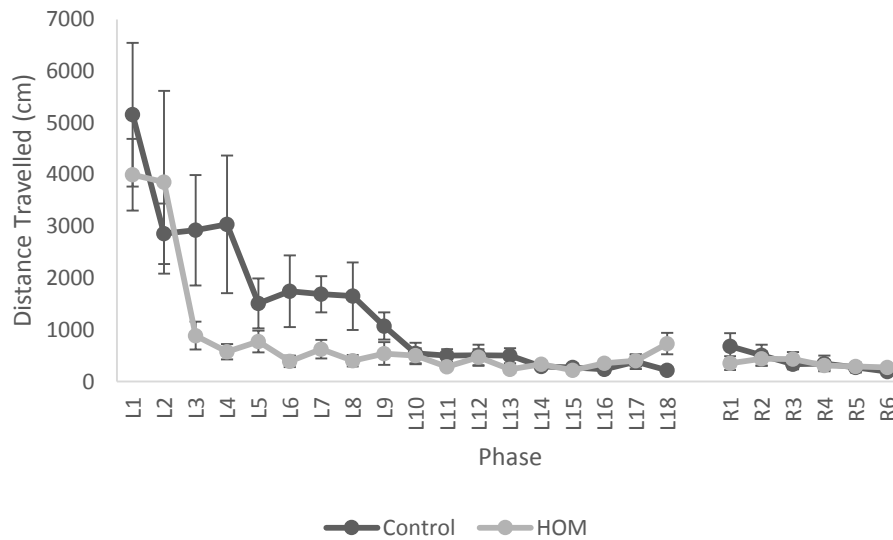


Figure 22. Line graph depicting the distance travelled (in centimeters) of the HOM and control groups in the reversal learning paradigm. Error bars represent ± 1 standard error of the mean.

Ramekin visits. There was a significant effect of trial in the learning ($F(17, 255)=4.81$, $p<.001$) and reversal phases ($F(5, 75)=3.25$, $p=.010$), showing a decrease in ramekin nose-dipping over trials. There were no significant differences between the HOMs and the controls on frequency of ramekin nose dips in either the learning or the reversal phase, nor was there an interaction between trial and genotype (see *Appendix G*).

Effects of EH. Distance travelled. There was a significant effect of trial when the EH subjects were compared to the controls in the learning trials ($F(17, 289)=11.43$, $p<.001$), with a decrease in distance over trials. There was also a significant effect of environment, as the WT EHs ($M=674.41$, $SD=1139.38$) had a significant decrease in distance travelled during the learning phase when compared with the controls ($M=1396.40$, $SD=2255.40$) ($F(1, 17) = 10.57$, $p=.005$). The EH rats ($M=170.54$, $SD=155.97$) also showed a significant decrease in distance travelled during the reversal phase versus the controls ($M=390.57$, $SD=505.25$) ($F(1, 17) = 6.68$, $p=.019$) (see *Figure 23*). There was no interaction between trial and environment.

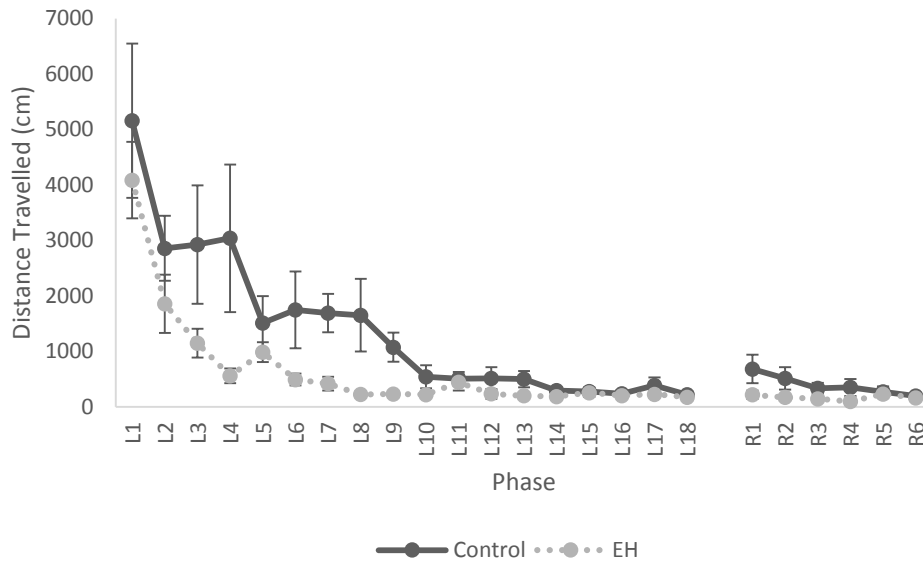


Figure 23. Line graph depicting the distance travelled (in centimeters) of the EH and control groups in the reversal learning paradigm. Error bars represent ± 1 standard error of the mean.

Ramekin Visits. There was a significant effect of trial in the learning phase ($F(17, 289)=9.54, p<.001$) with a decrease in ramekin dipping over trials. There was a significant effect of environment, where the EH subjects ($M=7.86, SD=12.23$) showed a significant decrease in frequency of ramekin visits in the learning phase compared with the controls ($M=21.37, SD=9.50$) ($F(1, 17)=8.80, p=.009$) (see Figure 24). No significant differences were found in ramekin visits during the reversal phase, nor was there an interaction between trial and environment in either phase.

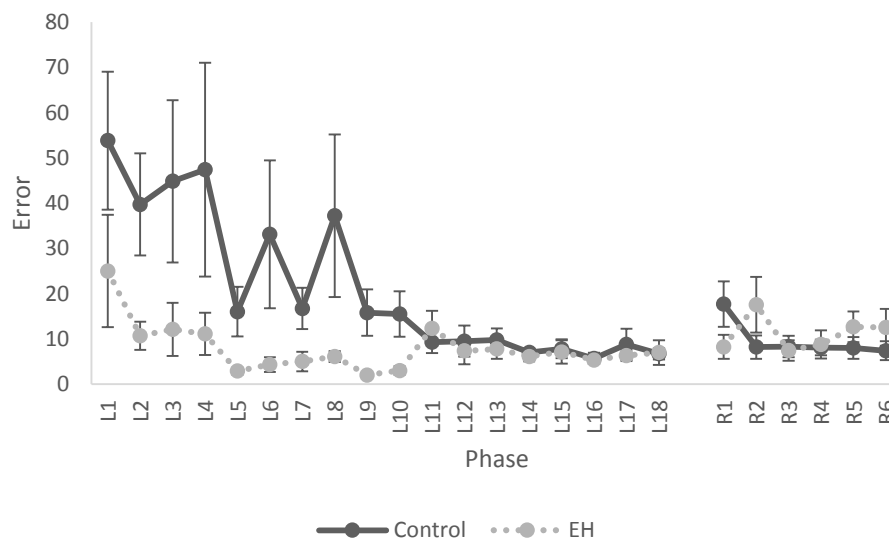


Figure 24. Line graph depicting the number of nose dips into the ramekins (error) of the EH and control groups in the learning and reversal phase of the reversal learning paradigm. Error bars represent ± 1 standard error of the mean.

learning phase ($F(17, 306)=9.95, p<.001$) as well as a significant effect of environment, where the WT MD rats ($M=781.07, SD=1260.79$) showed a significant decrease in distance travelled compared with the controls ($M=1396.40, SD=2255.40$) ($F(1, 18)=8.13, p=.010$). There was also a significant effect of trial in the reversal phase ($F(5, 90)=2.77, p=.023$). Finally, there was a significant trial-environment interaction ($F(17, 306)=1.86, p=.021$), where the improvement in the MD group occurred in the beginning of the learning phase (see *Figure 25*). There were no significant differences in environmental treatment seen in the reversal phase.

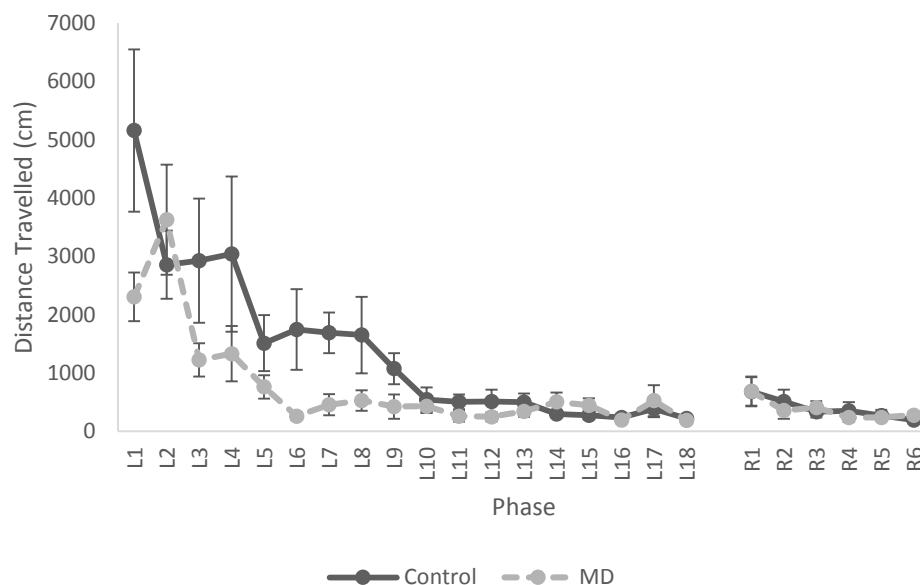


Figure 25. Line graph depicting the distance travelled (in centimeters) of the MD and control groups in the reversal learning paradigm. Error bars represent ± 1 standard error of the mean.

Ramekin Visits. There was a significant effect of trial in the learning ($F(17, 306)=4.56, p<.001$) and reversal phases ($F(5, 90)=2.78, p=.022$), with a decrease in ramekin dips over trials. There were no significant environmental differences or trial-environment interactions in ramekin visits in the learning and reversal phases (see *Appendix G*).

Probe trial. WT controls. A paired samples t-test showed that the WT control group spent significantly longer in the target quadrant ($M=30.57, SD= 16.48$) as compared to the

opposite quadrant ($M=11.66$, $SD=7.67$) during the post-learning phase probe trial

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($t(9)=3.47$, $p=.007$) (see Figure 26).

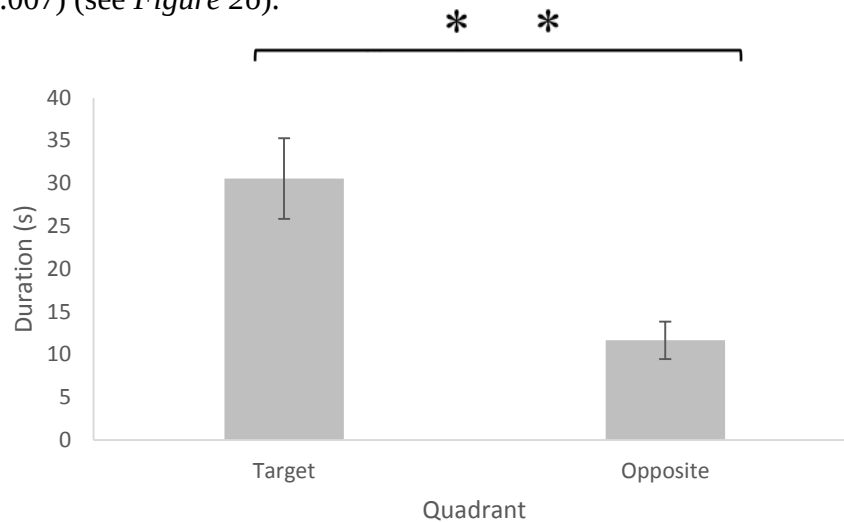


Figure 26. Bar graph depicting the duration of time (in seconds) spent in the target compared with the opposite quadrant during the post-learning phase probe trial in the WT controls. Error bars represent ± 1 standard error of the mean.

HOM control. There were no significant differences in time spent in the target versus the opposite quadrant during the probe trial in the HOM group (see Appendix G).

Effects of EH. A paired samples t-test was conducted, showing that the WT EH group spent significantly longer in the target quadrant ($M=18.66$, $SD= 6.86$) as compared to the opposite quadrant ($M=8.89$, $SD=4.63$) during the post-learning phase probe trial ($t(9)=3.54$, $p=.008$) (see Figure 27).

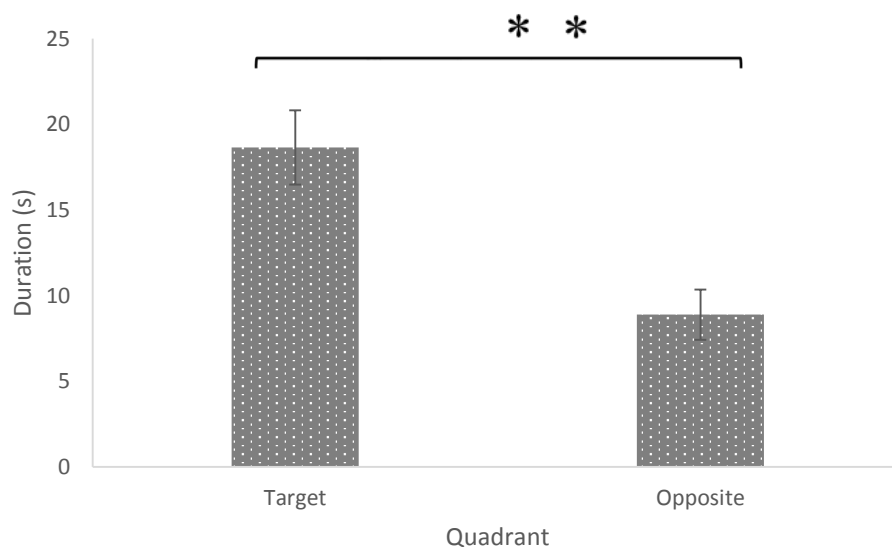


Figure 27. Bar graph depicting the duration of time (in seconds) spent in the target compared with the opposite quadrant during the post-learning phase probe trial in the WT EHs. Error bars represent ± 1 standard error of the mean.

versus the opposite quadrant in the MD group (see *Appendix G*).

Anxiety-Related Parameters

HOM Controls. Body elongation. There was a significant effect of trial in the learning ($F(17, 255)=4.15, p<.001$) and reversal phases ($F(5, 75)=5.3, p<.001$), with a decrease in elongation over trials. There were no significant genetic differences during the learning phase, however the HOMs ($M=4.10, SD=3.57$) showed an increase in duration of body elongation in the reversal phase when compared with the controls ($M=1.91, SD=2.35$) ($F(1, 15)= 8.41, p=.011$). Finally, there was a significant interaction between trial and genotype in the learning phase ($F(17, 255)=1.93, p=.016$) where the HOMs show an increase in elongation in the later trials (see *Figure 28*).

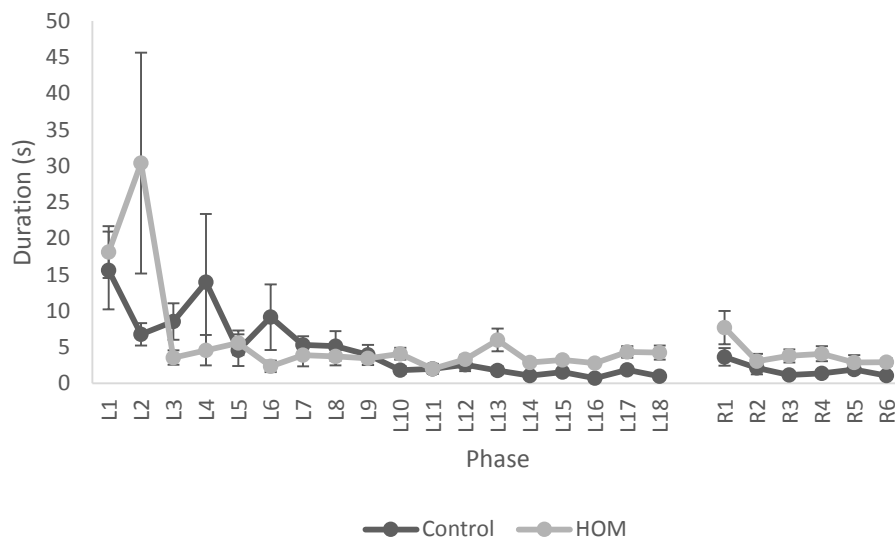


Figure 28. Line graph depicting the duration of time (in seconds) of body elongation in the HOMs and controls. Error bars represent ± 1 standard error of the mean.

Body Contraction. There was a significant effect of trial in the learning ($F(17, 255)=9.23, p<.001$) and reversal phases ($F(5, 70)=3.06, p=.015$), where the subjects showed a decrease in duration of contraction over trials. In terms of genetic effects, there were no significant differences between the WT controls and HOM controls in duration (in seconds) of body contraction during the learning phase, whereas the HOMs ($M=4.51, SD=3.80$)

showed a significant increase in duration of contraction during the reversal phase, versus 60 the controls ($M=1.57$, $SD=1.45$) ($F(1, 14) = 28.38$, $p<.001$) (see *Figure 29*). There was no significant interaction between trial and genotype.

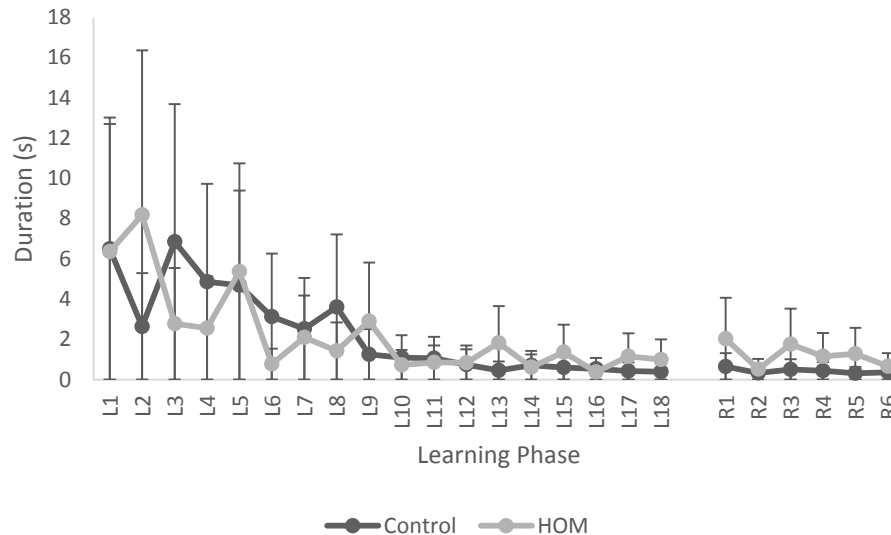


Figure 29. Line graph depicting the duration of time (in seconds) of body contraction in the HOM subjects and controls. Error bars represent ± 1 standard error of the mean.

Immobility. There was a significant effect of trial in the learning ($F(17, 255)=8.22$, $p<.001$) and reversal phases ($F(5, 75)=4.29$, $p=.002$), with a decrease in immobility over trials. In terms of genetic effects, no significant differences were found in duration of immobility during the learning phase. However, the HOMs ($M=20.34$, $SD=14.87$) showed a significant increase in immobility compared with the controls ($M=12.01$, $SD=13.40$) during the reversal phase ($F(1, 15) = 8.79$, $p=.010$) (see *Figure 30*). There was no significant interaction between trial and genotype.

Thigmotaxis. The time spent in the center of the arena was first divided by the total time spent in the whole arena to calculate the percentage of time the subject spent in the center. There was a significant effect of trial in the learning phase ($F(17, 255)=4.81$, $p<.001$), where there was an increase in percentage of time spent in the center of the arena across trials. There was a significant effect of genotype, where the HOMs ($M=50.82$, $SD=28.09$) showed a significant decrease in percentage of time spent in the center of the field in the

learning phase when compared with the controls ($M=64.22$, $SD=28.25$) ($F(1, 15)=6.55$, $p=.022$). The HOMs ($M=72.48$, $SD=21.84$) also showed a significantly decreased duration in the center in the reversal phase when compared to the controls ($M=92.64$, $SD=13.66$) ($F(1, 15)=12.30$, $p=.003$) (see *Figure 31*). There was no significant interaction between trial and genotype.

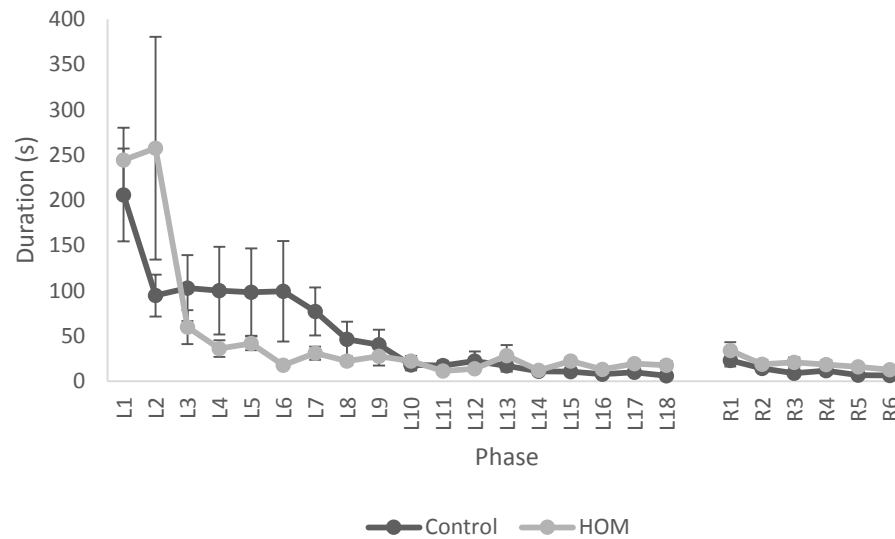


Figure 30. Line graph depicting the duration of immobility in the HOM subjects and controls. Error bars represent ± 1 standard error of the mean.

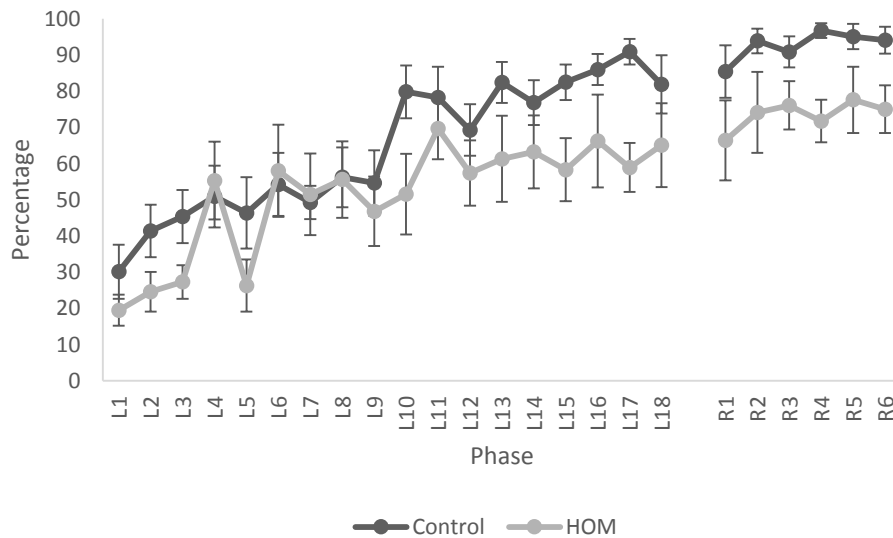


Figure 31. Line graph depicting the percentage of time spent in the center of the open field (thigmotaxis) in the HOM subjects and controls. Error bars represent ± 1 standard error of the mean.

Effects of EH. Body elongation. There was a significant effect of trial in the learning

($F(17, 289)=15.80$, $p<.001$) and reversal phases ($F(5,85)=3.42$, $p=.007$), with a decrease in

elongation duration over trials. There were no significant environmental differences 62

between the EH and control groups in duration of body elongation in the learning or reversal phase of the task. Finally, there was a significant interaction between trial and environment ($F(17, 289)=2.27, p=.003$), whereby the EH rats show an initial increase in elongation in the learning phase (see *Figure 32*).

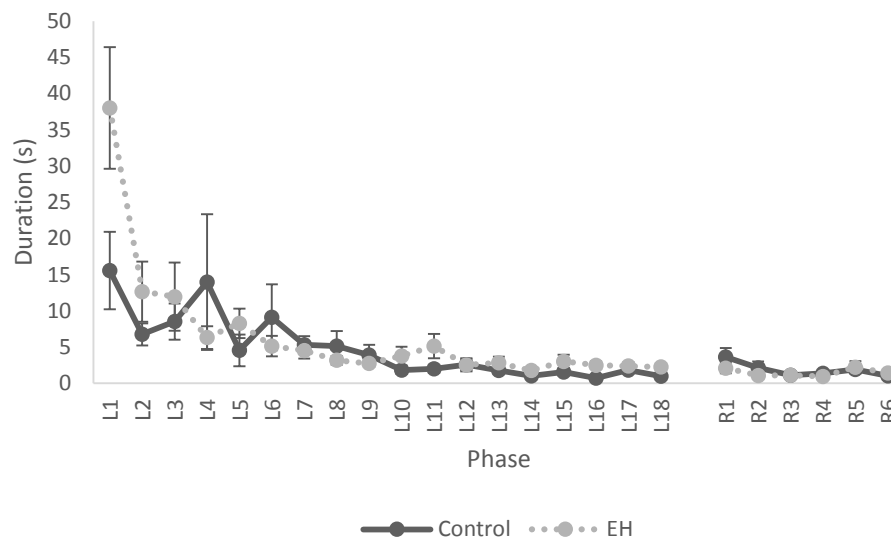


Figure 32. Line graph depicting the duration of time (in seconds) of body elongation in the WT EHs and controls. Error bars represent ± 1 standard error of the mean.

Body contraction. There was a significant effect of trial in the learning phase ($F(17, 289)=9.34, p<.001$), with a decrease in duration of body contraction over trials. In terms of environmental effects, there were no significant differences between the EH and control rats on duration of body contraction in the learning phase, however, in the reversal phase, the EH subjects ($M=2.48, SD=2.59$) had a significantly longer duration of body contraction than the controls ($1.57, SD=1.45$) ($F(1, 16)=6.96, p=.018$) (see *Figure 33*). There was no significant interaction between environment and trial.

Immobility. *WT EHs.* There was a significant effect of trial in the learning ($F(17, 289)=15.80, p<.001$) and reversal phases ($F(5, 85)=2.43, p=.042$), with a decrease in immobility over trials. In relation to environmental effects, there were no significant differences between the EH and control subjects in duration of time spent immobile in the

learning or reversal phase. There was a significant interaction between trial and environmental treatment in the learning phase ($F(17, 289)=3.21, p<.001$), where a decrease in immobility occurred after the initial trials in the EH subjects compared with the controls (see *Figure 34*).

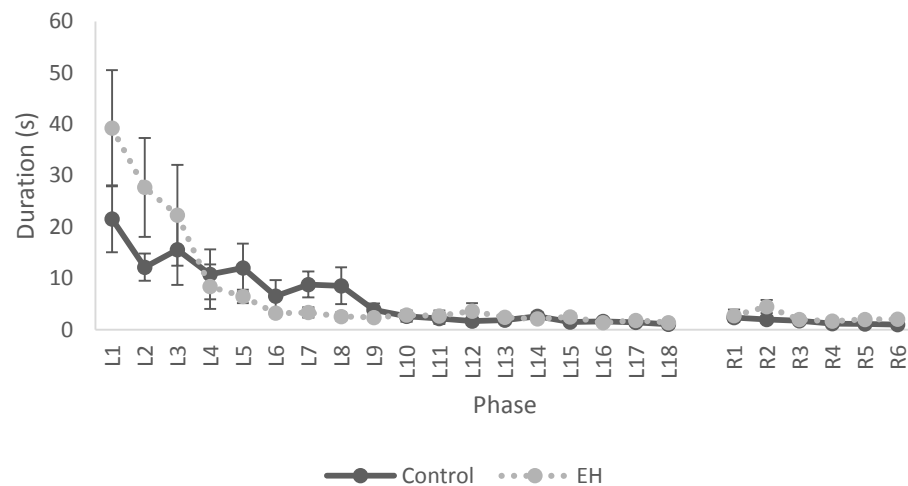


Figure 33. Line graph depicting the duration of time (in seconds) of body contraction in the EH subjects and controls. Error bars represent ± 1 standard error of the mean.

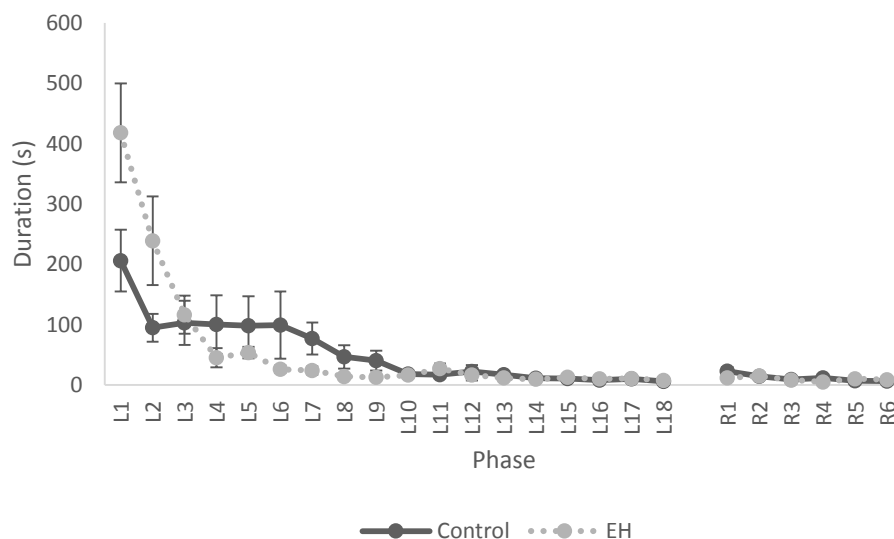


Figure 34. Line graph depicting the duration of time (in seconds) of immobility in the WT EHs and controls. Error bars represent ± 1 standard error of the mean.

Thigmotaxis. There was a significant effect of trial in the learning phase ($F(17, 289)=21.72, p<.001$), with an increase of percentage in the center across trials. There was a significant effect of environment, where the EH group ($M=72.96, SD=26.77$) spent a

significantly higher percentage of time in the center of the arena in the learning phase compared with the controls ($M=64.22$, $SD=28.25$) ($F(1, 17) = 4.84$, $p=.042$). Lastly, there was a significant interaction between environment and trial in the learning phase ($F(17, 289)=2.15$, $p=.006$), where this increase occurred after the initial trials (see *Figure 35*).

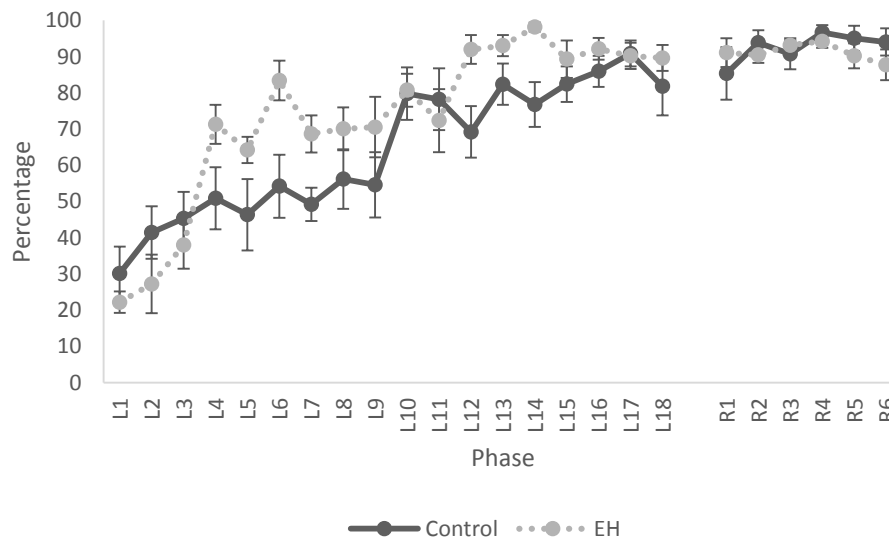


Figure 35. Line graph depicting the percentage of time spent in the center of the open field (thigmotaxis) in the EH subjects and controls. Error bars represent ± 1 standard error of the mean.

Effects of MD. Body elongation. There was a significant effect of trial in the learning ($F(17, 306)=4.59$, $p<.001$) and reversal phases ($F(5, 90)=2.96$, $p=.016$), with a decrease in body elongation over trials. In terms of environment, there were no significant differences in the learning phase, however, the MD group ($M=4.64$, $SD=6.53$) had a significant increase in body elongation in the reversal phase compared with the controls ($M=1.91$, $SD= 2.35$) ($F(1, 18) =8.57$, $p=.009$). Finally, there was a significant environment-trial interaction in the learning phase ($F(17, 306)=1.93$, $p=.015$) where the MD subjects show an increase in elongation during the later trials (see *Figure 36*).

Body contraction. There was a significant effect of trial in the learning phase ($F(17, 306)=8.96$, $p<.001$), with a decrease in duration of body contraction over trials. There were no significant environmental effects, or trial-environment interactions in body contraction in

the learning or reversal phases in the WT MDs when compared with the controls (see 65
Appendix H).

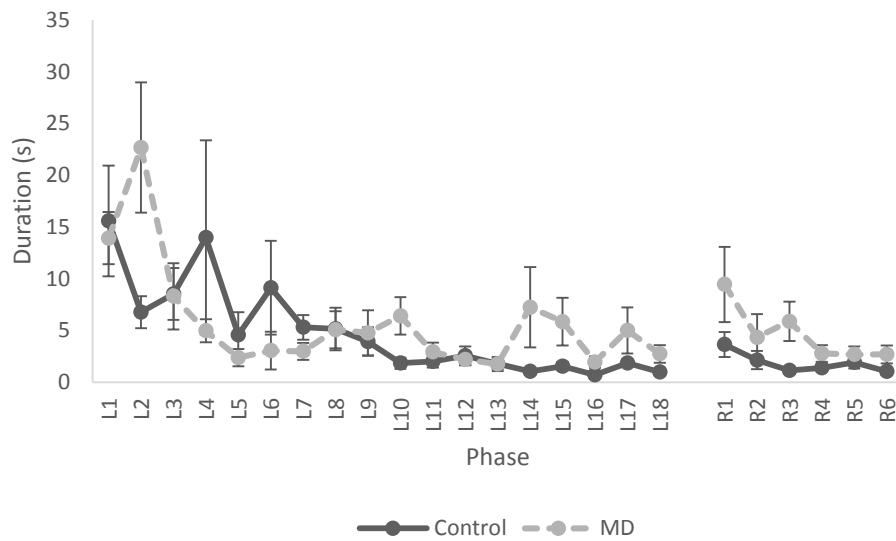


Figure 36. Line graph depicting the duration of time (in seconds) of body elongation in the WT MDs and controls. Error bars represent ± 1 standard error of the mean.

Immobility. There was a significant effect of trial in the learning ($F(17, 306)=10.02$, $p<.001$) and reversal phases ($F(5, 90)=4.23$, $p=.002$), with a decrease in immobility over trials. There were no significant environmental effects of immobility in the learning and reversal phases in the WT MDs compared with the controls. Finally, there was a significant environment-trial interaction in the learning phase ($F(17, 306)=2.15$, $p=.006$), where the MD subjects showed a decrease in immobility after the initial trials in the first half of the learning phase (see *Figure 37*).

Thigmotaxis. There was a significant effect of trial in the learning phase ($F(17, 306)=14.49$, $p<.001$), with an increase of percentage of time in the center across trials. There was a significant effect of environment, where the MD group ($M=72.66$, $SD=25.42$) had a significant increase in the percentage of time spent in the center of the field compared to the controls in the learning phase ($M=64.22$, $SD=28.25$) ($F(1, 18)=4.72$, $p=.050$) (see *Figure 38*). There was no significant interaction between trial and environmental treatment.

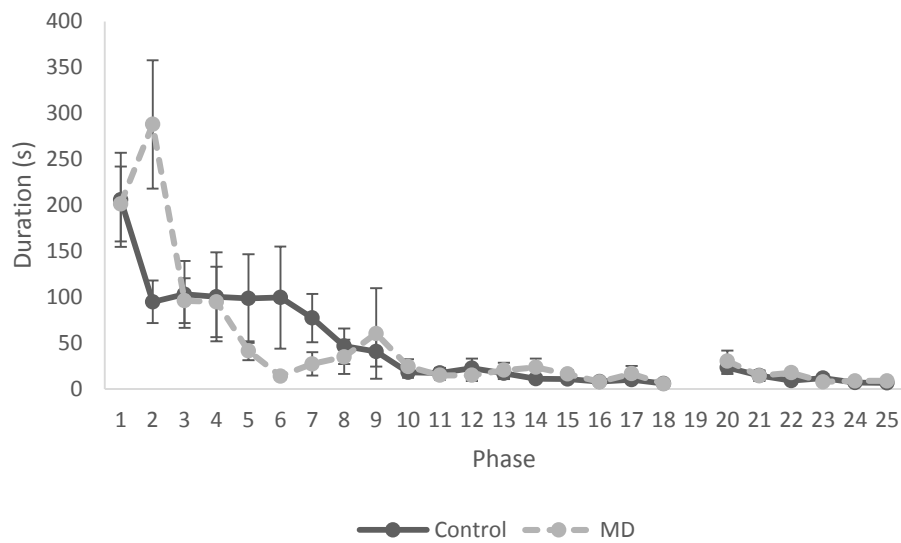


Figure 37. Line graph depicting the duration of time (in seconds) of immobility in the WT MDs and controls. Error bars represent ± 1 standard error of the mean.

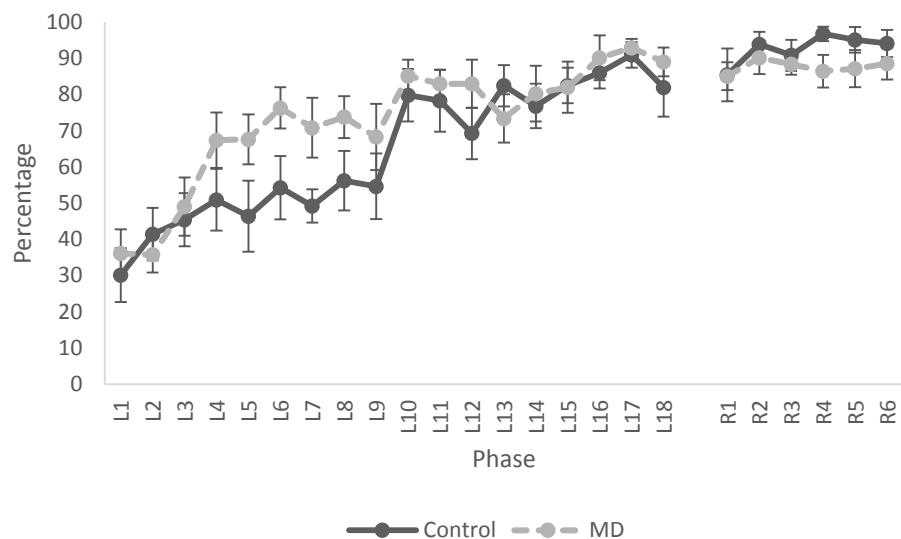


Figure 38. Line graph depicting the percentage of time spent in the center of the open field (thigmotaxis) in the MD subjects and controls. Error bars represent ± 1 standard error of the mean.

Stereotypy-Related Parameters

HOM controls. *Head weaving.* There were no significant effects of trial, genotype or trial-genotype interactions in the HOM controls in the learning or reversal phase (see Appendix I).

Rotation. There was a significant effect of trial in the learning ($F(17, 255)=7.56$, $p<.001$) and reversal phases ($F(5, 75)=4.29$, $p=.001$), with a decrease in rotation over trials.

There were no significant genetic effects, or interaction between genotype and trial in either phase (see *Appendix I*). 67

Effects of EH. Head weaving. There was a significant effect of trial in the learning phase ($F(17, 289)=1.86, p=.021$), with an increase in head weaving across trials. There were no significant environmental effects of head weaving in the learning phase, however the EH group ($M=90.14, SD=9.66$) showed a significant increase in frequency compared to the controls ($M=81.78, SD=15.40$) in the reversal phase ($F(1, 17) =4.50, p=.049$). Finally, there was a significant trial-environment interaction ($F(17, 289)=1.87, p=.021$), where the increase in head weaving in the EH group occurred in the later trials (see *Figure 39*).

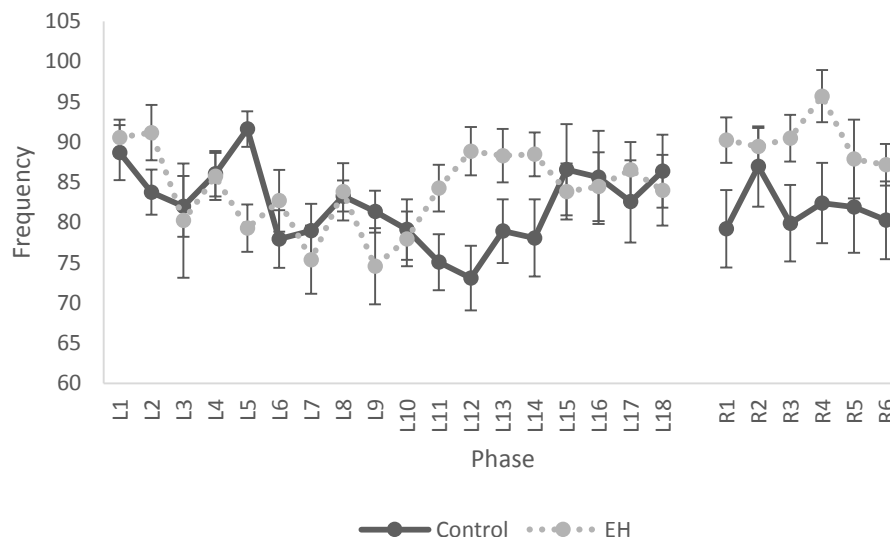


Figure 39. Line graph depicting the frequency of head weaving in the EH and control subjects. Error bars represent ± 1 standard error of the mean.

Rotation. There was a significant effect of trial in the learning phase ($F(17, 289)=13.00, p<.001$), where frequency of rotation decreased across trials. There were no significant environmental effects, or interactions between trial and environment in the learning or reversal phase (see *Appendix I*).

Effects of MD. Head weaving. There was a significant effect of trial in the learning phase ($F(17, 306)=1.93, p=.016$), with an increase in head weaving across trials. In relation to environmental effects, the MD group ($M=89.03, SD=9.07$) had a significant increase in

head weaving in the learning phase compared with the controls ($M=82.17$, $SD=12.66$) ($F(1, 18)=13.77$, $p=.002$). The MD subjects ($M=95.30$, $SD=8.67$) also showed an increase in head weaving in the reversal phase compared to the controls ($M=81.78$, $SD=15.40$) ($F(1,18)=15.80$, $p=.001$). Finally, there was a significant interaction between trial and environment in the learning phase ($F(17, 306)=1.74$, $p=.035$), where the increase in head weaving in the MD group occurs in the later trials (see *Figure 40*).

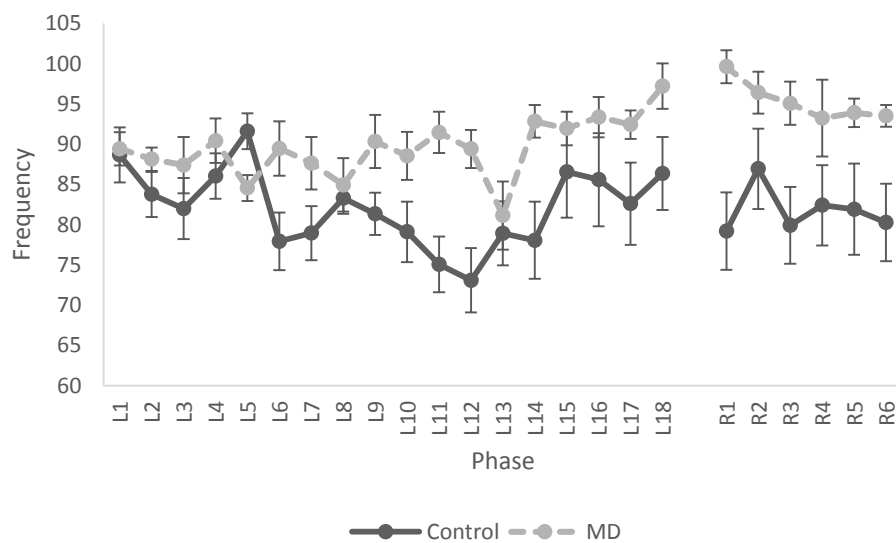


Figure 40. Line graph depicting the frequency of head weaving in the MD and control subjects. Error bars represent ± 1 standard error of the mean.

Rotation. There was a significant effect of trial in the learning ($F(17, 306)=8.37$, $p<.001$) and reversal phases ($F(5, 90)=2.84$, $p=.020$), with a decrease in rotation over trials. There were no significant environmental effects in frequency of rotation in the learning phase. In the reversal phase, however, the MD rats ($M=35.42$, $SD=50.67$) showed a significant increase in rotation frequency when compared with the controls ($M=25.36$, $SD=41.49$) ($F(1, 18)=19.93$, $p<.001$) (see *Figure 41*).

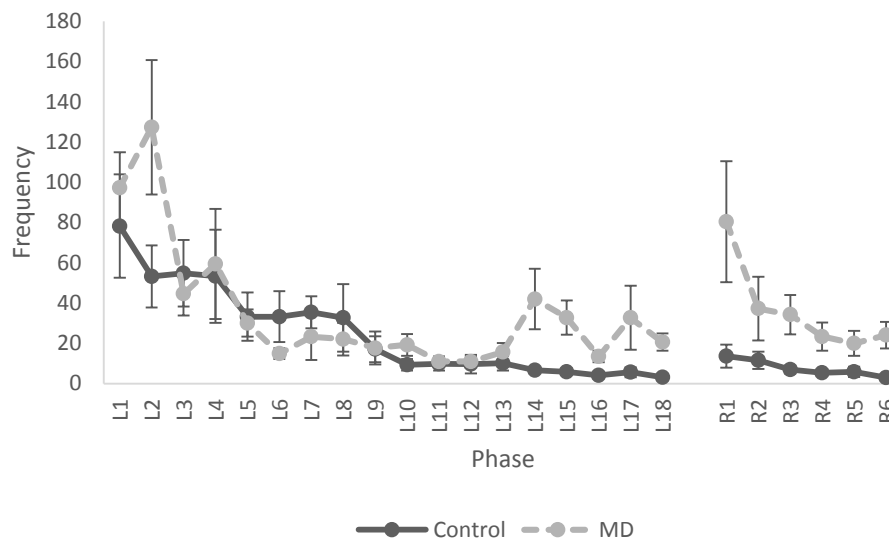


Figure 41. Line graph depicting the frequency of rotation in the MD and control subjects. Error bars represent ± 1 standard error of the mean.

Discussion

The purpose of the current research was to evaluate two different gene-environment interaction models: the diathesis-stress and the plasticity model, in relation to ASD-like behaviour in animals. To that end, heterozygous (HET) and homozygous (HOM) SERT knockout and wildtype (WT) rats were exposed to either a positive (EH) or a negative (MD) early environment. Both models predict that the HET and HOM KO MD rats will show significantly more ASD-like behaviours compared to the other groups. However, while the vulnerability model predicts that the HET EH and WT EH control groups are more or less the same, the plasticity model predicts that the HET (and HOM) SERT EH rats show significantly less symptoms, compared to the WT EH groups. Consequently, the results from each experiment will be discussed in relation to whether or not they provide support for either of these predictions (see *Table 2* and *Table 3* for results summaries for the social interaction, approach/avoidance, scent marking and hole-dipping tasks).

Table 2

Early Handling Results Summary

<i>Test</i>	<i>G</i>	<i>E</i>	<i>G x E</i>
<i>Social Behaviour</i>			
SI Active Frequency	HOM>WT		
SI Active Duration			
SI Passive Frequency	HOM>WT/HET		
SI Passive Duration	WT/HOM>HET		
SAA-Novel Rat Duration	HOM>WT		
SAA-Novel Rat Frequency			HOM EH>HOM Control
SAA-Novel Object Frequency	WT>HOM	EH>Control	WT EH>WT Control
<i>Social Communication</i>			
Number of Social	HET/HOM>WT	EH>Control	HET/HOM EH>HET/HOM Control
Number of Non-Social	HOM>HET/WT	EH>Control	HET/HOM EH>HET/HOM Control
Number of Total	HET/HOM>WT	EH>Control	HET/HOM EH>HET/HOM Control
% Social		EH>Control	
<i>Stereotypy</i>			
Novel Hole Dips	HOM>HET	EH>Control	WT/HOM EH>WT/HOM Control
Repeat Hole Dips		EH>Control	

Table 3

Maternal Deprivation Results Summary

<i>Test</i>	<i>G</i>	<i>E</i>	<i>G x E</i>
<i>Social Behaviour</i>			
SI Active Frequency			
SI Active Duration			
SI Passive Frequency			
SI Passive Duration	HET<WT		HOM MD<HOM Control
SAA-Novel Rat Duration	HET<HOM		
SAA-Novel Rat Frequency			
SAA-Novel Object Frequency			
<i>Stereotypy</i>			
Novel Hole Dips		MD>Control	
Repeat Hole Dips			

In the present study, two different paradigms were used to assess social cognition: social interaction and social approach/avoidance. One of the main findings is that the HOM MD rats have a significant reduction in some aspect of social behaviours, particularly in the duration of passive contact. Unlike active social contact, which is a measure of social interest, and involves one, or both of the subjects initiating contact, passive social contact measures the general need for physical contact (File & Pope, 1974; Latane, Cappell, & Joy, 1970). This indicates that while the HOM MD subjects show a normal social interest in their conspecifics, they may not require the prolonged physical contact that is a typical aspect of rat behaviour (Latané, Cappell, & Joy, 1970). A reduction in social investigation has previously been found in MD rats, although, unlike the current research, this effect was found in the social approach/avoidance task, rather than the social interaction task (Girardi, Zanta, & Suchecki, 2014). As there were no significant results for the MD group in the social approach/avoidance task in the current study, it seems that this task measures different aspects of social behaviour. Although one could argue that the task may not be sensitive enough to identify these impairments, perhaps due to the use of the T maze instead of the three chambered apparatus, a gene*environment interaction was found with this task (see below). In any case, this reduction in social contact is similarly reflected in ASD in humans, as autistic individuals often find physical contact, such as hugging and touching, particularly unpleasant (Voos, Pelphrey, & Kaiser, 2013).

While the HOM MD rats exhibited an impairment in passive social behaviour, the HOM EH rats showed an improvement in other aspects of social behaviour, specifically in the social approach and avoidance paradigm. This paradigm is designed to measure more active social investigation, and differs from the social interaction task in that there is choice within the stimuli, and the social stimulus is stationary, meaning that the duration and

frequency of visits to these stimuli is entirely determined by the experimental rat.

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Therefore, while no effects were seen in the EH subjects in the social interaction task, the increased frequency of visits to the social stimulus in the HOM EH rats indicates that there is an increase in social investigation in these subjects. An increased preference for the novel social stimulus has previously been found in EH rats (Todeschin et al., 2009), although these subjects also showed a decrease in social investigation during the social interaction task, which was not found in the current study, but further highlights the different behavioural measures in the interaction versus the approach/avoidance tasks.

Overall, these findings provide support for the plasticity model, as the KO HOMs show an impairment in social behaviours when exposed to maternal deprivation, while the KO HOMs that experienced early handling showed some improvement. The vulnerability model cannot be supported by these results, as it was predicted that early handling would have little to no effect on the behavioural outcome.

Along with these findings, there was also a relatively consistent increase in social behaviour in the SERT KO HOMs across the social tasks, especially in relation to the frequency of active and passive social interactions, which was unexpected, as a decrease in social behaviours was previously found in homozygous SLC6A4 knockout mice (Moy et al., 2009). However, while Moy and colleagues found a reduction in time spent with the stranger mouse in the social interaction task, they did not find any impairment in frequency of sniffing in the three chamber social approach task, with the HOM mice showing increased sniffing of the novel rat when compared to the novel object, which reflects the increased frequency of visits to the novel rat in the social approach avoidance paradigm. Additionally, while HOM rats have been found to have a reduction in social play behaviour, other aspects of social interaction and social exploration have been found to be unimpaired. In fact, an increase in following and chasing behaviour has previously been found in these subjects (Homberg et al.,

2007b). Therefore, while the HOMs may be impaired in some aspects of social behaviour, 73 they also show improvement in other facets of these behaviours. Moreover, while the majority of research has focused on the negative behavioural characteristics associated with the HOMs, they also show improvements in other behavioural aspects, such as decision-making in the rodent version of the Iowa Gambling Task (Homberg et al., 2008), as well as inhibitory control and impulsivity (Homberg et al., 2007a). These behavioural improvements highlight the potential role of the SERT genotype as a plasticity gene, which is not inherently positive or negative, but dependent on the aspect of behaviour that is measured, as well as the environment that the individual experiences. Consequently, while the increased social contact in the HOM controls was unexpected in the current experiment, it is an interesting finding and should be explored further in future, by investigating a broad range of social behaviours in the SERT KO model, to determine whether additional improvements are also apparent.

Finally, while the HOMs showed an improvement in social behaviours, the opposite pattern was seen in the HETs. This was unusual, as the HETs have generally been found to either reflect the behaviours present in the WTs, or to show similar behavioural characteristics to the HOMs, but to a lesser extent, due to gene-dosage effects (Kalueff et al., 2007). However, in this case, the HETs show a completely different pattern to both the WTs and the HOMs. While this cannot be fully explained as yet, there has been some suggestion that the 50% reduction in SERT in the HETs, compared with the complete loss of SERT in the HOMs, results in fundamentally different alterations to particular aspects of behaviour, including social behaviours (Moy et al., 2009). Therefore, the HETs and the HOMs may be showing very different patterns of social behaviours due to these different underlying mechanisms, with a positive behavioural shift in the HOMs, and an impairment in the HETs.

Interestingly, the WT EH subjects showed an increased interest in the non-social stimulus in phase two of the social approach/avoidance task when compared with the WT

controls. This indicates that while the HOM EHs benefited from the environmental treatment in this task, in the WTs it appears to have a detrimental effect. As with the effect in the HOM EHs, this finding was specific to the frequency of visits, rather than duration, and was not seen in the social interaction task, which again highlights the fact that different social paradigms assess different aspects of sociability, and multiple measures are important when attempting to identify the effects of environmental treatments. While it was predicted that early handling would result in improvements in all aspects of behaviour, behavioural impairments have previously been found following the handling procedure (Raineke, Lucion, & Weinberg, 2014; Todeschin et al., 2009). For instance, Todeschin and colleagues found a decrease in socially investigative behaviours, as well as social memory, in their EH subjects, which suggests that the ameliorating effects of this treatment are task dependent and do not extend to all behavioural characteristics. Finally, as only the HOMs appeared to show significant improvements in social behaviours after early handling, with the WT EHs showing some impairments, this provides further support for the plasticity model, as the behavioural improvements are only apparent when the plasticity gene and positive environmental treatment are combined.

Lastly, in the social approach and avoidance task, the only effect that was present in phase three was an increase in frequency of visits to the familiar rat in the EH subjects, which suggests a reduction in social behaviours. This finding could indicate the presence of an impairment to selective attention in these subjects, where they are unable to differentiate between the novel and familiar rats. However, this seems unlikely, as Todeschin and colleagues (2009) found that their EH subjects were able to distinguish between a novel and familiar social stimulus, and EH rats also show an improvement in latent inhibition, as well as no general memory or attentional deficits (Weiner, Schnabel, Lubow, & Feldon, 1985). As previously discussed, both positive and negative behavioural outcomes have been found after

early handling, depending on the behavioural parameters that are being measured (Raine et al., 2014), therefore, it may be that early handling results in a general impairment in phase three of the social approach and avoidance paradigm, regardless of genotype.

The importance of using multiple measures of social behaviours is highlighted in the current research, as while the social interaction and social approach/avoidance tasks both found an increase in social behaviours in the HOMs, and a decrease in the HETs, the findings differed in terms of the EH and MD rats. As previously mentioned, the social interaction task provides a measure of both active interest, and comfort with physical contact, while the approach/avoidance task enables choice within the stimuli, while also providing a measure of interest in social novelty. Furthermore, there are also different social measures within the approach/avoidance task, with phase two measuring general sociability, and phase three measuring the preference for social novelty (Moy et al., 2004). Because of the heterogeneous nature of social behaviours, it is important that a number of different social measures are used in order to fully discern any impairments or improvements that are present.

Limitations of the Social Behavioural Measures

Firstly, while a non-toxic hairspray was used to differentiate the subjects in the social interaction task for automatic recording, the subjects showed some aversion to being sprayed, which may have increased anxiety in the social interaction task. While this colour differentiation was necessary for later analysis, it would be ideal to use an alternative in future, to reduce anxiety in the subjects. This potential increase in anxiety is not likely to have had a significant effect on results, as all rats were subject to this manipulation, although it cannot be completely ruled out the degree to which the rats were affected by the hairspray, as it may in part be determined by genotype, the environmental manipulation, or an interaction between them.

Secondly, it would be useful to use a more detailed behavioural assessment of social behaviours, as previous studies have quantified social behaviours into multiple categories, such as; mounting, social grooming, sniffing, attacking, submission, crawling over/under, chasing and approach (Garau, Martic, Sala, & Balada, 2000; Patin, Lordi, Vincent, & Caston, 2005). If the behavioural measures were to be quantified further than active versus passive, this could enable the categorization of even more refined differences between the experimental groups. It would also be interesting to investigate play behaviour, particularly because the HOM rats have previously been found to show a decrease in this aspect of social behaviours (Homberg et al., 2007b), therefore it would provide a valuable measure of whether EH can overcome these impairments.

Thirdly, there has been some suggestion that the social approach and avoidance task has a lack of ecological validity, as it does not fully reflect real-world interactions, because only the experimental subject is mobile (Wohr & Scattoni, 2013). However, when used in combination with the social interaction task, this paradigm can provide a different measure of social approach, where the subject can choose between multiple stimuli, enabling the rat to show preference for either the social or non-social stimulus. Furthermore, rather than only looking at duration of time spent with the stimuli, the frequency of visits was also measured. Frequency of visits, measured by movement of the nose, provides an indicator of sniffing, which is a measure of curiosity, whereas duration may overestimate the sociability of the subject, particularly if the subject remains stationary in close proximity to the novel rat, but may not be showing active interest (Moy et al., 2004).

However, despite using both these measurements, none of the subjects in the control group showed an increased preference for the social stimulus compared with the non-social stimulus in phase two and three of the social approach/avoidance task. This was unexpected, as the subjects in the environmental treatment groups did show this social preference. The

lack of significance in the control group appears to be a result of the high variability within 77 this data, as visual analysis indicates that the WT and HOM controls did have a preference for the novel rat in phase two over the novel object, but this did not reach significance. The HETs did not show this pattern, although, as discussed, they appear to show a general reduction in social behaviours. While it is expected that this pattern would be seen in the WT and HOM controls if this experiment were to be repeated with a larger number of subjects, it should be noted that Moy and colleagues (2004; 2008) also failed to find preference for the novel mouse in a number of strains, suggesting that the task itself may not be particularly sensitive. This is especially the case in phase three, where the novel and familiar rodents are both present, a finding which is supported by the current results, as none of the groups showed a significant preference in this phase.

Measures of Communication

As the USV data could not be analysed to a satisfactory extent, the findings from the scent marking task will be used to determine the effects of the manipulations on communication. There were no significant findings in the MD group and, as scent marking behaviour has not been previously investigated in MD rats, the lack of significant difference between the MD and control rats provides preliminary evidence that scent marking behaviour may not be impaired in MD subjects. While the MDs showed no effect, the HETs and HOMs in the EH group showed a significant increase in social, non-social and total scent markings when compared with the controls, indicating an overall increase in communication in these subjects. Furthermore, when percentage of social scent marking was analysed, the EH subjects showed a significant increase when compared with the controls. Therefore, it appears that the early handling treatment increases scent marking in general, with the greatest increase in overall scent marking occurring in the HET and HOM subjects. As the number of scent markings around a particular odour provides an indication of both communication and

social motivation (Silverman et al., 2010), these results provide evidence for an improvement in communicative and social behaviours in the HET and HOM EH subjects. To the author's knowledge, no previous research has investigated the impact of early handling on communicative behaviours in adult rats, consequently, these results provide preliminary support for an increase in social communication after EH treatment. In fact, relatively few studies have investigated scent marking in rats (compared to scent marking in mice) and the results of the present study suggest that more research is needed in this respect.

As with the findings in the social measures, these results also partially support predictions made under the plasticity model, as the combination of early handling treatment and the SERT genotype has resulted in overall improvement in communication, in comparison with either of these factors in isolation. However, the plasticity model is not fully supported, as there were no significant findings in the MD group. The lack of findings in the MD group also means that the vulnerability model is not supported by this data.

Limitations in the Measures of Communication

Ideally, when attempting to record USVs, a soundproof environment should be used to reduce the likelihood of irrelevant noise affecting the data. However, as the social interaction experiment occurred at the same time as the USV analysis, and required the use of EthoVision tracking, the room that was used for this was not particularly sound proof. This led to reduced sensitivity in the USV recording, as there was a large amount of extraneous background noise-such as the wind, construction, and other people in the lab. If this task were to be repeated, it would be ideal to use a sound-proofed apparatus that filtered out unnecessary external noise, to increase the sensitivity of USV recording. As it was, the USV recordings contained a large number of artefacts, where background noise was often detected as calls, while a large number of calls went undetected, thus this data could not be properly analysed and was subsequently discarded.

rats is important, as it enables the gathering of urine during the oestrus phase. However, it was often very difficult to find subjects in the oestrus phase of the cycle, and urine from females in the transitional phase was sometimes required, due to time constraints. While it is unlikely that this would have severely impacted the interest of the males in this odour, as social males should always show more interest in female urine than lemon essence, the non-social odour, it would be useful to compare the scent markings of the urine of females in and out of the oestrus phase, in order to fully determine whether significant differences in scent marking occur. Furthermore, to combat the issue of having to use urine from a female in the oestrus phase of the reproductive cycle, future research could look at the use of the urine of an unfamiliar male conspecific, as rodents use scent markings for territorial marking as well as reproductive displays (Arakawa et al., 2008).

Finally, the scent markings were manually counted, which increases the likelihood of experimenter error, particularly due to the similarities in appearance of scent markings, footprints and sneeze markings. If the experiment were to be repeated, it would be useful to use an automated measure of counting, for example the use of a bacterial colony counting program, such as OpenCFU (Geissmann, 2013).

Measures of Stereotypy

Hole Dipping

The HOMs and WTs in the EH group showed an increase in novel hole-dipping, and the EH subjects showed a general increase in repeat hole-dipping overall. In comparison, the MD subjects showed a general increase in novel hole-dips, and no change in repeat dipping.

While the increase in repeat hole-dips in the EH group was unexpected, it does not seem likely that this is a result of increased stereotypy in this group. Instead, as they showed an increase in both novel and repeat dipping, and there was no increase in percentage of

repeat hole dips (*results not shown*), it is likely that these subjects have an overall increase 80 in hole-dipping behaviour, as a result of increased exploratory behaviour. As there were a total of 16 holes in the task, once the rats had dipped into most, if not all, of the holes present, they then began to repeat dip into the holes they had already investigated, which resulted in the increase in repetitive dips. Furthermore, as EH rats show a decrease in anxiety (Levine et al., 1967; Liu et al., 1997; Plotsky & Meaney, 1993), they spend more time in the center of the hole-board apparatus than more anxious rats (Levine et al., 1967), and thus would have more access to the center holes, resulting in an increase in overall dips.

Against the predictions of both the plasticity and vulnerability theories, the MD rats showed an increase in novel hole dips when compared with controls, and no increase in repeat hole dips. Previous studies, however, have found an increase in exploration and locomotor activity in a novel environment in rats exposed to maternal deprivation during the early post-natal period (Marco et al., 2007; Rentesi et al., 2013; Wang et al., 2015), which may account for the increase in novel hole-dips in this group. Additionally, Narita and colleagues (2010) also found an increase in hyperactivity in their subjects treated with thalidomide, which suggests that an increase in hyperactivity might result from adverse environmental treatment. Finally, as the MD rats showed an increase in novel, but not repeat, hole-dips, this also provides support for the increased response to novelty that has previously been found in these subjects (Rentesi et al., 2013).

In conclusion, the results from the hole-dipping task do not support the plasticity theory, as both the MD and EH groups showed an increase in novel hole-dips, and the EH group also showed an increase in repeat hole-dips, while the MD group did not. Nor do these results provide support for the vulnerability model, as the MD rats did not show the greatest degree of repeat behaviours.

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Interestingly, while the HOM EH and WT EH rats showed an increase in novel hole-dips, most likely due to an increase in exploratory behaviours, this increase did not occur in the HET EH subjects. As with the findings from the social-related tasks, the HETs appear to be showing a different pattern to both the WT and HOM subjects in terms of hole-dipping behaviours. As previously discussed, it may be that the complete lack of SERT in the HOMs leads to vastly different social and hole-dipping behaviours compared with the HETs, who only have a 50% reduction in SERT activity. Importantly, as the HETs showed similar scent marking behaviours to the HOMs, these differences appear to affect some, but not all behaviours.

Limitations. The findings from the hole-dipping task could either indicate that maternal deprivation did not result in an increase in repetitive behaviours, or that the hole-dipping task is not a completely reliable measure of these behaviours. The hole-dipping task was selected for use in the current experiment as it has previously been used to measure repetitive behaviours (Makanjuola et al., 1977; Moy et al., 2008), however, both changes in levels of anxiety and a general increase in exploratory drive may confound the measurement of repetitive behaviours. As the rats are in a confined environment, with no other stimulation except for the holes in the hole-board floor, there were a large number of hole-dips overall. While there does not currently seem to be a highly reliable measure of repetitive behaviours available, in the future, it might be more valuable to use a different measure of repetitive behaviours, in a more stimulating environment, where the subjects are able to participate in behaviours other than hole-dipping alone.

Reversal Learning Paradigm

Given the time required for this task, only four groups could be tested in this paradigm: WT controls, WT EH and WT MD as well as HOM controls. The results show that the HOM controls had an overall higher level of anxiety (measured as body elongation,

thigmotaxis and immobility), with little change in learning and stereotypical behaviours, 82 compared to WT controls. Within the WT animals, the WT EH subjects showed a significant improvement in learning across all parameters measured compared with WT controls, and some stereotypical behaviour. Finally, the WT MD rats had an overall increase in stereotypical behaviours, with some improvement in learning, and varied anxiety-related behaviours.

The EH subjects showed the greatest improvement in learning, with a decrease in distance travelled, memory errors and a significant increase in time spent in the target quadrant during the probe trial. The number of nose dips into the ramekins was used as the measure of working memory errors, as nose-dips into the incorrect ramekin indicates reference memory errors, while repeat nose-dips into the same ramekin is an indicator of a working memory error. Therefore, these results indicate that early handling results in an improvement in spatial learning and working memory. The improvement in spatial learning is supported by previous research, as EH has been found to reduce the deficits in spatial learning that occur with increased age (Meaney, Aitken, Van Berkel, Bhatnagar & Sapolsky, 1988), and to increase reference, working and spatial memory, as well as behavioural flexibility (Plescia et al., 2014).

Along with general improvements in spatial learning, the EH subjects also showed a slight increase in some repetitive behaviours, particularly an increase in head weaving during the reversal trials. This indicates that these subjects may have some increase in stereotypy, although not to the extent of the MD subjects. This experiment should be repeated with HET and HOM EH subjects, in order to determine whether the combination of the SERT genotype and the positive environmental treatment would result in a decrease or even elimination of these repetitive behaviours.

The MD subjects, on the other hand, showed an increase in head weaving in both 83
the learning and reversal phase, as well as an increase in rotation during the reversal phase. The presence of stereotypical behaviours in animals subjected to adverse environmental conditions has been replicated in numerous studies (Dantzer, 1991; Mason & Rushen, 2008; Powell et al., 1999), showing that environmental treatment alone, without the presence of underlying risk/plasticity genes, can result in significant behavioural impairments in terms of stereotypy.

While the MD subjects showed an increase in repetitive behaviours, they also showed a decrease in distance travelled in the learning phase, indicating that they have an improvement in spatial recognition (Wang et al., 2015). Although this result was unexpected, as previous studies have found an impairment in reversal learning in MD rats (Enthoven et al., 2008; Han, Wang, Xue, Shao, & Li, 2011; Wang et al., 2015), MD subjects usually show no change in spatial learning compared with controls (Enthoven et al., 2008; Levy et al., 2003). Additionally, some improvement in spatial learning in MD subjects has also previously been found (Lehmann et al., 1999; Pryce, Bettschen, Nanz-Bahr, & Feldon, 2003). Importantly, the MD subjects that showed an improvement in previous research, also experienced the 24 hour separation event on PND9, and, as with the current experiment, the mother was taken away from the home cage for this time (Lehmann et al., 1999). This suggests that this particular protocol of MD may result in an improvement in spatial learning abilities, along with the impairments social and repetitive behaviours that are present.

Lastly, the MD subjects showed an increase in body elongation during the reversal trials, indicating an increase in anxiety, but a decrease in thigmotaxis during the learning trials, which supports a reduction in anxiety. However, as the MD subjects also show a decrease in distance travelled during the learning phase, the reduction in thigmotaxis is most

likely a result of taking a more direct route to the target location, and is an indicator of an 84 improvement in spatial learning.

Finally, the HOM controls showed an impairment across all measures of anxiety, with an increase in body elongation, indicating that they were investigating stimuli by stretching, rather than moving towards the object (Grewal, Shepherd, Bill, Fletcher, & Dourish, 1997), and immobility, an indicator of freezing behaviour (Brandao et al., 2008). The HOMs also showed a reduction in time spent in the center of the maze in both the learning and reversal phases, which may be an indicator of thigmotaxis, where the subjects are spending more time in the peripheral zone of maze, due to heightened fear (Barnett, 1975). As thigmotaxis is usually measured in an open field, rather than a large maze, and the subjects needed to cross through the center section in order to reach the target in the reversal learning task, the rats would have had an increased motivation to enter the center, thus there were competing factors between anxiety and motivation to reach the target. Finally, three HOM subjects had to be excluded from the reversal paradigm, for failing to explore the maze and locate the target after consecutive trials, due to high anxiety. Therefore, although the HOMs appear to have general improvements in social behaviours, they show a heightened stress response, a finding which is supported by previous literature (Kalueff et al., 2007; Olivier et al., 2008).

In summary, as only the WT's in the environmental treatment groups could be run on the reversal paradigm, due to time constraints, this makes it impossible to determine whether the results support either the plasticity or the vulnerability model. Nevertheless, even with the WT's and the HOM controls, significant environmental and genetic effects were apparent. Overall, the HOM subjects showed an increase in anxiety during these tasks, while the MD group showed a slight increase in anxiety, and a substantial increase in repetitive behaviours. Lastly, in support for the ameliorating effects of positive environmental treatment, the EH subjects showed a significant improvement in spatial learning, as well as a slight decrease in

anxiety. In order to fully determine whether the plasticity or vulnerability models were supported, this experiment needs to be replicated using all nine experimental groups. 85

Limitations

While the use of automatic scoring is very useful in terms of data collection and reducing experimenter error, the use of body contraction as a measure of anxiety was somewhat ambiguous. Both the EH and HOM subjects showed an increase in body contraction, however it is difficult to determine whether this is an accurate measure of increased anxiety, as for instance grooming behaviour and rearing would also result in a reduction in body size. Therefore, if this experiment were to be repeated, it would be useful to combine the contraction measure with manual scoring of grooming and rearing behaviours, in order to better determine what behaviours this measure reflects.

Secondly, as time constraints prevented the use of SERT KO MD subjects in this task, it can only be tentatively speculated that the stereotypical behaviours that were present in the WT MDs would be enhanced in subjects with the knockout genotype, and this experiment needs to be replicated with these subjects in order to confirm these speculations. Further research into the development of stereotypical behaviours is highly important, to provide further insight into the underlying mechanisms involved, as it appears that environmental factors have a large influence on the development of repetitive behaviours. It is also important to repeat this experiment with the SERT KO EH subjects, in order to determine whether the head weaving behaviour that was observed would be reduced when the environmental treatment and knockout genotype were combined.

Finally, it is likely that the reversal learning task was too simple for the rodents, and did not provide a true measure of cognitive flexibility, as the rats only had to move a short distance between the initial target and the reversed location. In line with this, animals showed only a short increase in distance travelled when the location was reversed, and usually after

the second trial the performance was comparable to before the switch. It is therefore 86

important to increase the complexity of the reversal learning phase in this task, by using a novel configuration of ramekins (such as a circle) during the reversal learning trials, which would force the subjects to learn a completely new target location, rather than merely switching between left and right. In this way, the second phase of the test becomes more of a switching than a reversal task, hopefully leading to a significant drop in performance and a longer re-learning phase. By increasing the complexity, it is likely that larger between-group differences could be found. Alternatively, it would be interesting to look at set-shifting, rather than reversal learning, as the measure of cognitive flexibility in future research. While reversal learning involves shifting responses within the same stimulus dimension, set-shifting requires the rat to alternate between stimuli with completely different characteristics (Butts, Floresco, & Phillips, 2013). For instance, Butts and colleagues used lights as the target stimuli in the initial phase of their lever pressing experiment, and then switched the target stimulus to a lever on a particular side of the maze in the next phase, where the rats would need to ignore the lights presented in order to receive the reward. Their rats, which were exposed to acute stress, showed an impairment on the set-shifting, but not the reversal learning task, indicating that the potential impairment in cognitive flexibility in the MD rats, and improvements in the EH rats, might be more apparent in a more complex task such as this.

Summary of Findings

In summary, findings from the social and communicative tasks provide some preliminary support for the plasticity theory, as an improvement in social and communicative behaviours was seen in the HET/HOM subjects that experienced early handling treatment, while an impairment in social behaviour was seen in the HOMs that experienced maternal deprivation. As there is evidence for both an increase in autistic-like behavioural impairments

in the HOM MD group, as well as an improvement in autistic-like behaviours in the HOM 87 EH group, the vulnerability model is not supported by these findings overall. These results support the findings by van IJzendoorn and colleagues (2012) that both positive and negative environmental factors result in permanent behavioural changes in individuals with the s/s and s/l 5HTTLPR genotypes.

General Discussion

As there was support for the plasticity model in some, but not all, of the behavioural measures used, this indicates that the effects of the combination of plasticity genes and environmental treatments, is dependent on the parameters that are being measured. Thus, although the data are supportive of the “*plasticity*” model put forward by Belsky (2009), it suggests that the reality is in fact more complex. Not only are the effects of positive and negative environments moderated by the “*plasticity*” genes, the moderation also depends on the actual tests (or parameters) measured. For instance, the HOM EH subjects showed a significant improvement in social behaviour in phase two of the approach/avoidance task, however no effects were seen in this group in the social interaction task.

EH resulted in improvements in a number of behavioural characteristics in the current research, such as social, communicative and learning behaviours, while impairments were also found, as the EH group showed an increased interest for the non-social stimulus in the social approach avoidance task. Similarly, while treatment with maternal deprivation resulted in a number of behavioural impairments, in social and stereotypical behaviours, some improvements were also seen, in spatial learning in the reversal learning paradigm. In their review of the early handling procedure, Rainecki and colleagues (2014) caution researchers against viewing early-life environmental treatments as inherently adverse or beneficial, as the outcome of these treatments is dependent on a number of factors, including the type of behaviour that is being investigated. Indeed, this all-or-nothing view of these environmental

treatments reflects the bias that led to the development of the vulnerability model, where 88 only the effects of negative environmental factors were assessed. In sum, while the results from this research generally support the role of early handling as a positive influence, and maternal deprivation as a negative event, these effects are behaviourally dependent and should not be viewed as all-encompassing. The same can be said for genotypic effects, as the SERT KO HOMs showed a general improvement in social behaviours, but an increase in anxiety, while the HETs showed a decrease in social behaviours, but an improvement in communication. As 30% of Caucasians have the s allele polymorphism in the *SLC6A4* serotonin transporter gene, the equivalent serotonin transporter deficiency of the SERT KO HETs, it is highly unlikely that this mutation would continue to be present in the population if its effects were entirely negative, however, the majority of research continues to focus on the detrimental effects of this mutation in both humans and rats (Homberg & Lesch, 2011). In fact, Homberg and Lesch propose that the anxiety-related impairments present in SERT KO rats, and humans with the s allele, may be completely counterbalanced by the improvements in cognitive function that are also found in these individuals. Therefore, when investigating the effect of genetic and environmental factors on behaviour, it is important to look at both the negative and positive effects of these factors, for a more comprehensive view of the effects, and to reduce the inherent negative bias that is present in the literature.

Limitations

There were a few factors in the current research that may have affected the findings. Firstly, the early handling protocol involved handling the dams for 15 minutes per day from PND2-14, while the pups remained in the home cage. This technique differs from previous methods, as most researchers handled the pups individually while the mother was removed from the home cage (Levine et al., 1956; Pryce et al., 2003; Todeschin et al., 2009).

However, while the early handling procedure differs substantially between studies, in terms

of timing, duration and frequency of separation events, it appears that the behavioural effects of this technique are relatively consistent across the literature, provided that the EH protocol is brief and repeated over consecutive days (Todeschin et al., 2009). Thus, while the procedure used in the current research might differ slightly from previous studies, it is unlikely that the results would differ dramatically, had a slightly different handling technique been used. 89

Conversely, the effects of maternal deprivation appear to be entirely dependent on the protocol that is used, with dramatic differences in behaviours being observed depending on the timing, duration and frequency of the separations, as well as the gender of the pups, and whether the pups are separated from their littermates, or just the dam (Levy et al., 2003; Rentesi et al., 2013). For instance, when 24 hours of MD was performed on PND14, an increase in the stress response was seen in the open field task (Penke et al., 2001), whereas a decrease has also been found when MD was performed on PND11 (Suchecki, Palma, & Tufik, 2000). No change in spatial learning was found in pups that were artificially reared (Levy et al., 2003), while an improvement in spatial learning was found in rats exposed to MD on PND9 (Lehmann et al., 1999). Additionally, an increase in locomotor activity occurred in subjects that experienced 24 hours of MD on PND9 (Rentesi et al., 2013), while a decrease was found in subjects exposed to two separate 12 hour separations on PND9 and PND11 (Takase, Yamamoto, & Yagami, 2012), although it has been shown that multiple shorter separations are not necessarily equal to a single 24 hour separation (Lehmann et al., 1999). The effects of MD also seem to be dependent on whether the offspring can see and hear the dams whilst separated (Haigh, Stookey, Bowman, & Waltz, 1997). Finally, an increase in maternal care has been found in pups that are placed in a novel environment during the separation event, resulting in an overall decrease in stress reactivity compared with those that remain in the home cage (Daskalakis et al., 2011). The current study exposed the

pups to 24 hours of maternal deprivation on PND9, as there is some consensus that this 90 protocol results in the most drastic behavioural effects (Ellenbroek & Riva, 2003; Marco et al., 2015; Rentesi et al., 2013). Nevertheless it is important to note that the findings from this research will most likely differ from the findings of other researchers, if they were to use a different MD protocol. Importantly, however, while the behavioural effects differ between studies, it does appear that the MD technique generally results in permanent behavioural changes, despite these differences, highlighting the significant influence of environmental factors on the developmental process (Fenoglio, 2006; Levy et al., 2003; Plescia et al., 2014).

Secondly, the rats were not kept on a reverse light-dark cycle, meaning that behavioural testing occurred in the inactive phase of their nocturnal cycle. In mice, behavioural differences have been found when experimentation was carried out in the light versus the dark phase, with an improvement in habituation and memory, as well as a decrease in inhibition being observed when testing occurred during the dark phase (Roedel, Storch, Holsboer, & Ohl, 2006). These findings indicate that behavioural differences may also be present in rats when testing occurs in the dark compared with the light cycle, thus, future research should look at comparing these conditions.

Finally, while only male rats were used for this study, due to the increased risk of ASD development in males (Dawson et al., 2002), previous studies have found different effects of EH and MD on male and female subjects (Lehmann et al., 1999; Todeschin et al., 2009). For instance, when MD occurs on PND9 in rats, an improvement in spatial learning has been found, with a greater effect in MD males than females (Lehmann et al., 1999), and an improvement in latent inhibition was found in EH males, but not females (Weiner et al., 1985). These findings indicate that male rats may have an enhanced susceptibility to the effects of postnatal environmental manipulations, most likely due to the differences in maternal care that they receive (Lehmann et al., 1999; Moore & Morelli, 1979). Dams are

able to differentiate between male and female pups due to the production of androgen in 91 males, and they show increased licking of the anogenital region in males compared with female pups, which is an indication of an increase in maternal care (Moore & Morrelli, 1979). As these environmental manipulations are largely dependent of the level of maternal care that the pups receive, it appears that there may be a stronger effect in males (Moore & Morelli, 1979l; Plescia et al., 2014). Following from this, it is possible that the ameliorating effects of early handling, and the detrimental effects of maternal deprivation that were seen in the current research, would not be as pronounced in female rats, although the experiment would need to be replicated with female subjects in order to confirm this.

Applications for ASD

It is often difficult to translate findings from an animal study into relevant human applications, despite the usefulness of animal researching in determining the cause-and-effect relationship of environmental treatment. As most rodent research into ASD looks specifically social behaviours, rather than the triad of impairments in this disorder (Belzung et al., 2005), the battery of tests used in the current research were specifically selected to measure a range of autistic-like behaviours in the rat model, which could then be translated for clinical use.

The current research is one of the first studies that has thoroughly investigated the role of a combination of genetic and environmental factors in ASD. Consequently, these findings provide some valuable insight into the important role of early environmental factors in combination with the SERT KO genotype on the development of autistic-like behaviours in rats. As previously stated, the genetic and environmental factors underlying ASD are heterogeneous and complex, and the findings from the current study should not be taken as a complete and definitive model of the development of ASD in a rodent. However, the current research has shown that the combination of MD and SERT KO genes does provide a reliable model of autistic-like impairments, as deficits in social behaviours, as well as increased

stereotypy, are seen in these subjects. The finding that MD increases stereotypical behaviours is important, as there are very few animal models that reflect the repetitive behaviours in ASD. Whether the combination of the SERT KO genotype and MD treatment increases the severity of these repetitive behaviours still remains to be seen, although it is an excellent target for future research. Additionally, whether impairments in communication are also present is uncertain at this point, as there were no significant findings for the MD group in the scent marking task, and the USV data could not be used due to extraneous background noise. While it seems likely that deficits in communication are also present in this model, the USVs of SERT KO MD subjects should be properly recorded and analysed in a sound-proof environment in order to confirm this.

The current findings support those from previous literature, in regards to the important role of both serotonin and early postnatal factors in the development of the ASD phenotype. As permanent behavioural changes were seen in those subjects that experienced MD and EH in the early postnatal period, this indicates that this is a critical developmental period in rodents. The increased sensitivity to the influence of environmental factors in the early postnatal period in rats appears to mimic the sensitivity in humans during the late gestational period (Marco et al., 2015). While there are differences in the developmental timelines of humans and rodents, these results show that early-life stressful events, in combination with serotonin dysregulation, play a role in the development of autism. Along with the influence of environmental factors, the current research provides evidence for the important role of serotonin in a number of processes during important periods of development, as the SERT KO rats showed permanent behavioural changes, both positive and negative, depending on environmental influences. As an imbalance in serotonin looks to be a significant contributing factor to autistic-like behaviours, this supports the use of

serotonergic drugs in the treatment of ASD, a treatment which has had some success in the 93 past (McPheeters et al., 2011).

To conclude, by thoroughly investigating the role of genetic and environmental factors, both positive and negative, in the development of autism, it is hoped that novel and improved therapeutic techniques and interventions can thus be developed, as well as targeted screening methods. This development is very important, particularly because it is currently extremely difficult to diagnose ASD in the first two years, despite the need for early intervention (DiCicco-Bloom et al., 2006). The finding that early handling can overcome some of the negative behavioural characteristics associated with the SERT KO genotype, indicates that although individuals may have a gene that is associated with ASD, such as the s allele, this will not necessarily result in the development of autism. In fact, if the gene in question is a plasticity gene, then it is possible that the presence of this gene may be advantageous to the individual, depending on what environmental factors they are exposed to during their early development. It is important that positive environmental factors are studied to a greater extent than is occurring in current research, with an extreme bias towards the negative, because it appears as though some beneficial factors might outweigh the negative effects of other genetic and/or environmental influences. Due to the importance of early environmental influences on the development of ASD, it may be that if targeted treatment is used very early in individuals with at risk genotypes, the behavioural impairments that are seen in this disorder could be reduced. The ultimate goal of ASD research is to discover treatment methods that are able to reduce the severity of this disorder, and enable autistic individuals to lead a more independent lifestyle, and it seems likely that further research into the beneficial effects of positive environmental factors is an important step towards achieving this goal.

Overall, there was evidence for the effect of genotype and environmental treatment in both the EH and MD groups, which would indicate that, unlike the predictions of the diathesis-stress theory, both negative and positive environmental treatment can have a permanent effect on the behavioural outcome of an individual, supporting the plasticity model. This is a highly important finding, as the majority of gene-environment research is focused on purely the negative effects of environmental factors, despite the mounting evidence that positive environmental factors also play an important role in development. By focusing on both the negative and the ameliorating effects of environmental factors, it is hoped that targeted early intervention and treatment techniques can better be developed for ASD.

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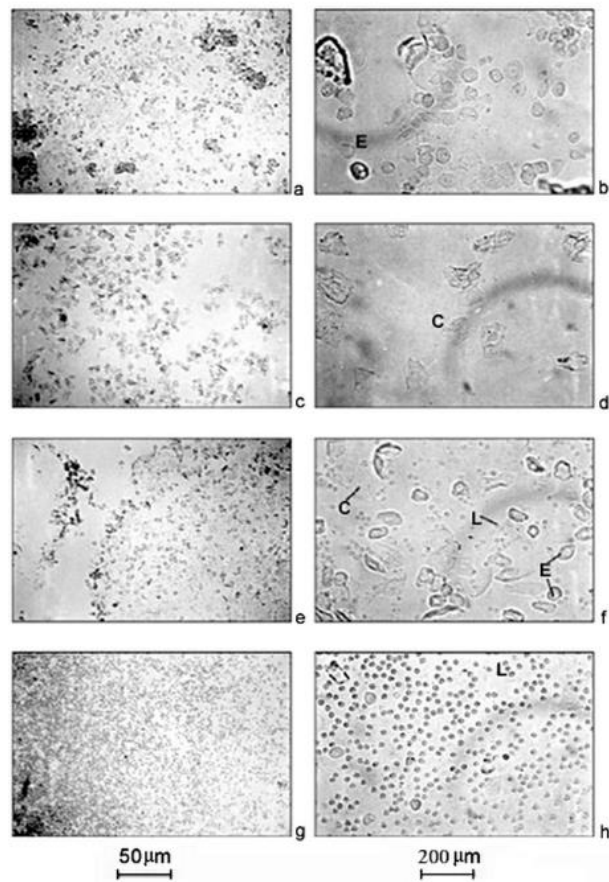


Figure A1. Image taken from Marcondes et al. (p612, 2002) indicating the cells present at each stage of the reproductive cycle in rats: proestrus (a & b), oestrus (c & d), metoestrus (e & f) and dioestrus (g & h). The presence of cornified cells during oestrus can be seen in slide d, denoted by the “C”

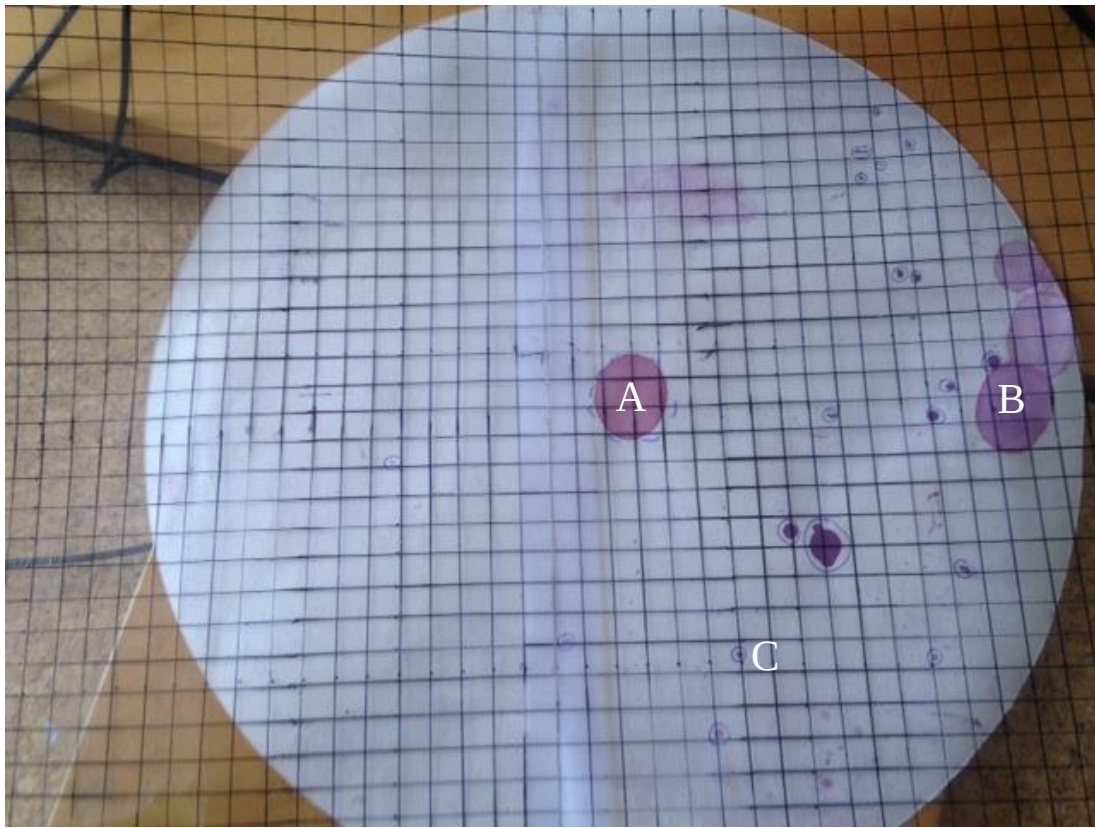


Figure B1. Example of a filter paper from the scent marking experiment-the purple colouration occurs after treatment with ninhydrin spray. The 1cm grid was used to count the number of scent markings. The “A” denotes the social odour (female urine), the “B” is an example of a urine pool, and the “C” is a scent marking.

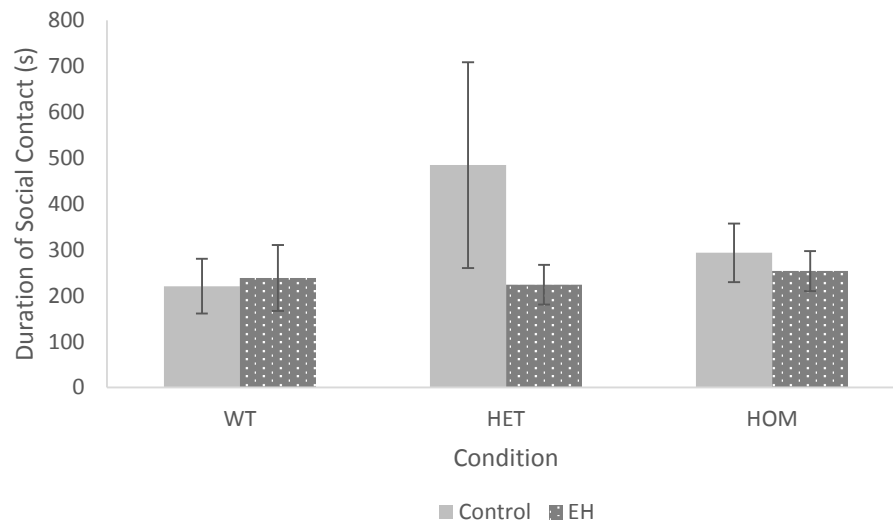


Figure C1. Bar graph depicting the time spent (in seconds) in *active* social contact in the EH and control groups during the social interaction task. Error bars represent ± 1 standard error of the mean.

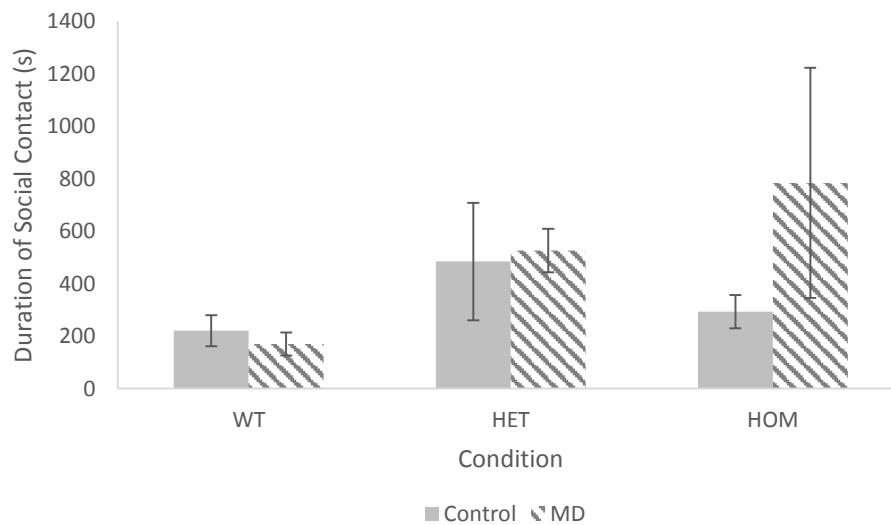


Figure C2. Bar graph depicting the time spent (in seconds) in *active* social contact in the MD and control groups during the social interaction task. Error bars represent ± 1 standard error of the mean.

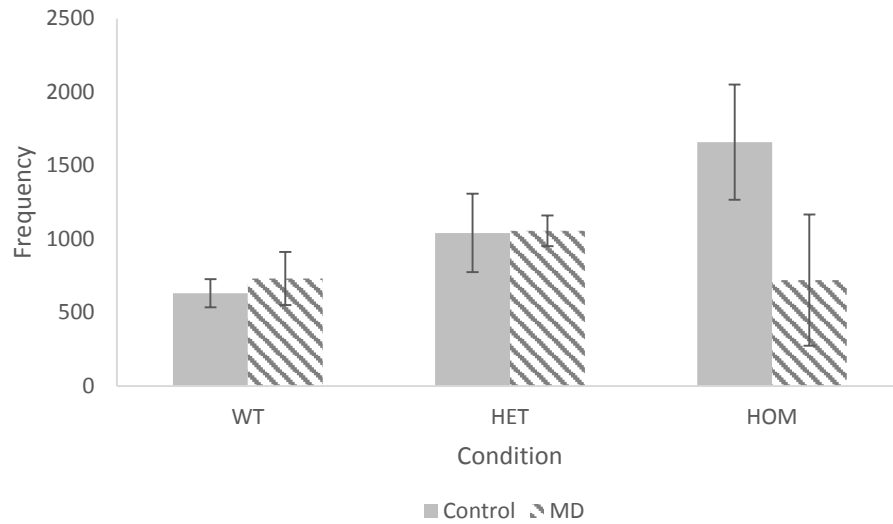


Figure C3. Bar graph depicting the frequency of *active* social contact in the MD and control groups during the social interaction task. Error bars represent ± 1 standard error of the mean.

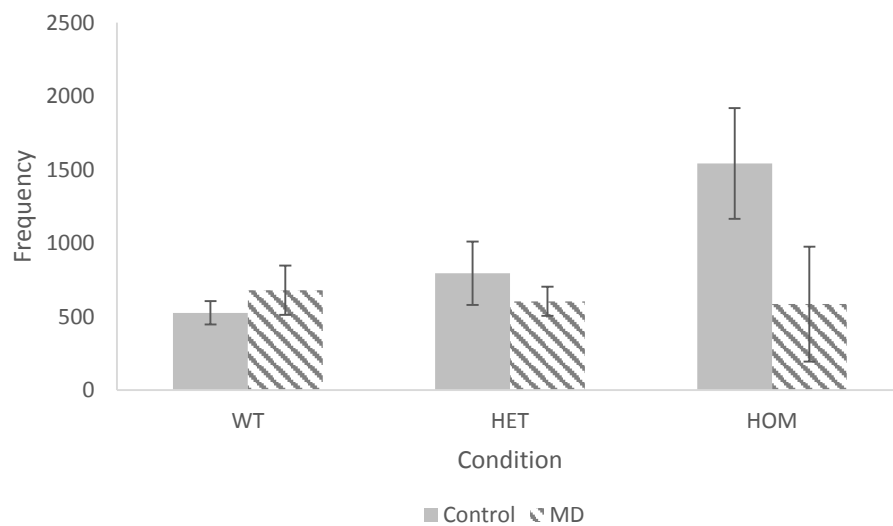


Figure C4. Bar graph depicting the frequency of *passive* social contact in the MD and control groups during the social interaction task. Error bars represent ± 1 standard error of the mean.

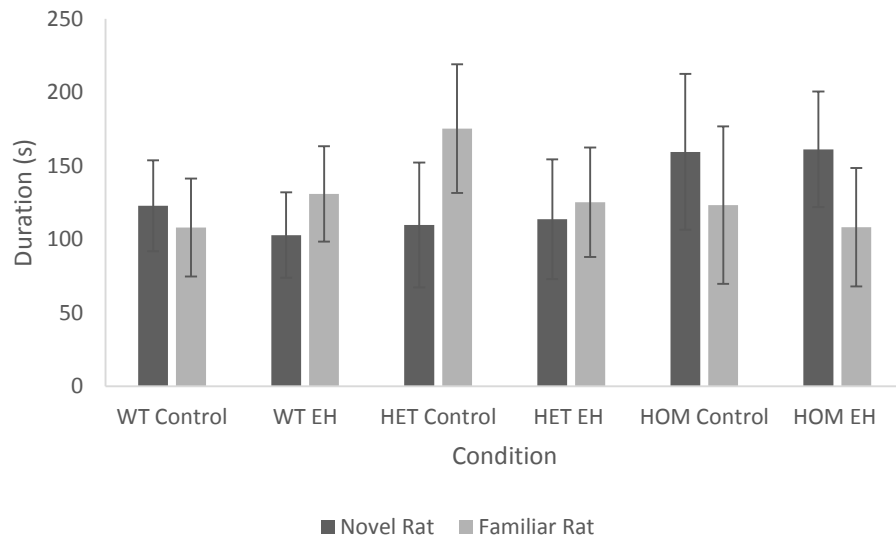


Figure D1. Bar graph depicting the duration of time (in seconds) spent in close proximity to the novel and familiar rat in the EH and control groups in phase three of the social approach and avoidance task. Error bars represent ± 1 standard error of the mean.

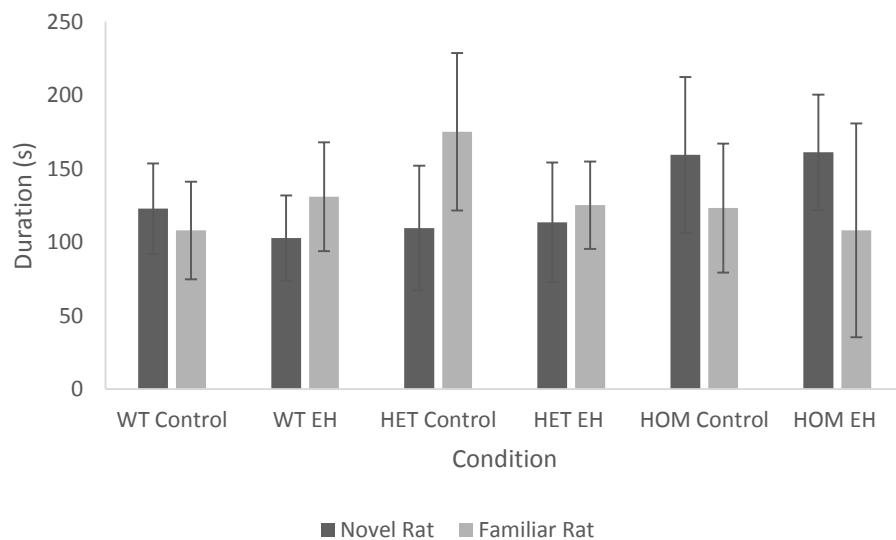


Figure D2. Bar graph depicting the duration of time (in seconds) spent in close proximity to the novel and familiar rat in the MD and control groups in phase three of the social approach and avoidance task. Error bars represent ± 1 standard error of the mean.

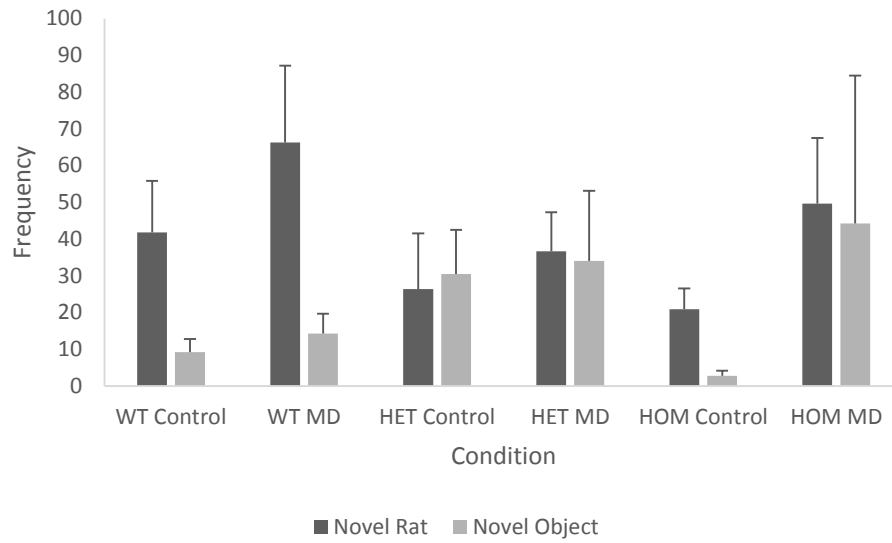


Figure D3. Bar graph depicting the frequency of visits to the novel rat and novel object in the MD and control groups in phase two of the social approach and avoidance task. Error bars represent +1 standard error of the mean.

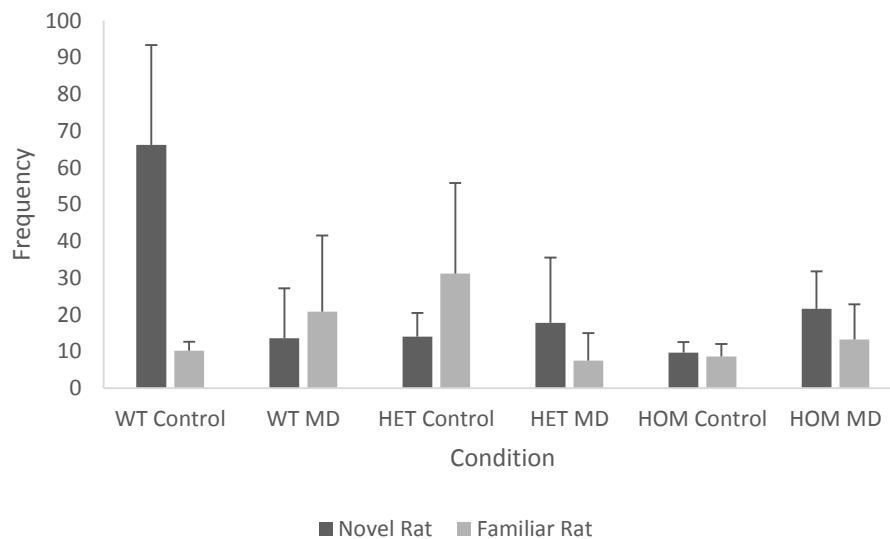


Figure D4. Bar graph depicting the frequency of visits to the novel and familiar rat in the MD and control groups in phase three of the social approach and avoidance task. Error bars represent +1 standard error of the mean.

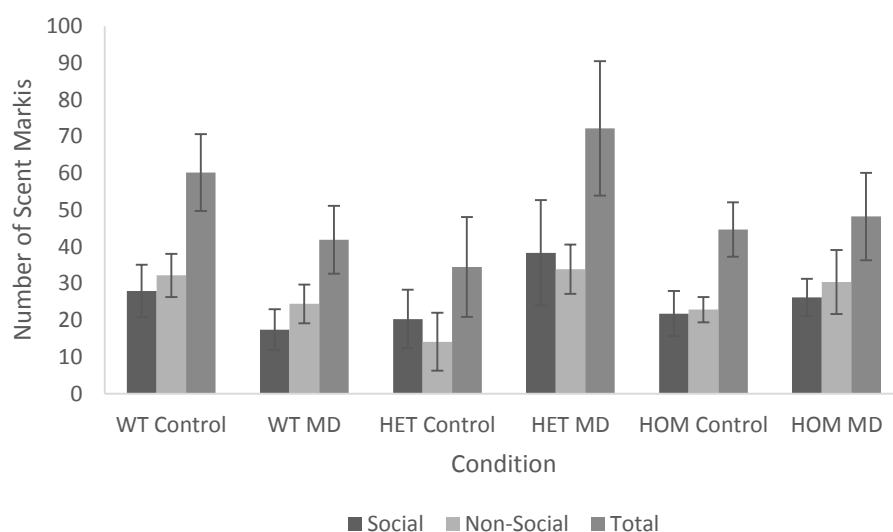


Figure E1. Bar graph depicting the number of scent markings in the MD and control groups during the scent marking task. Error bars represent ± 1 standard error of the mean.

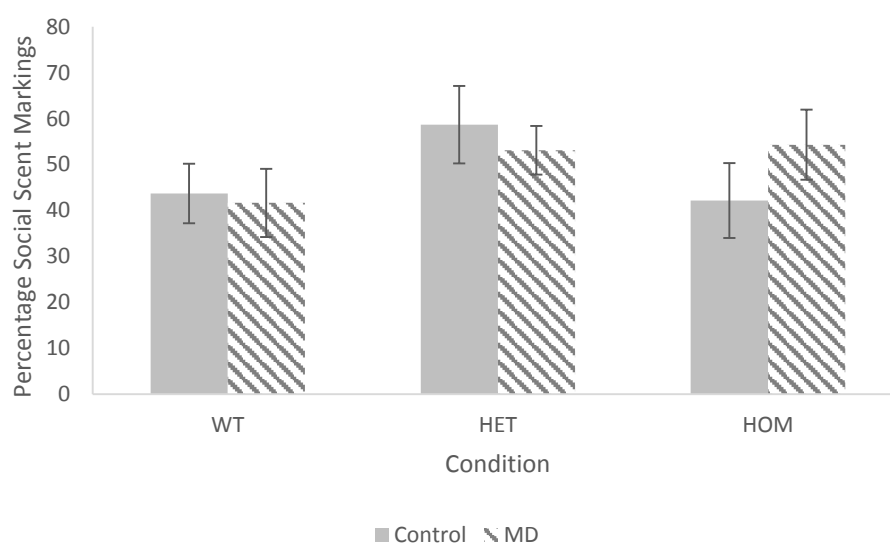


Figure E2. Bar graph depicting the percentage of social scent marking in the MD and control groups during the scent marking task. Error bars represent ± 1 standard error of the mean.

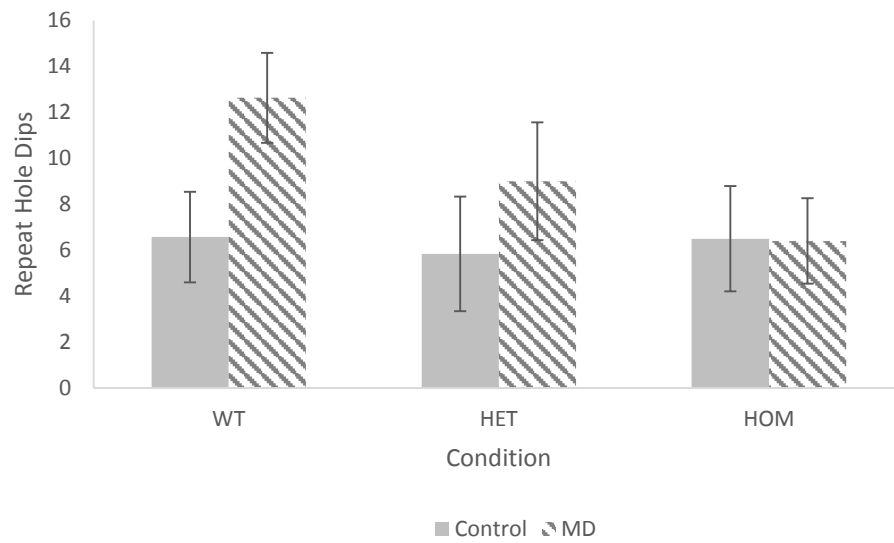


Figure F1. Bar graph depicting the number of *repeat* nose pokes/ hole dips in the MD and control groups in the hole dipping task. Error bars represent ± 1 standard error.

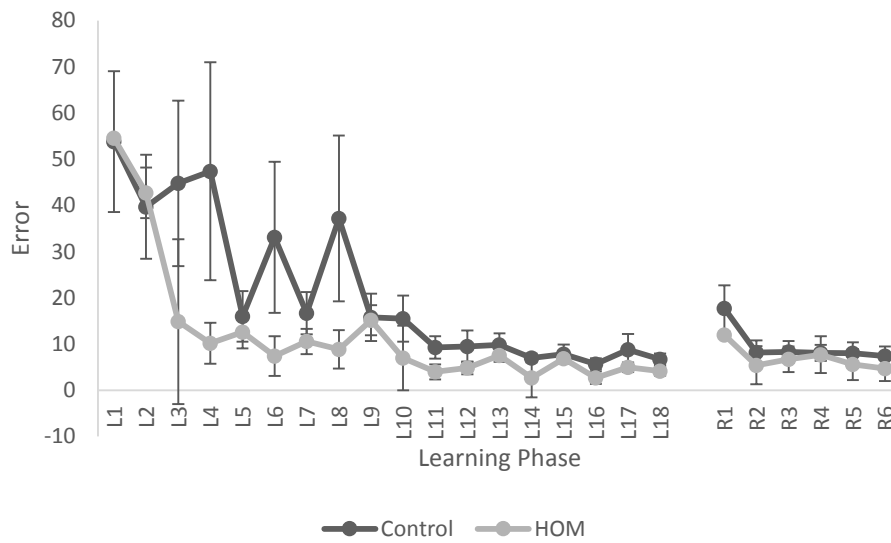


Figure G1. Line graph depicting the ramekin visits (error) of the HOM and control groups in the reversal learning paradigm. Error bars represent ± 1 standard error of the mean.

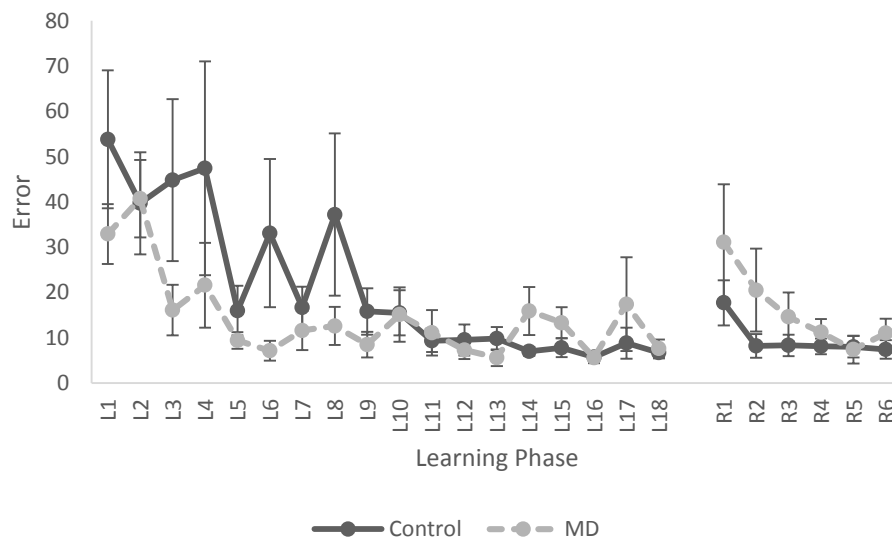


Figure G2. Line graph depicting the ramekin visits (error) of the MD and control groups in the reversal learning paradigm. Error bars represent ± 1 standard error of the mean.

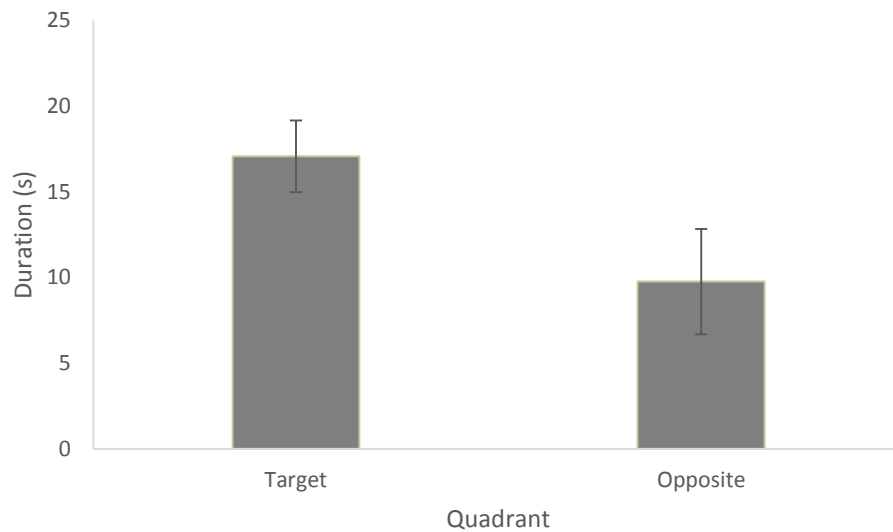


Figure G3. Bar graph depicting the duration of time (in seconds) spent in the target compared with the opposite quadrant during the post-learning phase probe trial in the HOM controls. Error bars represent ± 1 standard error of the mean.

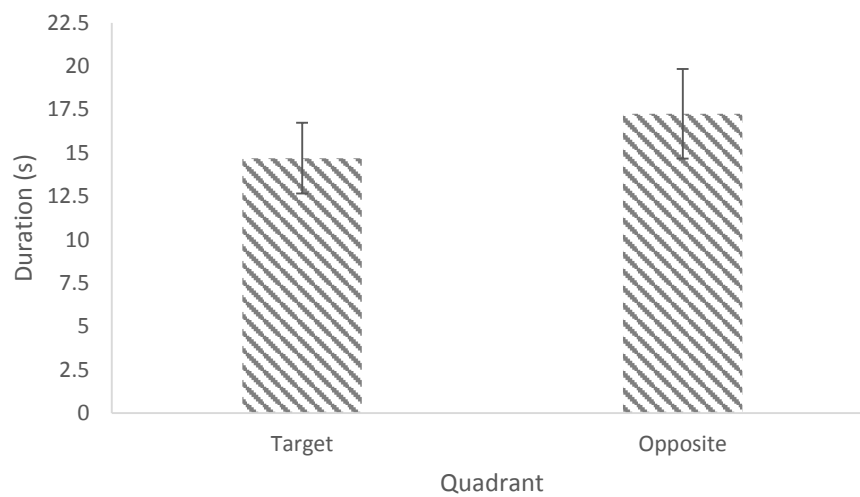


Figure G4. Bar graph depicting the duration of time (in seconds) spent in the target compared with the opposite quadrant during the post-learning phase probe trial in the WT MDs. Error bars represent ± 1 standard error of the mean.

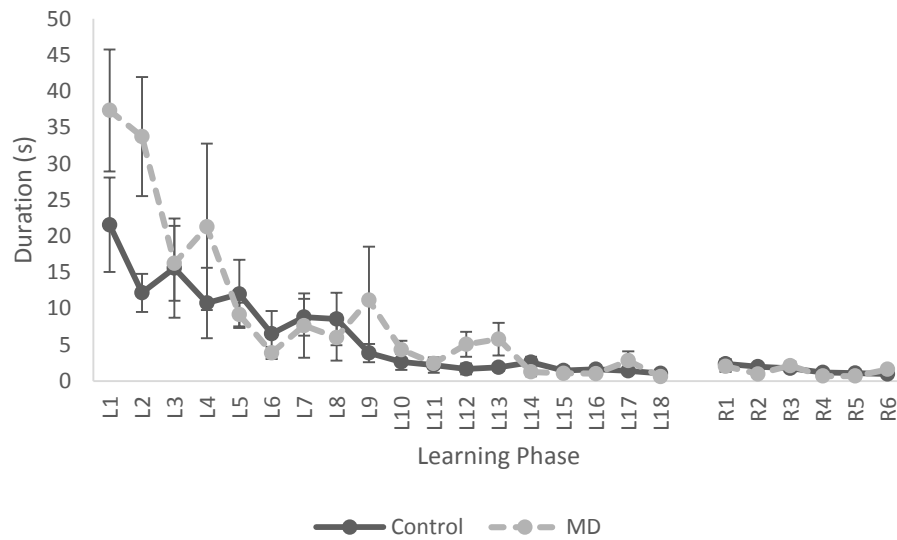


Figure H1. Line graph depicting the duration of time (in seconds) of body contraction in the WT MDs and controls. Error bars represent ± 1 standard error of the mean.

Appendix I

Reversal Learning Paradigm Stereotypy Data

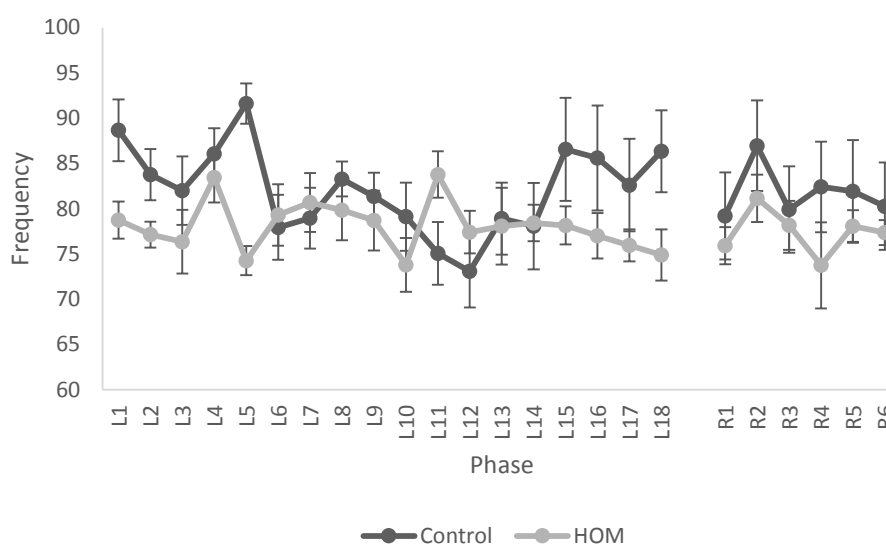


Figure I1. Line graph depicting the frequency of head weaving in the HOM and control subjects. Error bars represent ± 1 standard error of the mean.

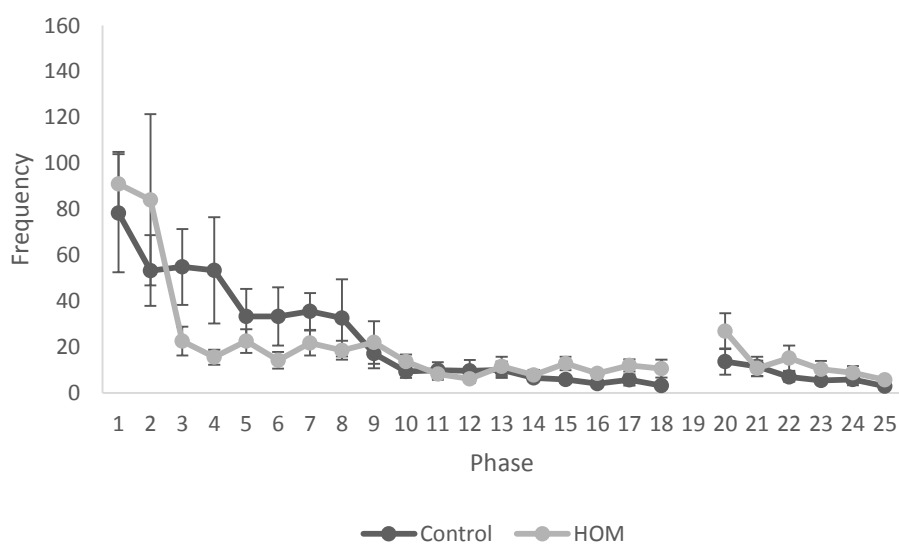


Figure I2. Line graph depicting the frequency of rotation in the HOM and control subjects. Error bars represent ± 1 standard error of the mean.

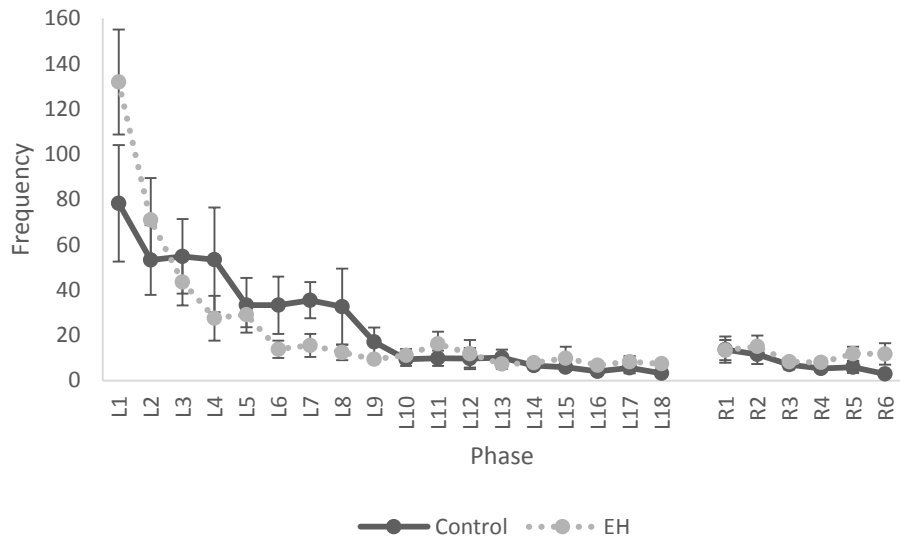


Figure I3. Line graph depicting the frequency of rotation in the EH and control subjects. Error bars represent ± 1 standard error of the mean.